



Transit Oriented Development for Future Cities



Key Messages to be Delivered.

Transit-Oriented Development (TOD) for Future Cities

- **Transit-Oriented Development (TOD)** promotes sustainable, walkable, and people-centered cities.
- **Transportation and land use** should be planned together to improve accessibility and reduce car dependence.
- **TOD models** help evaluate transit accessibility, urban development, and land-use planning.
- **Compact cities and the 15-minute city** support efficient and environmentally friendly urban growth.
- **Transportation analysis** can measure accessibility, equity, and network performance.
- **AI, smart cities, and autonomous mobility** will play an important role in future urban development.
- **Goal:** Create future cities that are **sustainable, accessible, equitable, and resilient** through integrated transportation and urban planning.

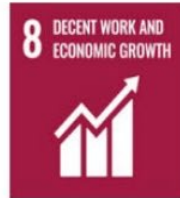
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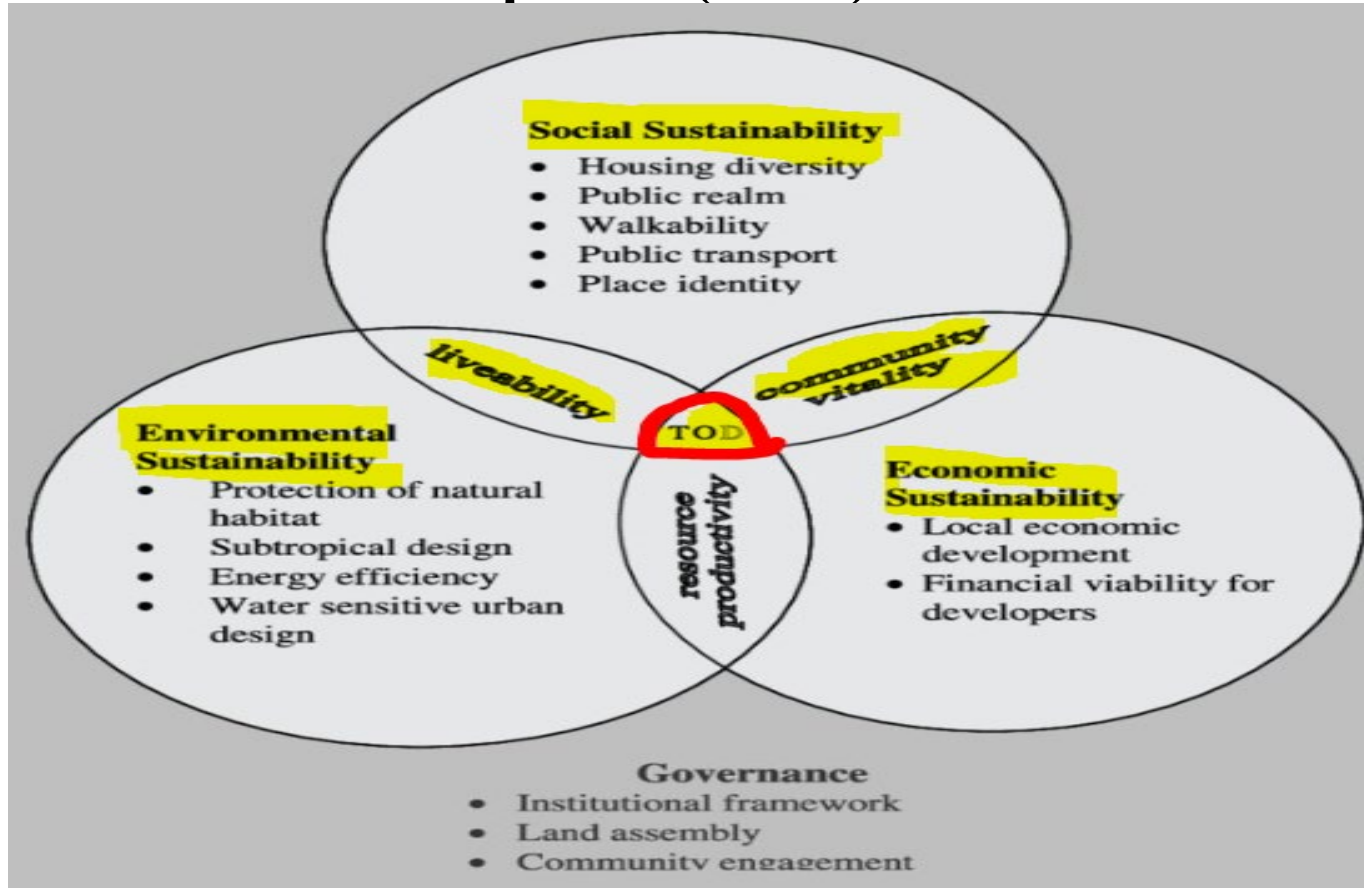


UN Sustainable Development Goals (SDGs)

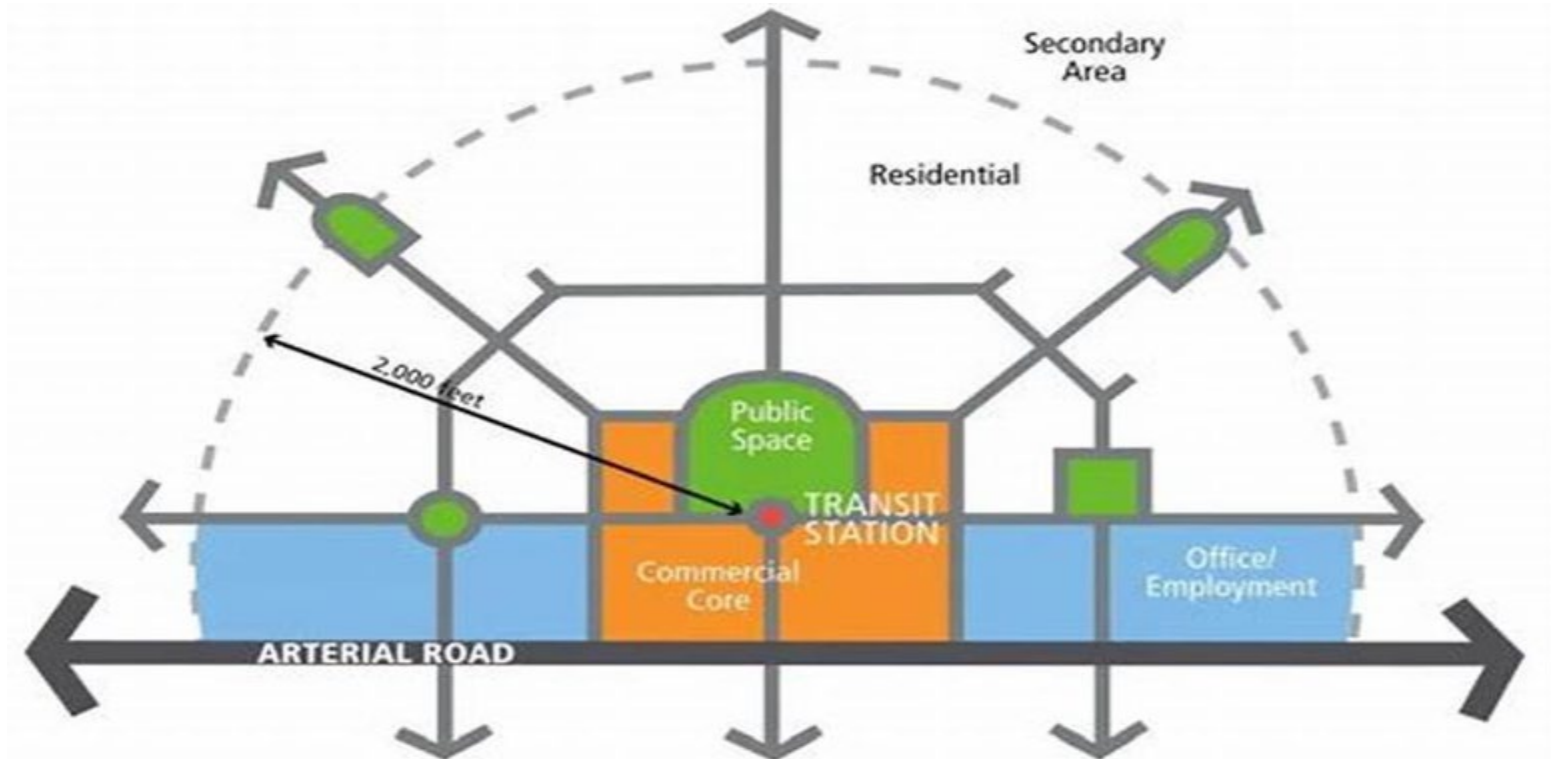
Global goals for sustainable development that the world must achieve by 2030



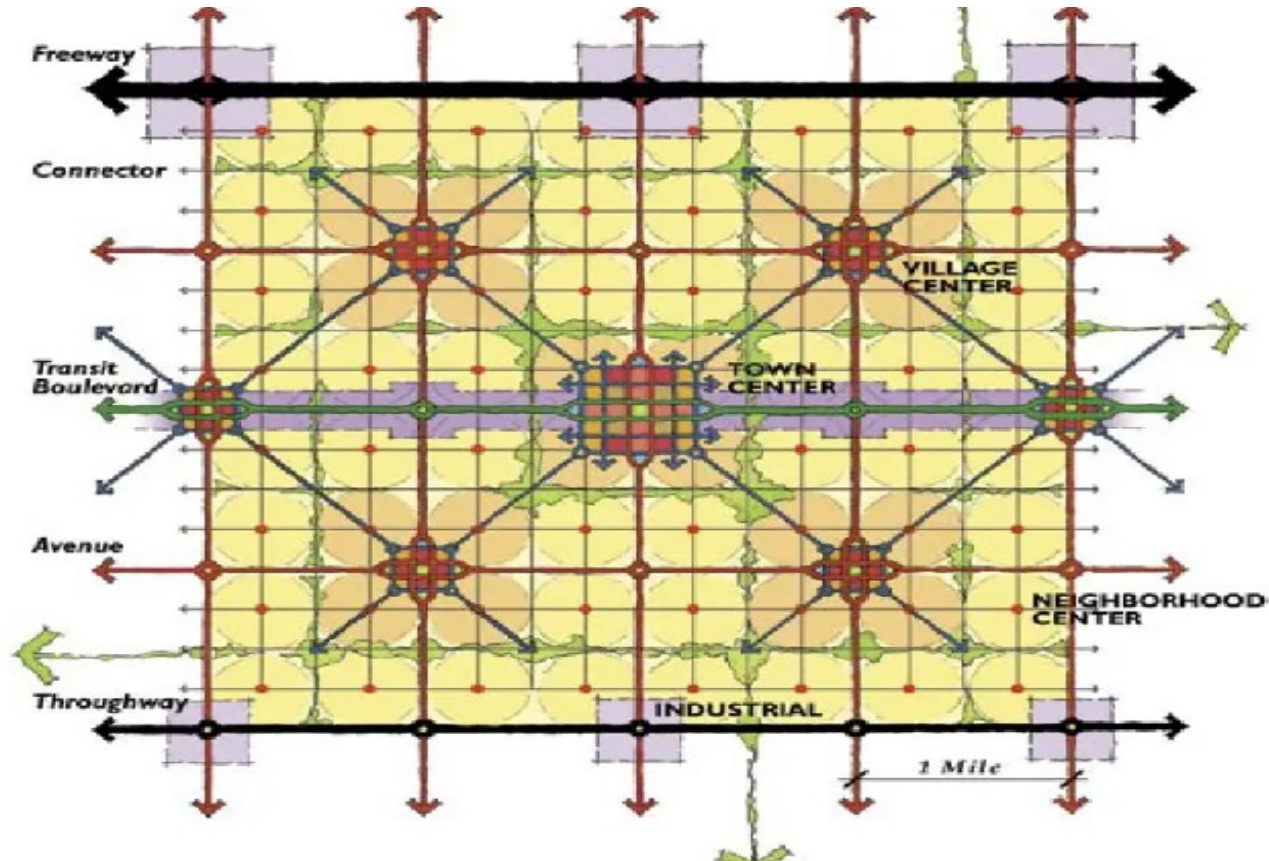
Framework for Sustainable Development Goals based on Transit Oriented Development (TOD) (S: Bhishna Bajracharya and Shahed Khan, 2006)



Transit Oriented Development (S: Peter Calthorpe 1993)

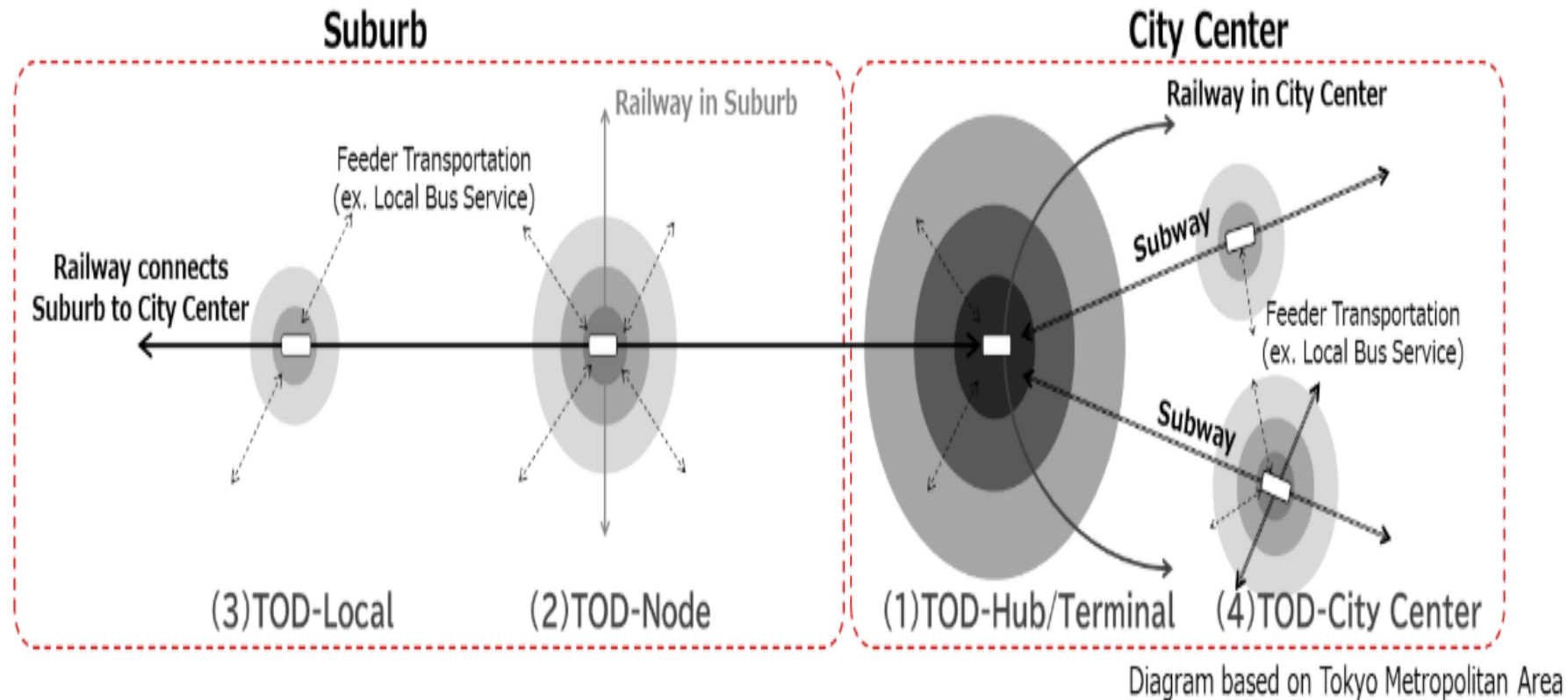


The Urban Network: A New Framework for Growth (S: Calthorpe and Fulton 2001)



4 Types of TOD in Japan

(S: [Transit Oriented Development \(TOD\) - Sharing our experience with the world - | U R 都市機構](#))





The Node and Place Model for TOD (S: L. Bertolini 1999, and J. Kim and N. Shin 2021)

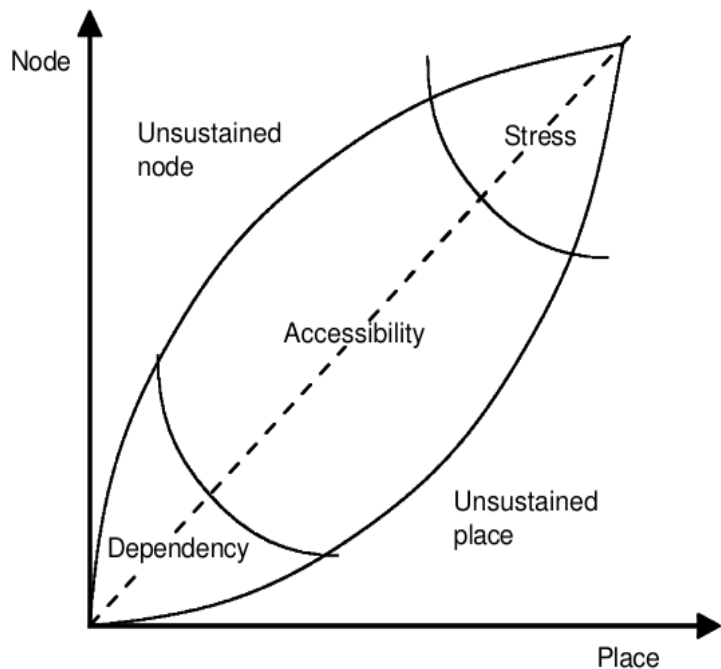
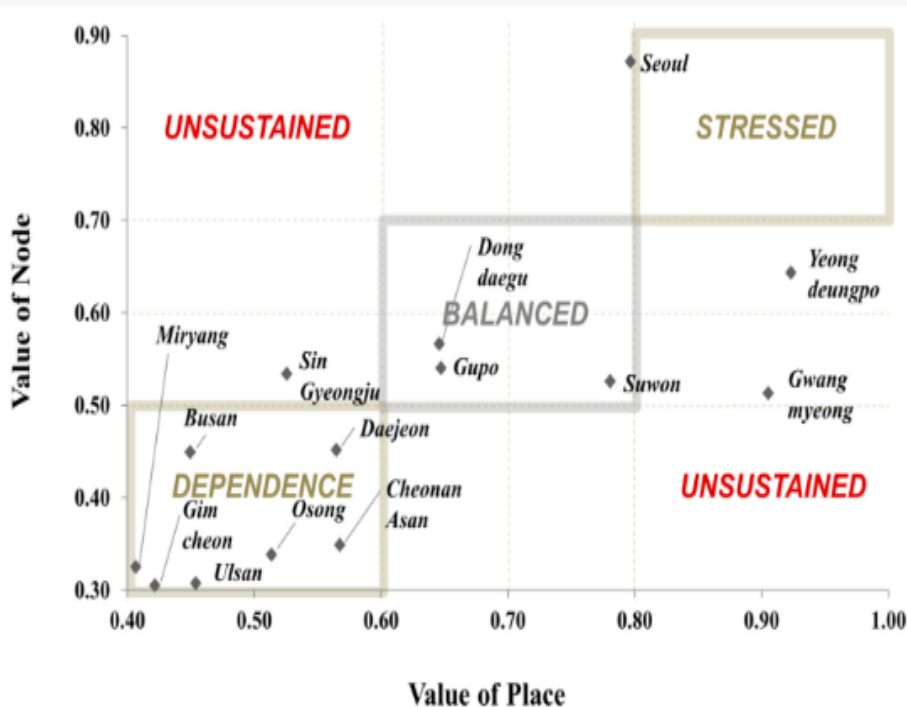


FIGURE 1. The node-place model.

Figure 2. Sustainability level based on place value (x-axis) and node value (y-axis).



Node for Transit Accessibility .vs. Place for Numbers of Jobs within Walking Distances of Stations (S: Wu, Lee and Levinson 2023)

The number of jobs within walking distance of a transit station is the best single predictor of transit ridership

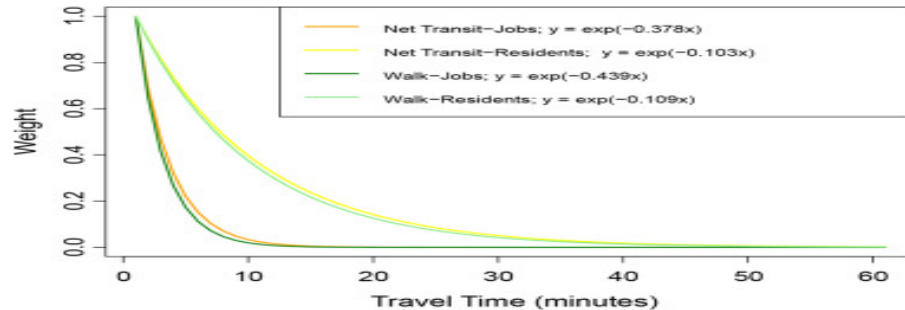
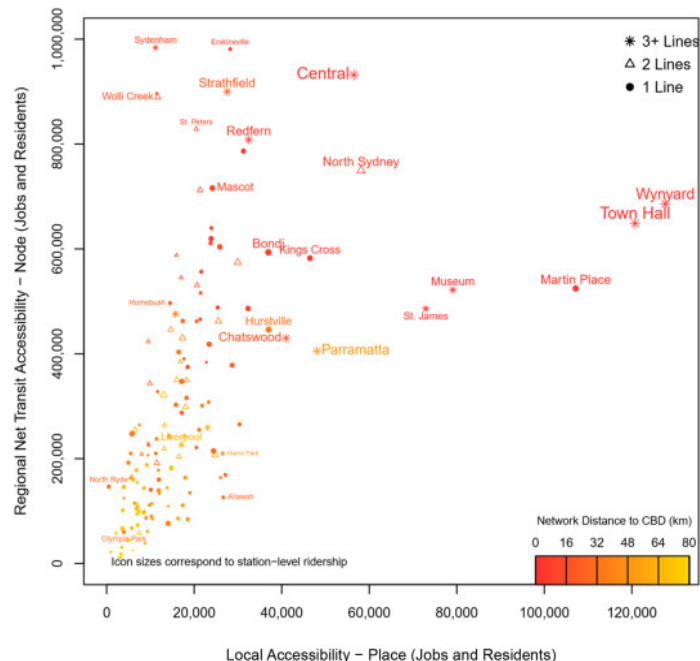


Table 5. Relative importance of access variables for station level transit ridership (all-day ridership).

Access Variables	Random Forest			Gradient Boosting			Linear Model			Average
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3	
(Discrete Travel Time)										
Walk 20 to residents	39.7%	–	21.3%	24.9%	–	13.3%	22.1%	–	10.2%	14.9%
Walk 10 to jobs	60.3%	–	41.2%	75.1%	–	50.9%	77.9%	–	81.3%	57.8%
NetTransit 30 to residents	–	76.0%	28.0%	–	68.2%	27.8%	–	96.9%	4.7%	20.2%
NetTransit 30 to jobs	–	24.0%	9.5%	–	31.8%	8.0%	–	3.1%	3.8%	7.1%

The Node-Place-Design Model for TOD (S: Y. Yang et al 2022)

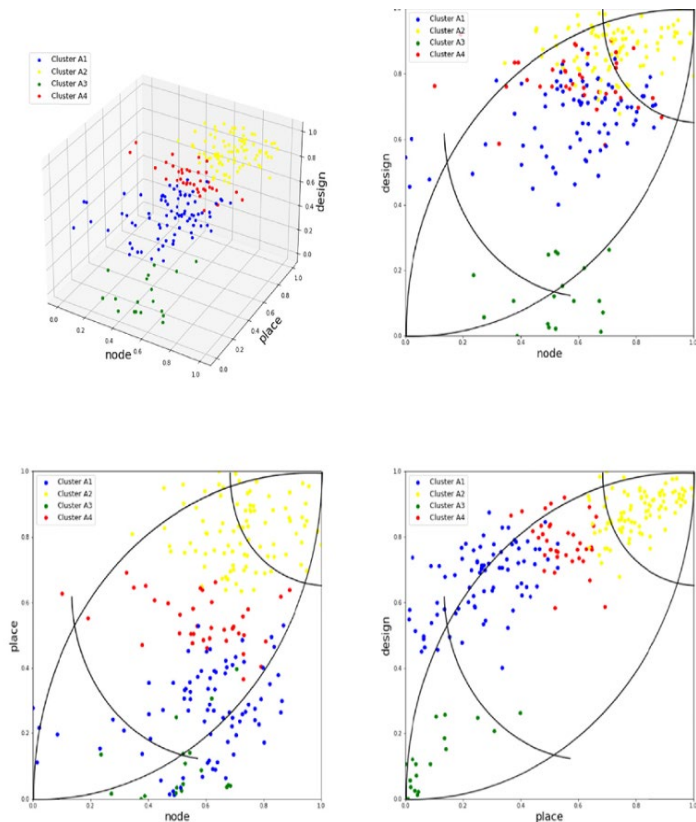


Fig. 8. Scatter plots of Guangzhou station clusters according to node, place, and design scores.

Table 1

TOD index components for NPD models and urban vibrancy.

Dimension	Indicator description	Measurement	Data source
Node	Directions offered by the station	X1 = Number of directions offered by the station (number of metro lines * 2 directions for each line)	Metro timetables
	Metro service	X2 = Average frequency of trains over the day	Metro timetables
	Bus service	X3 = Number of bus stops around the station	Bus lines and stops data from Gaode
	Betweenness centrality of the metro network	X4 = Local betweenness centrality weighted by straight-line distance between stations	Metro lines and stops data from Gaode
	Closeness centrality of the metro network	X5 = Local closeness centrality weighted by straight-line distance between stations	Metro lines and stops data from Gaode
Place	Population	Y1 = Population density determined from raster grid (/km ²)	Worldpop data
	Establishment facilities	Y2 = Number of establishment POIs	POI data
	Leisure facilities	Y3 = Number of leisure POIs	POI data
	Catering and restaurant facilities	Y4 = Number of catering and restaurant POIs	POI data
	Administration facilities	Y5 = Number of administration POIs	POI data
Design	Education facilities	Y6 = Number of education POIs	POI data
	Retail facilities	Y7 = Number of retail POIs	POI data
	Hotel facilities	Y8 = Number of hotel POIs	POI data
	Morphology	Z1 = Average building area (m ²)	Building data
	Compactness	Z2 = Average height of buildings (m)	Building data
Vibrancy	Pedestrian network for Topology	Z3 = Total length of pedestrian paths (m)	Street network data
	Road intersections for topology	Z4 = Number of various levels of road intersections	Street network data
	Station functionality	Z5 = Standard distance of the network to the station centre	Street network data
	Daily outflow	Number of trips from the station to the whole city	Mobile phone data
	Daily inflow	Number of trips from the whole city to the station	Mobile phone data
	Night-time activities	Average night-time light value for the year	Night-time light data

Network Criticality of Node-Place-Design Model (S: Zhang, Marshall, Manley 2019)

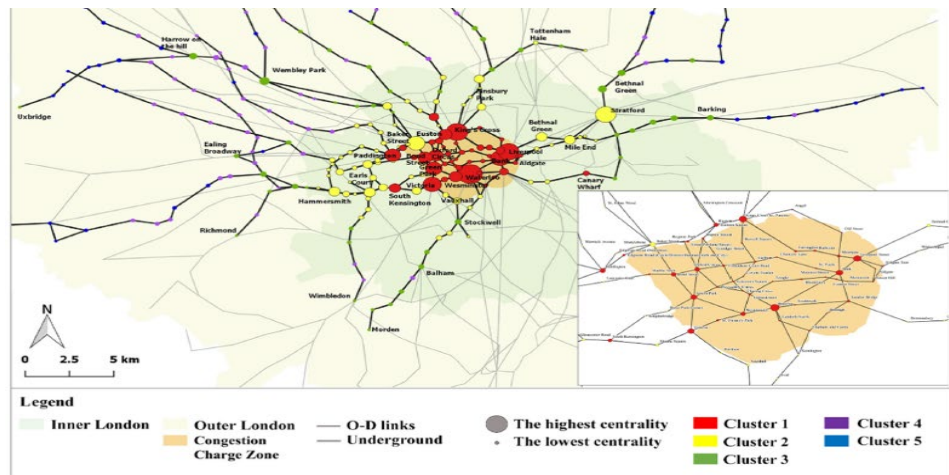


Fig. 5. The classification of metro station areas in Greater London based on the combination of criticality and node-place-design model (each station is represented by a dot, whose colour indicates the station's corresponding cluster, and whose size indicates the station's criticality level in the system).

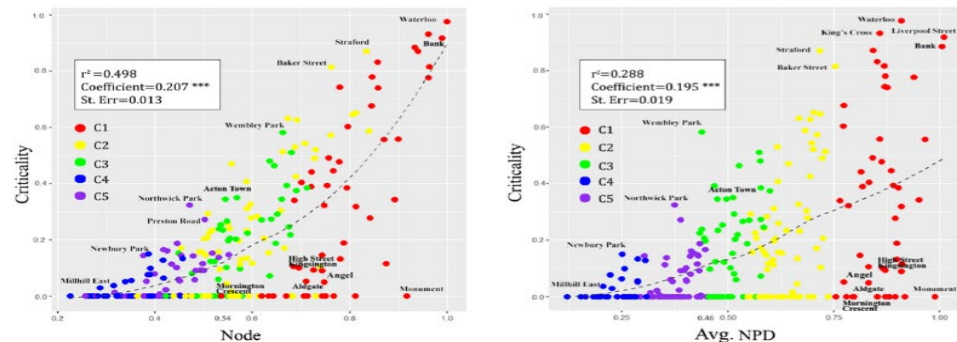
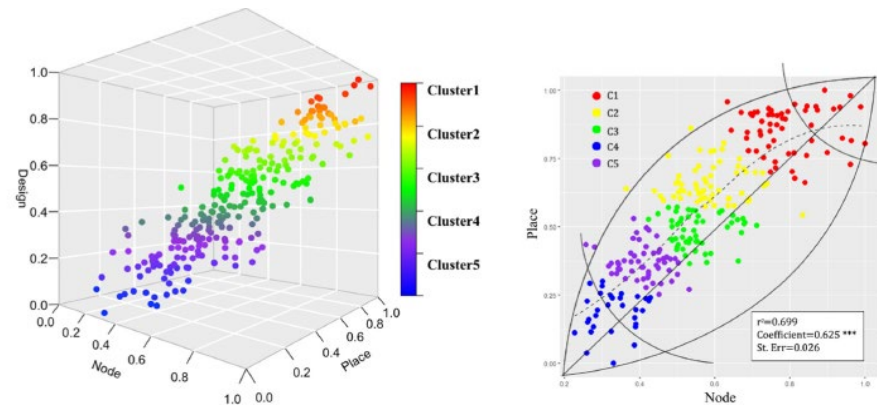
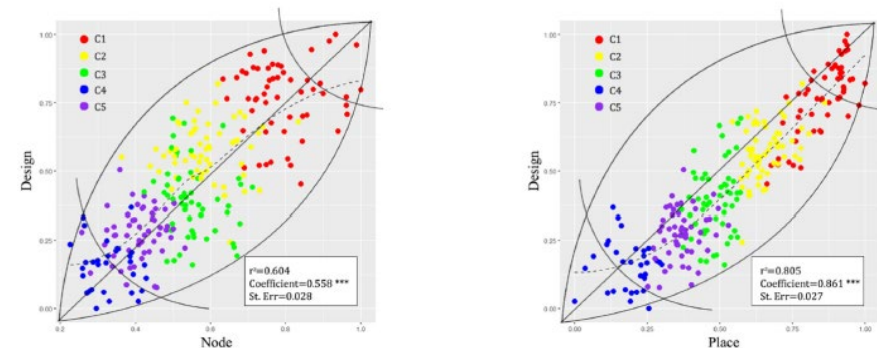


Fig. 6. Scatter plot of node-criticality and NPD-criticality.



(a) Scatter plot of node, place and design

(b) Scatter plot of node and place



(c) Scatter plot of node and design

(d) Scatter plot of place and design

Fig. 4. Scatter plot of London metro stations in the node-place-design model from a synthetic and three paired perspectives.

Review on the Node-Place Models for TOD (S: Kang and Miwa 2025)

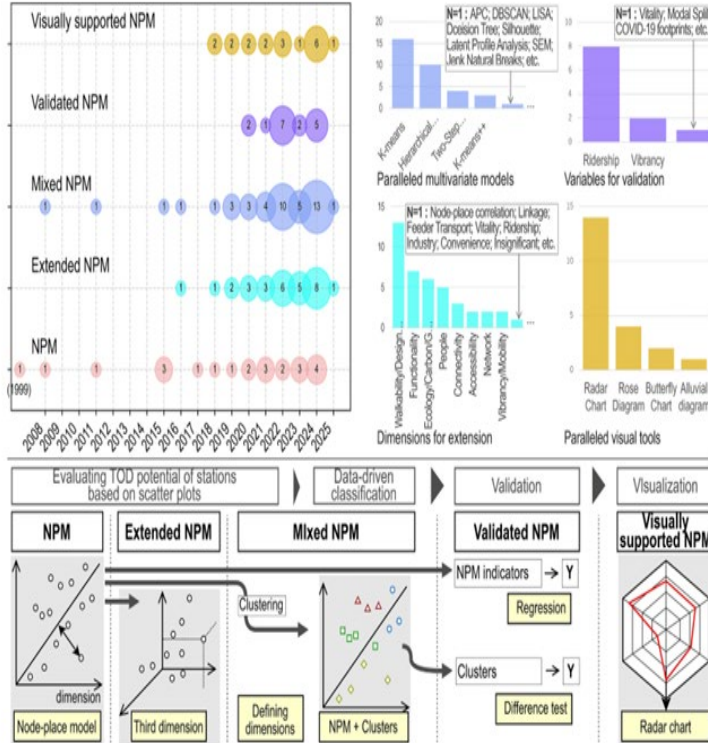


Fig. 1. Yearly publication trends and details based on methodological types.

Table 2
Indicators for methodological comparison.

Dimension and indicator	Explanation	CMA (a)		Nagoya (b)	
		Mean	SD	Mean	SD
Node dimension					
A1. Active transit accessibility	Number of workers accessible from the station within 30-min travel based on transit and pedestrian network	288,405	(399680)	941,349	(331115)
A2. Passive transit accessibility	Number of residents accessible to the station within 30-min travel based on transit and pedestrian network	344,237	(419282)	1,025,811	(398427)
R1. Railway directions	Number of directions of train departing the station	5.93	(5.94)	6.24	(6.06)
R2. Railway frequency	Frequency of trains departing the station	303.07	(406.01)	597.17	(671.05)
R3. Stations within 30-min travel	Number of stations accessible within 30-min travel on railway network	33.13	(66.46)	114.60	(110.44)
B1. Bus directions	Number of directions of bus within the station area	19.27	(26.69)	55.56	(35.93)
B2. Bus frequency	Frequency of buses serving within the station area	300.40	(500.90)	1073.03	(869.12)
B3. Bus stops	Number of bus stops within the station area	7.32	(6.15)	11.53	(6.94)
C1. Car parking capacity	Number of car parking lots within the station area	510	(1528)	1724	(2879)
C2. Proximity to expressways	Shortest network distance from the station to the expressway interchange (km)	5.36	(4.72)	1.60	(0.92)
T1. Proximity to city centers	Travel time from the station to Sakae or Nagoya based on transit network (minutes)	72.00	(52.80)	18.05	(10.18)
T2. Transit degree centrality	Number of directly connected nodes on transit network	0.0050	(0.0295)	0.0136	(0.0545)
T3. Transit betweenness centrality	Frequency that node functions as a bridge between other nodes	0.0097	(0.2219)	0.0046	(0.0267)
T4. Transit closeness centrality	Inverse of summed shortest distance to other connected nodes	0.0005	(0.0012)	0.0011	(0.0023)
T5. Ridership	Number of passengers using the station for a day	8090	(39696)	24,355	(80424)
Place dimension					
P1. Residents	Number of residents within the station area	7731	(6401)	16,530	(6486)
P2. Retail/hotel and catering	Number of workers within the station area in each economic cluster (P2, P3, P4, P5)	1854	(5774)	6306	(11576)
P3. Education/health/culture		1381	(2962)	4274	(5500)
P4. Administration and services		1988	(8455)	7423	(17432)
P5. Industry and distribution		996	(1864)	2662	(3478)
P6. Functional mix		0.91	(0.01)	0.92	(0.01)
$1 - \frac{\left(\frac{a-b}{d} - \frac{a-c}{d}\right)}{2} \text{ with } \begin{cases} a = \max(P1, P2, P3, P4, P5) \\ b = \min(P1, P2, P3, P4, P5) \\ c = \frac{P1 + P2 + P3 + P4 + P5}{5} \\ d = P1 + P2 + P3 + P4 + P5 \end{cases}$					
Walkability dimension					
W1. Street line density	Length of walkable streets divided by station area	21,376	(5955)	26,917	(4310)
W2. Intersection density	Number of street intersections divided by station area	113.96	(55.36)	140.67	(43.77)
W3. Average distance to residents	Average distance from the station to residents within the station area (m)	688.61	(67.90)	714.84	(51.71)
W4. Average distance to workers	Average distance from the station to workers within the station area (m)	727.60	(104.5)	762.44	(90.31)

*Each station area was determined as the area reachable within 10 min at a walking speed of 5 km/h, based on pedestrian network analysis.

A Node–Place Model to Explore the Land Use–Transport Development Dynamics of the I-287 Corridor (S: Baghestani et al 2023)

Figure 2. Framework of application of the node–place model for the I-287 Corridor.

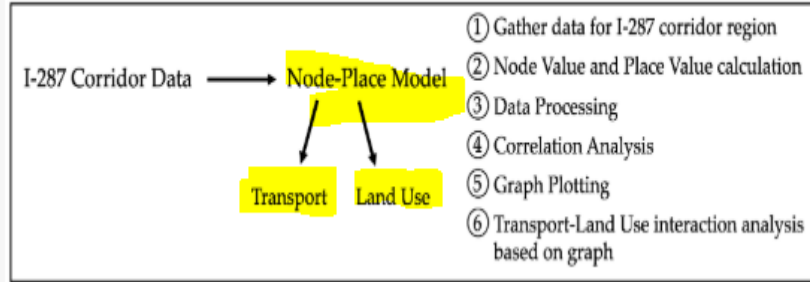


Figure 3. Study area: I-287 Corridor and adjacent areas with potential development lots.

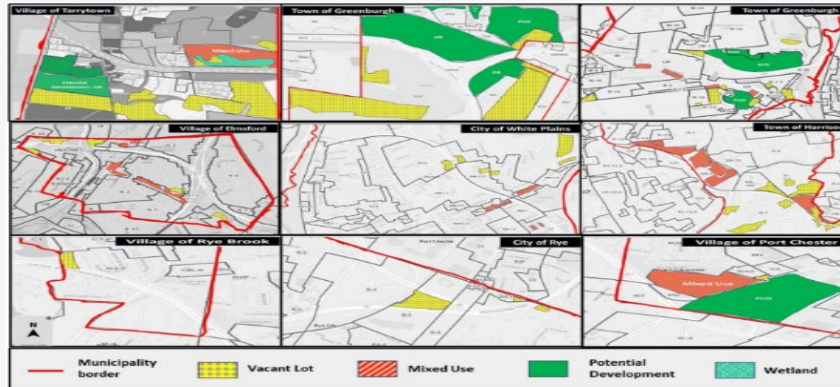
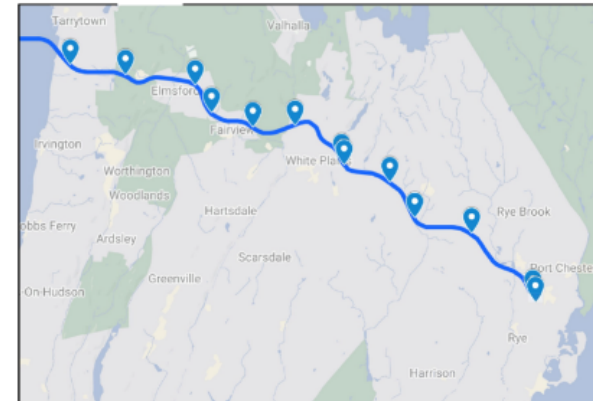
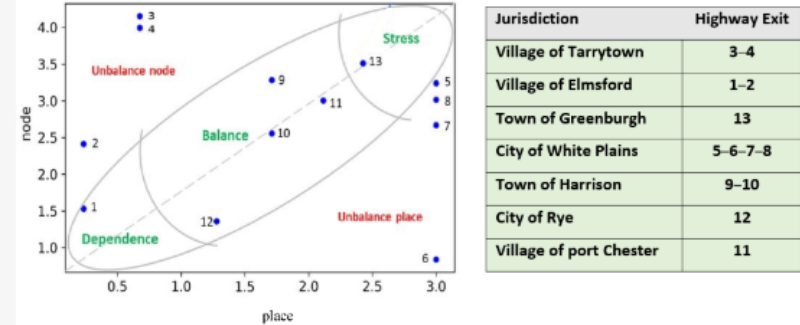


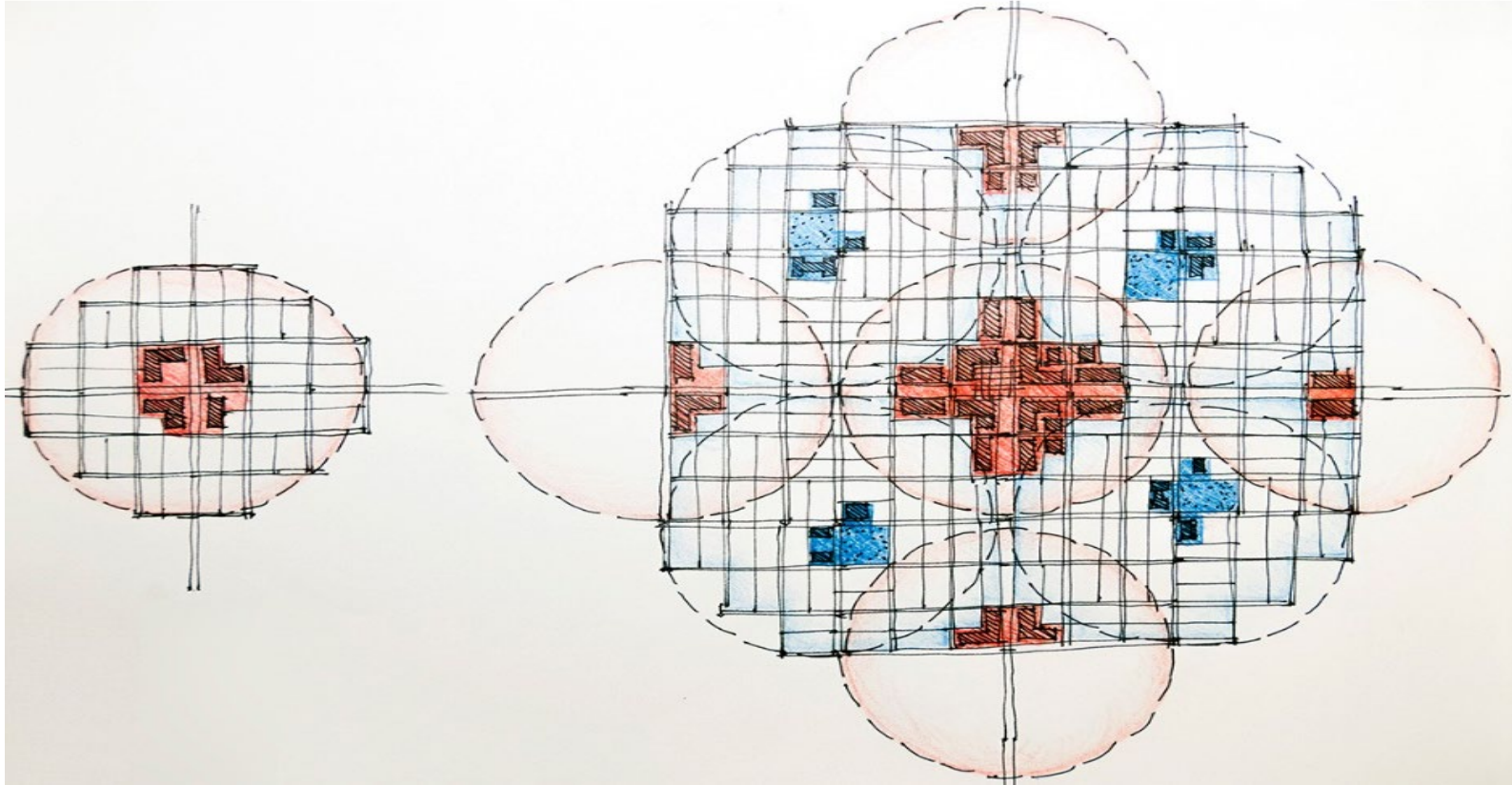
Figure 4. The output of the node–place model for the case of the I-287 Corridor exits.





Connecting Neighborhoods in a Continuous Urban Fabric

(S: Steve Mouzon 2015)



Urban Fabrics and Metabolism

(S: D. Shillinglaw, 2015 and Thomson and Newman 2017)



(Scream in London, United Kingdom)

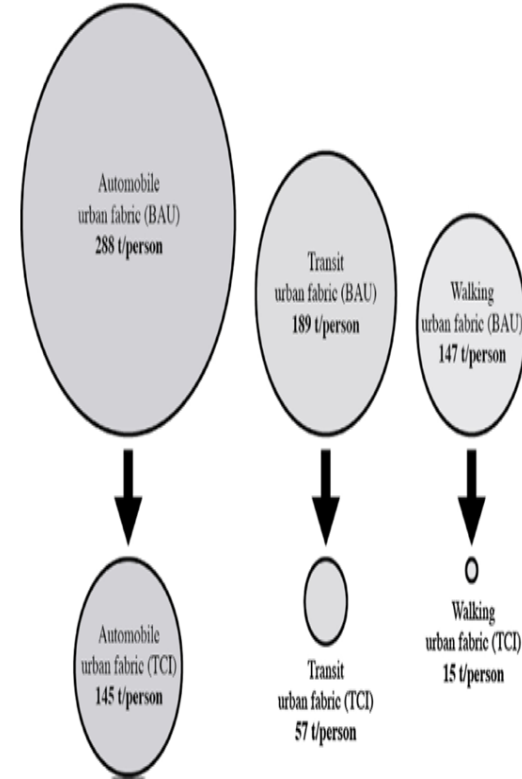
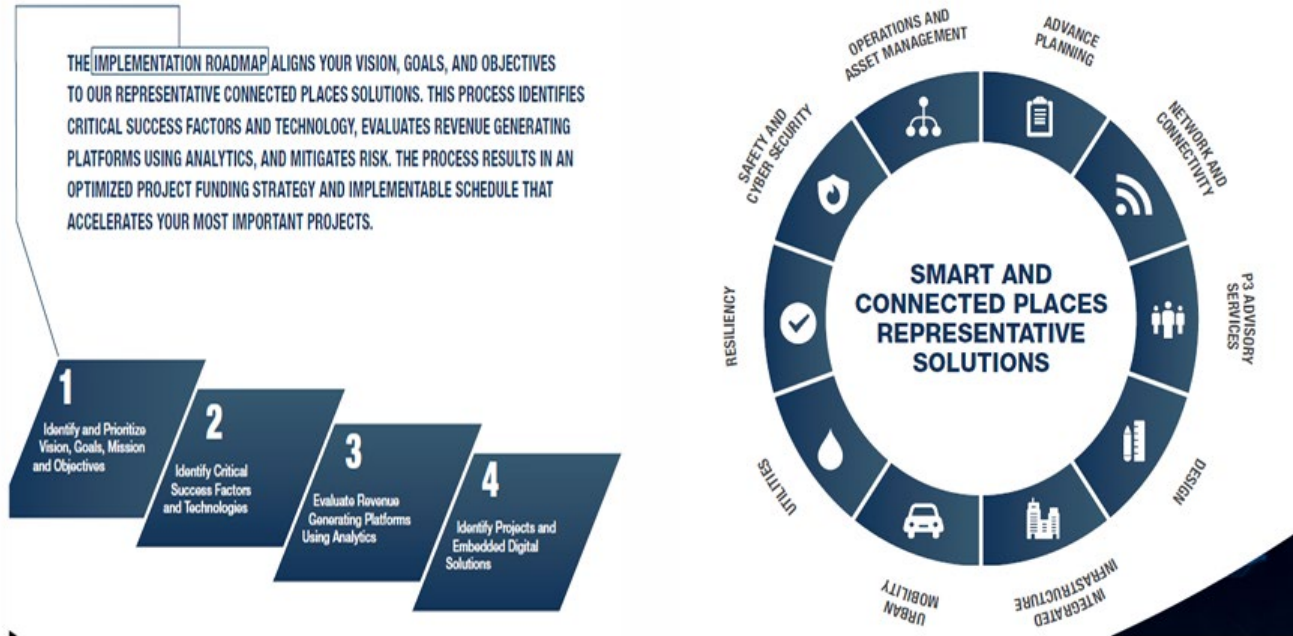


Fig. 4. Perth's basic raw material demand in terms of three urban fabrics plus Technology and Construction Innovation (TCI) Adapted from: Gardner and Newman (2013).

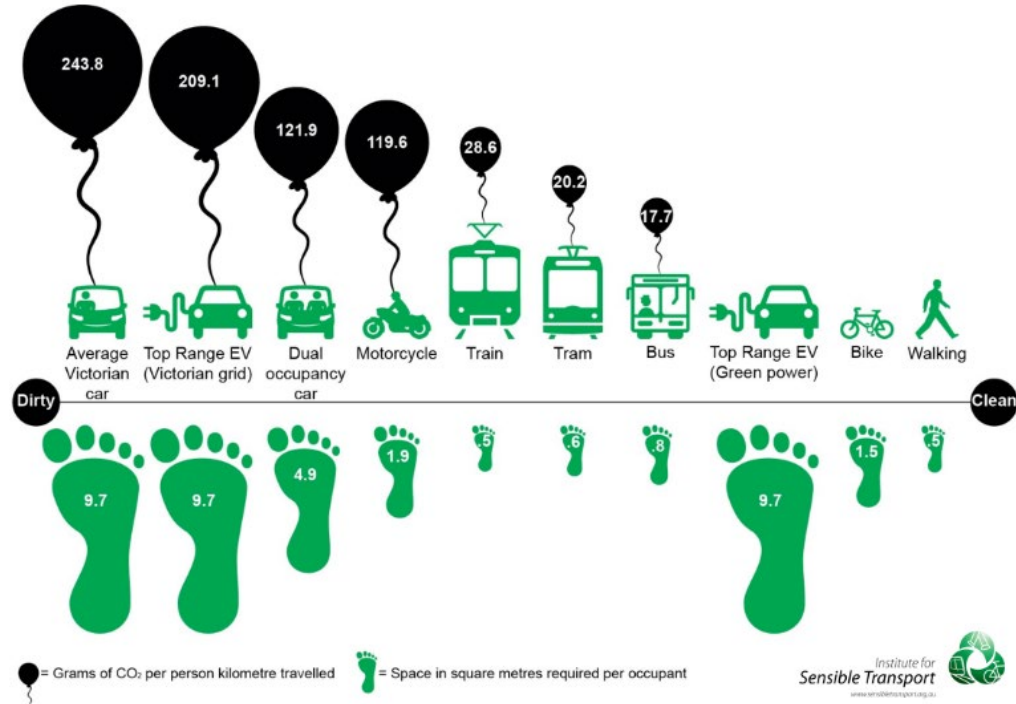
UN Sustainable Development Goals (SDGs) for Creating Value for People and Planet

Roadmap that outlines strategic steps and representative solutions for smart and connected urban spaces to achieve the Sustainable Development Goals (SDGs).



Transport and Climate Change

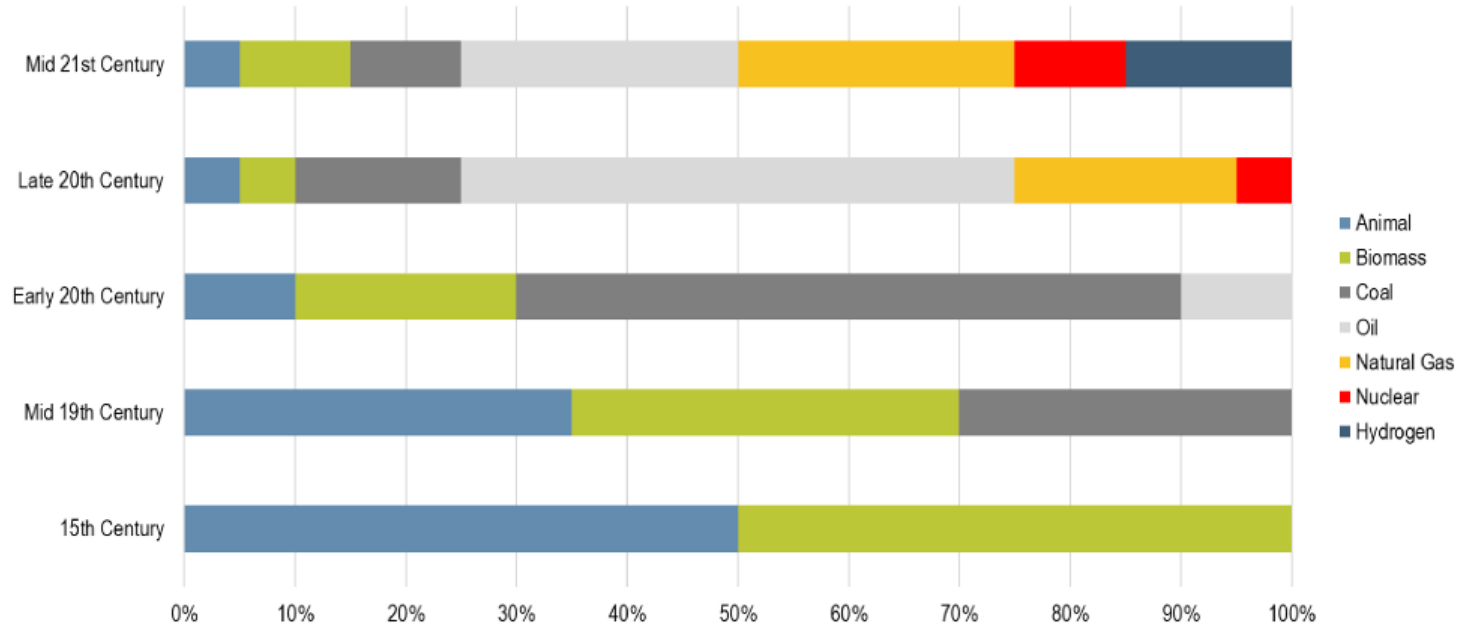
The potential to significantly reduce CO2 emissions by shifting from private cars to public transportation





Energy Types Used

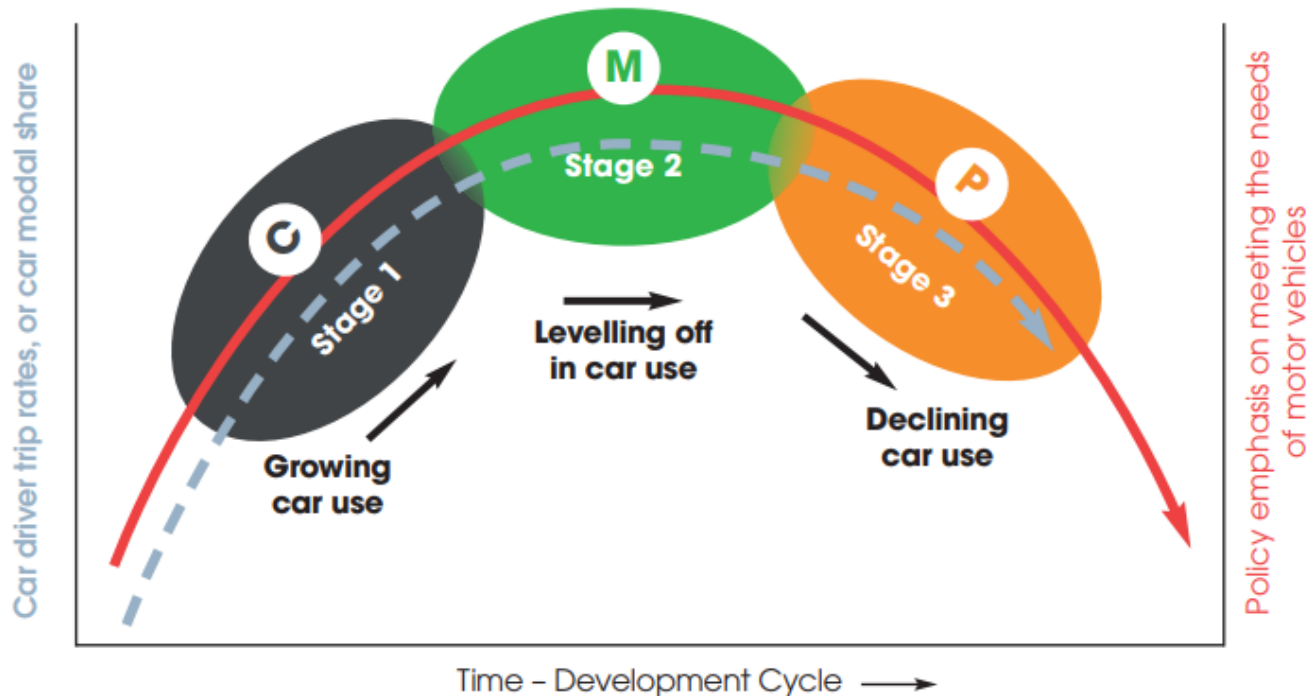
The transition of Energy Types Used from coal and oil to eco-friendly sources like hydrogen and biomass reflects the ongoing shift in the industrial era



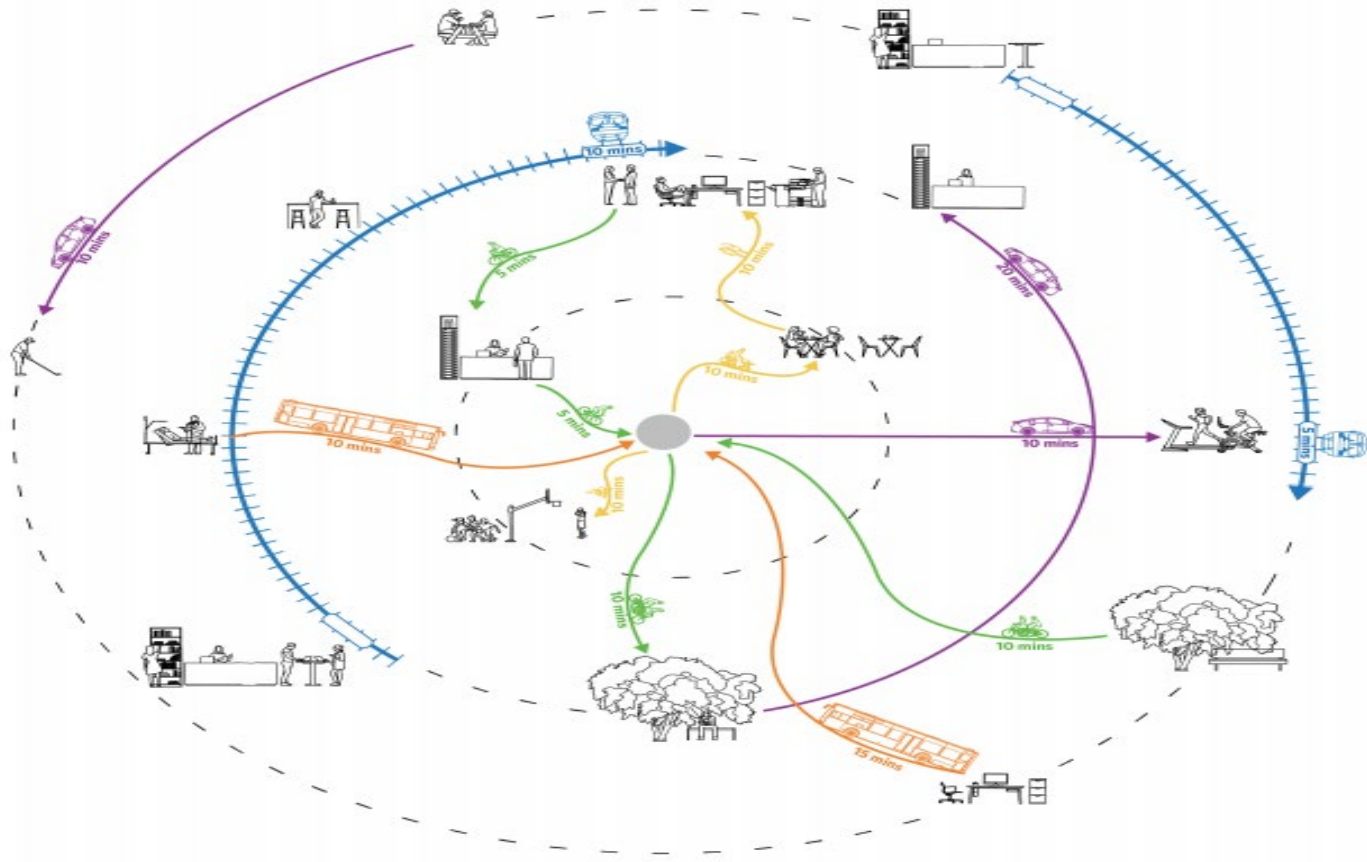


Shift in urban transportation policy, EU CREATE project

The Shift in urban transportation policy illustrates the evolution of major European cities

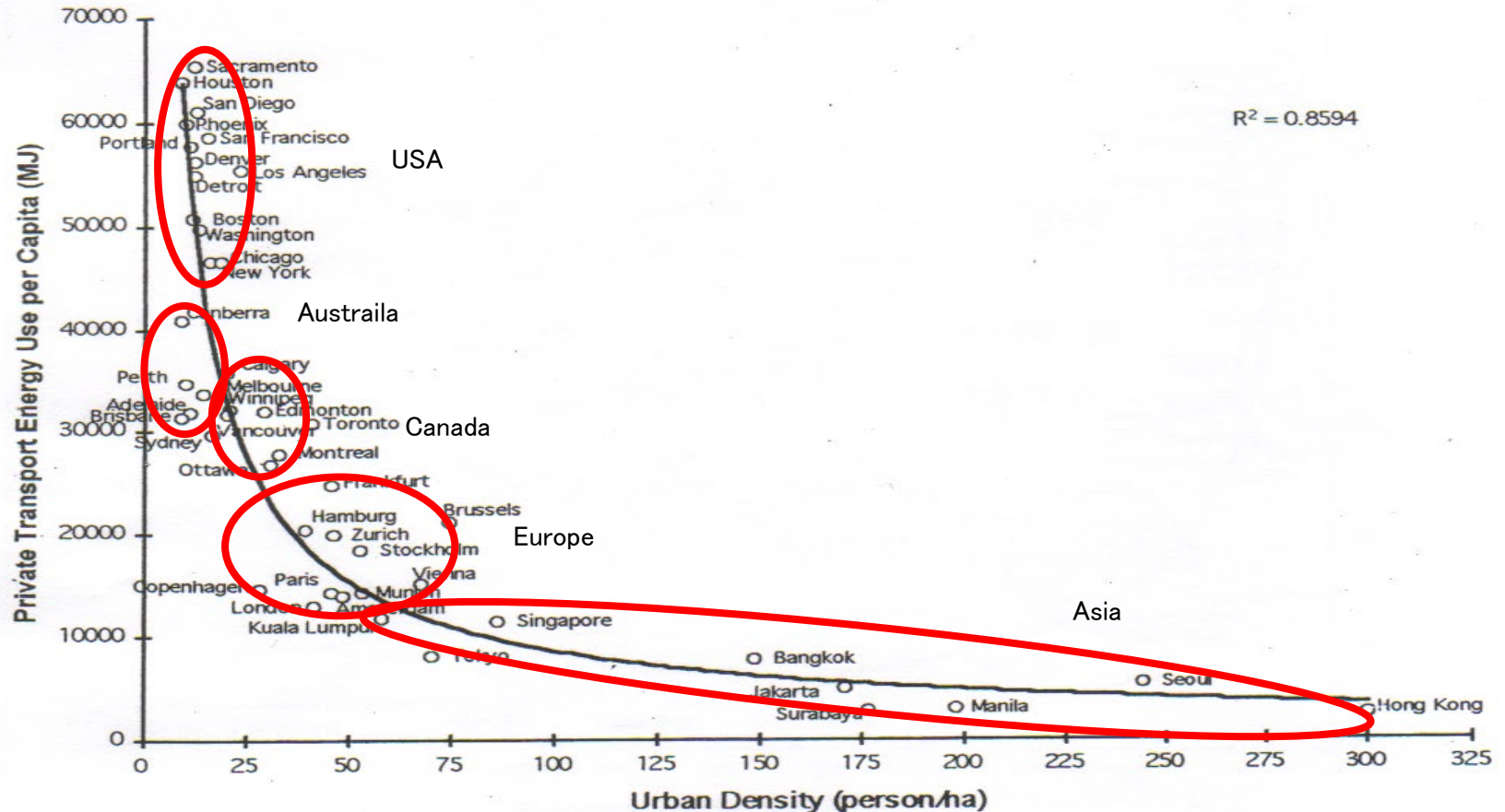


Access, Movement and Place



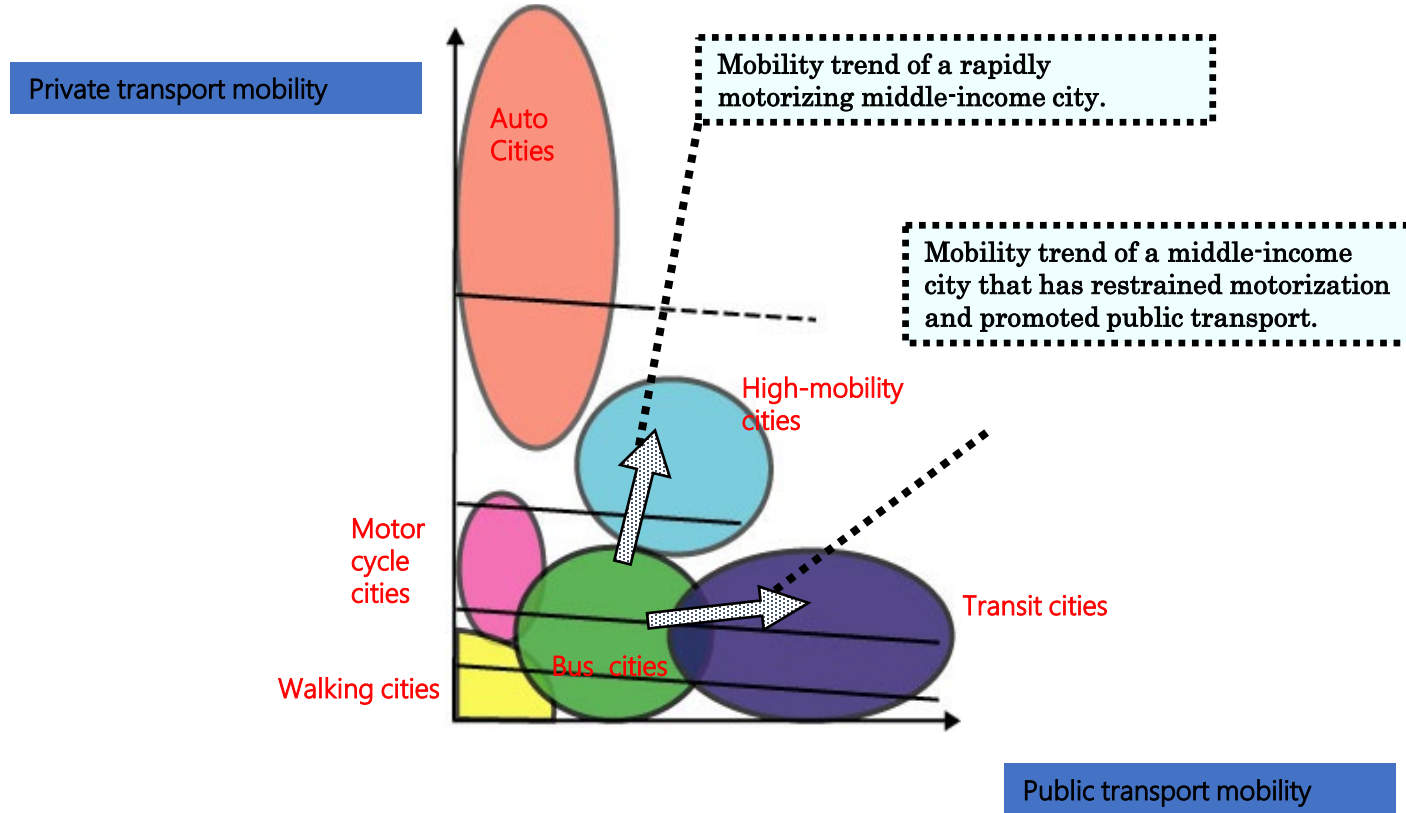
Private Energy Use **per Capita** and Urban Density

Source : Newman, P and Kenworthy, J. (1999) *Sustainability and Cities – Overcoming Automobile Dependence*



How to Transform Mobility Trends according to City Types and Urban Densities.

(Source: Barter, 1999)



As for LCS visions, we prepared two different but likely future societies

Vision A	Vision B
Vivid, Technology-driven	Slow, Natural-oriented
Urban/Personal	Decentralized/Community
Technology breakthrough Centralized production /recycle	Self-sufficient Produce locally, consume locally
Comfortable and Convenient	Social and Cultural Values
	 Akemi Imagawa

2. We need “Visions”

Towards an Urban Renaissance

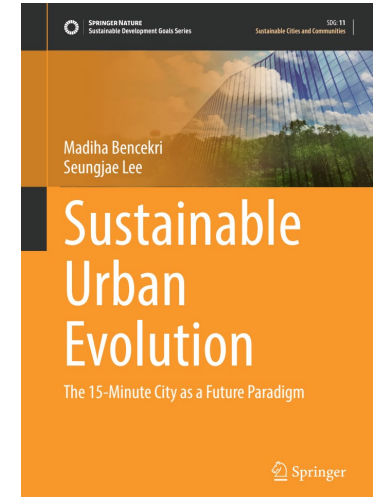
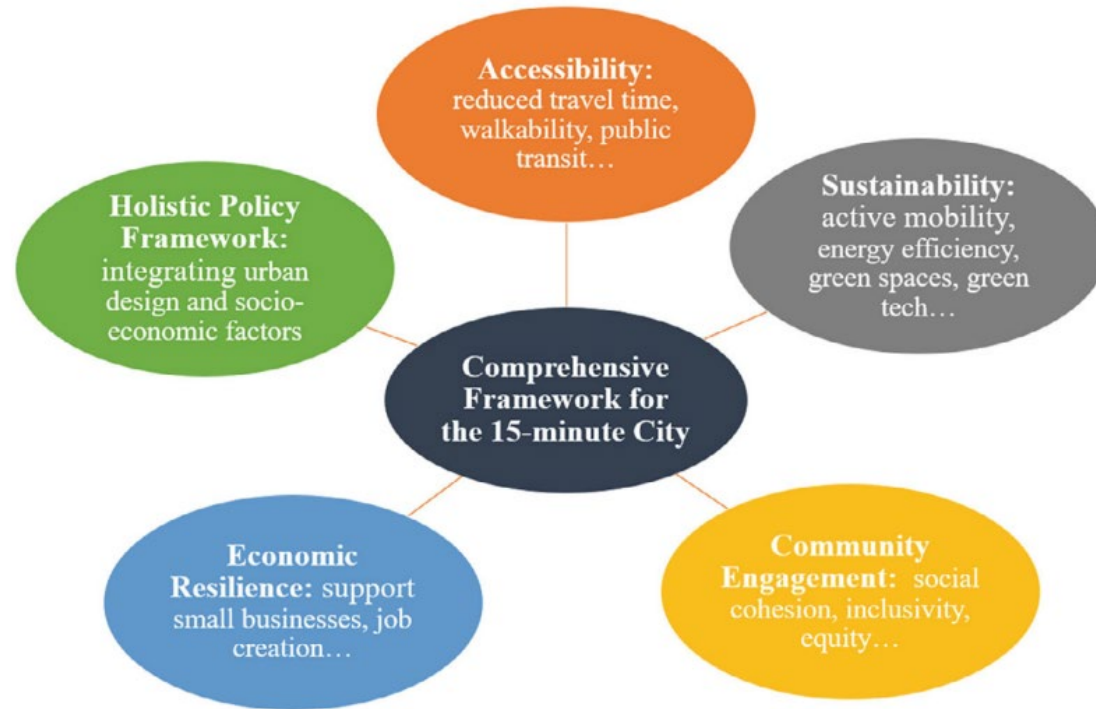
(Compact, Transit Oriented Development)

S:UK Urban Task Force, 'Towards an Urban Renaissance'



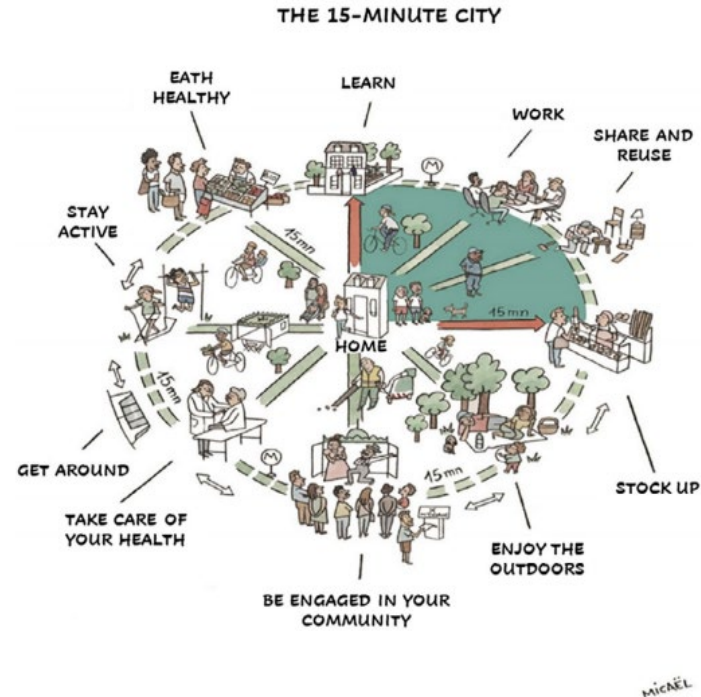
Sustainable Urban Evolution

Achieving **sustainable urban evolution** requires a framework, such as the 15-minute city concept, which addresses regional development and economic resilience

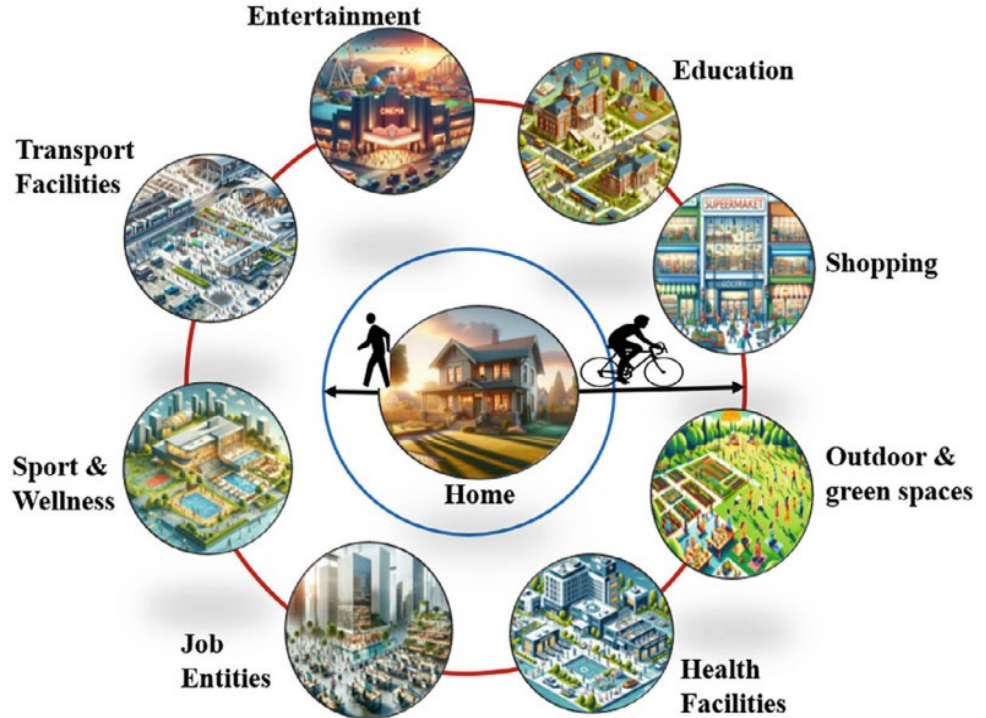


The 15-minute city

The **15-minute city** aims to improve urban accessibility by ensuring essential daily infrastructure, such as work and leisure, within a short walk or bike ride

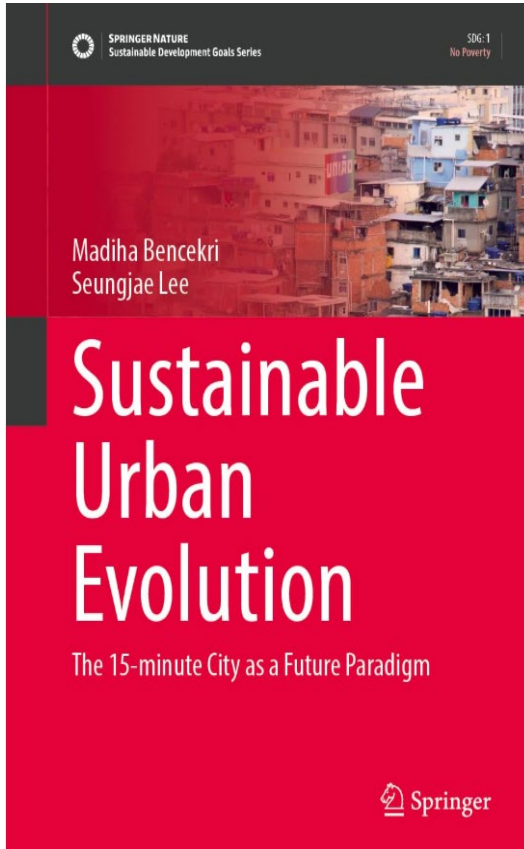


S: Crook, L. (2021)



S: Bencekri, M., & Lee, S. (2025)

How to Make it in the framework of Chrono Urbanism



- **Chapter 1: Urban Planning** lays the groundwork for exploring the historical development of urban planning and transportation systems, with an emphasis on the growing importance of sustainability in shaping modern cities.
- **Chapter 2: 15-Minute City Model** introduces the theoretical framework, benefits, and challenges of the 15-minute city, providing a clear understanding of its transformative potential.
- **Chapter 3: Time and Space Dimensions** examines the interplay of time and space in the 15-minute city model, focusing on optimizing spatial planning and temporal flexibility through innovative technologies and case studies.
- **Chapter 4: Community Engagement** highlights the essential role of community participation in the planning process, offering strategies and examples of successful community-led initiatives to foster inclusivity and equity.
- **Chapter 5: Smart 15-Minute Cities** discusses the integration of smart technologies into 15-minute city planning. Data-driven planning, intelligent infrastructure, and digital tools and applications can enhance sustainability, resilience, inclusivity, liveability, and well-being in urban environments.
- **Chapter 6: Applications of the 15-Minute City Model** focuses on implementing the model through strategies such as mobility hubs, enhanced walkability, and bikeability, while addressing social and economic concerns such as equity, inclusivity, and mental health.
- **Chapter 7: Guideline for a 15-Minute City** synthesizes key insights from previous chapters, addresses implementation challenges, and outlines a vision for adopting the 15-minute city model globally as a sustainable, inclusive, and resilient paradigm.

Chrono Urbanism: 15 Minutes Paris (source: Carlos Moreno, UN HABITAT, 2022)

Figure 6.6: The 15-Minute Paris



Diagram of 15 Minute City

(S:[Defining the 15-minute city | CNU](#))

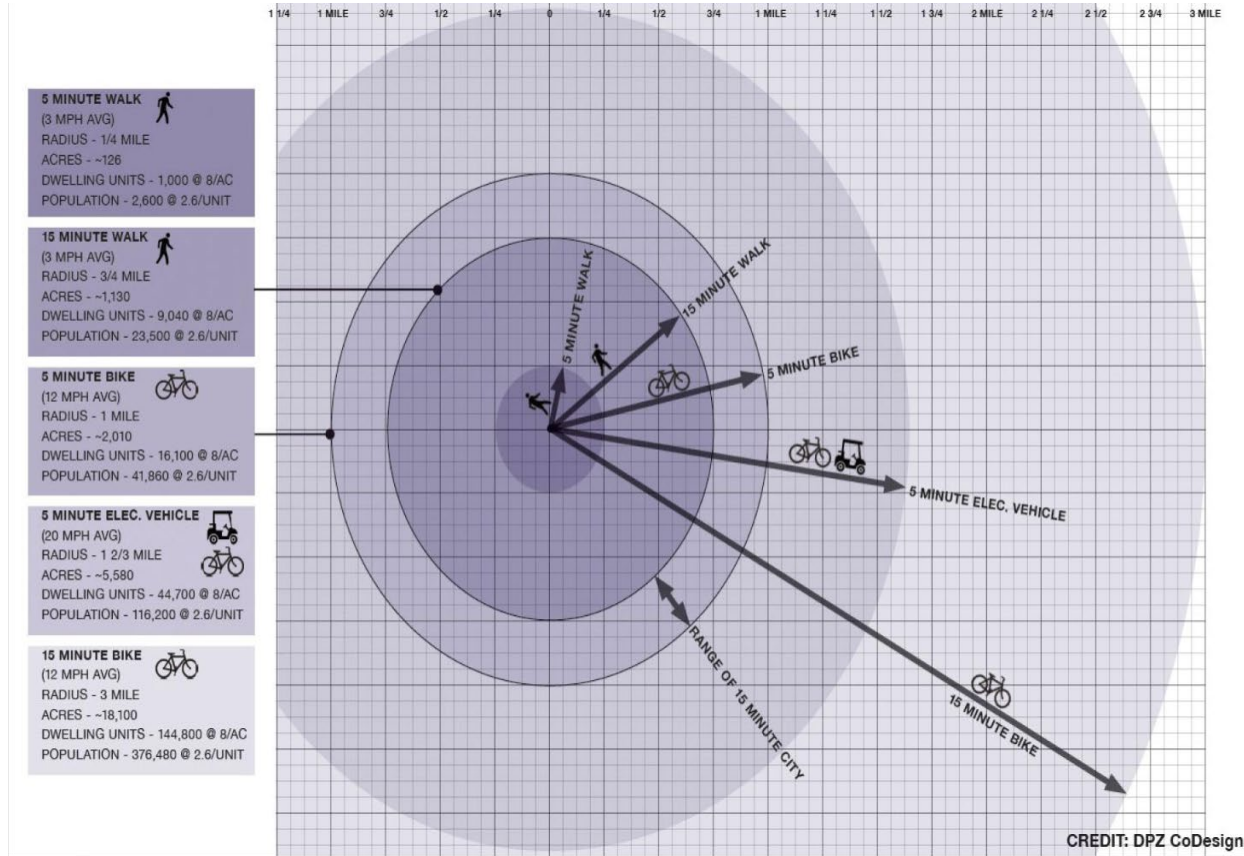


Diagram of 15 Minute City

(S:[Defining the 15-minute city | CNU](#))

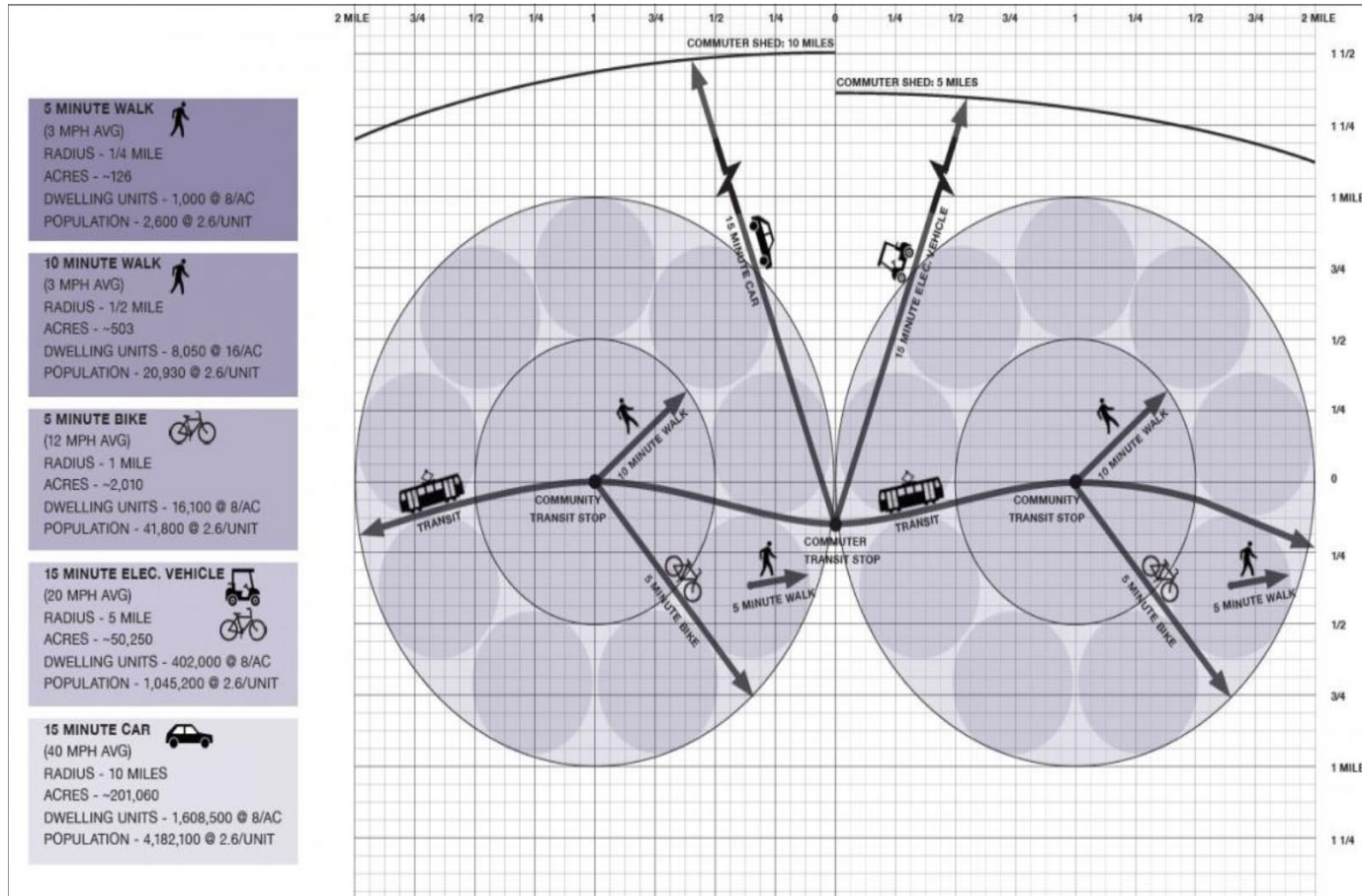


Diagram of 15 Minute City

(S:[Defining the 15-minute city | CNU](#))

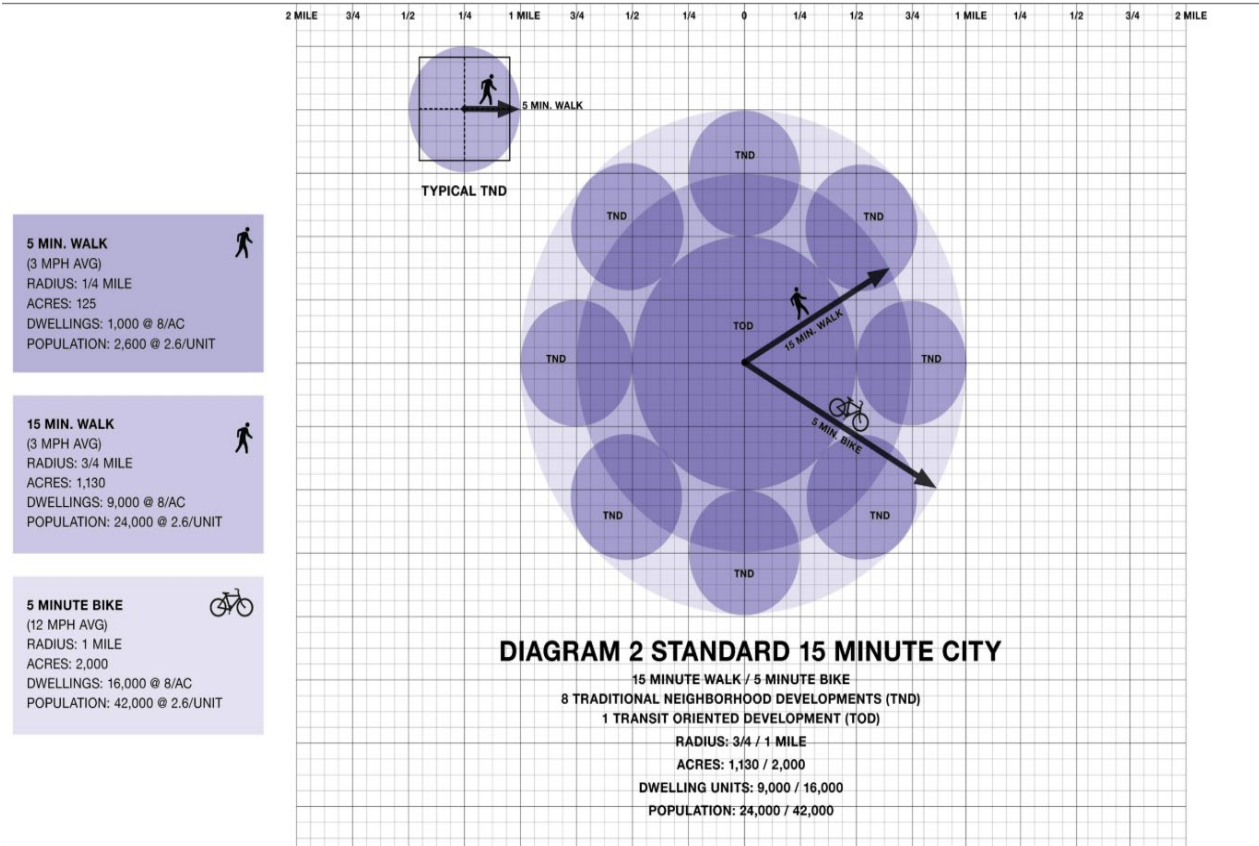
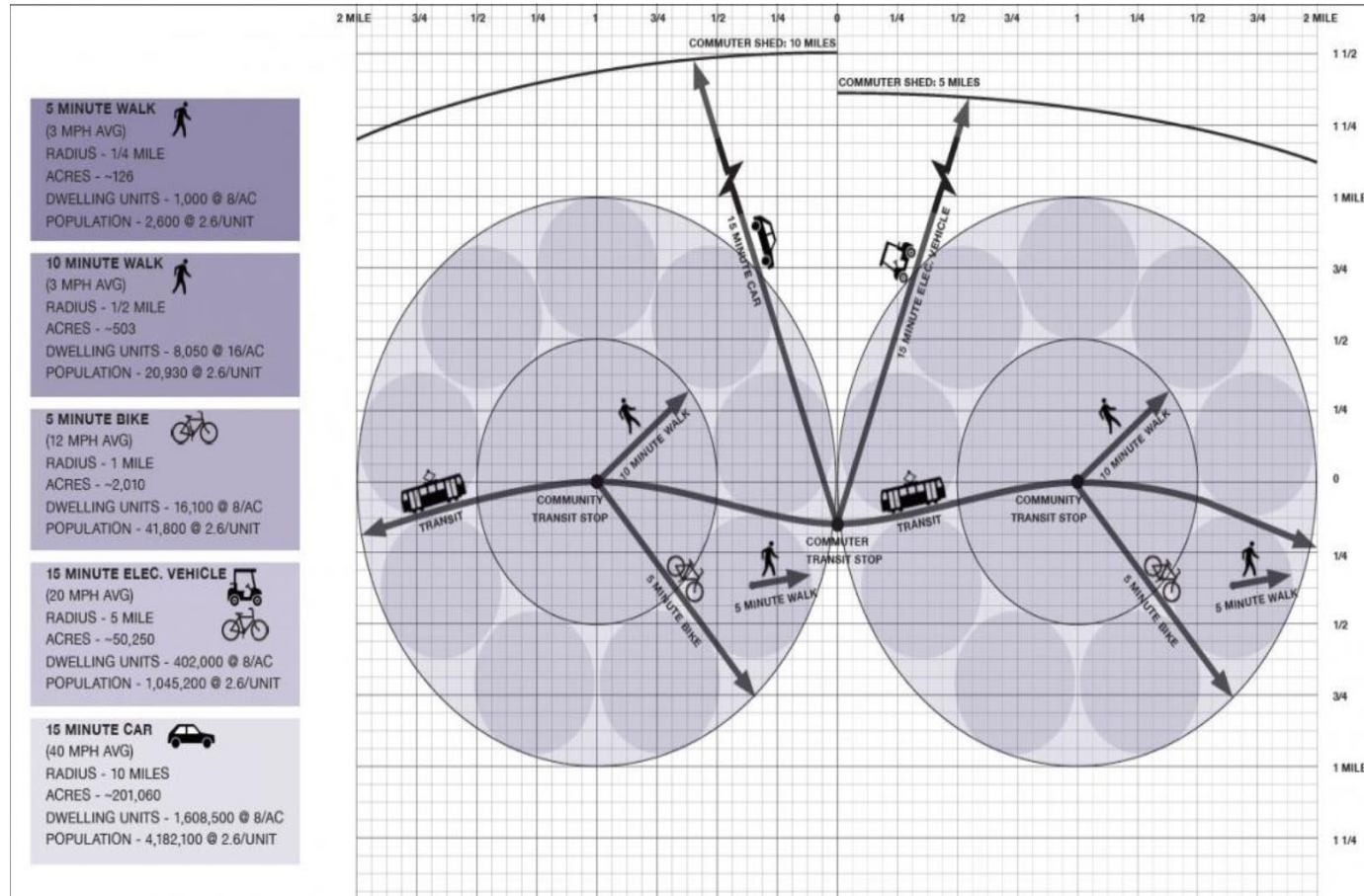


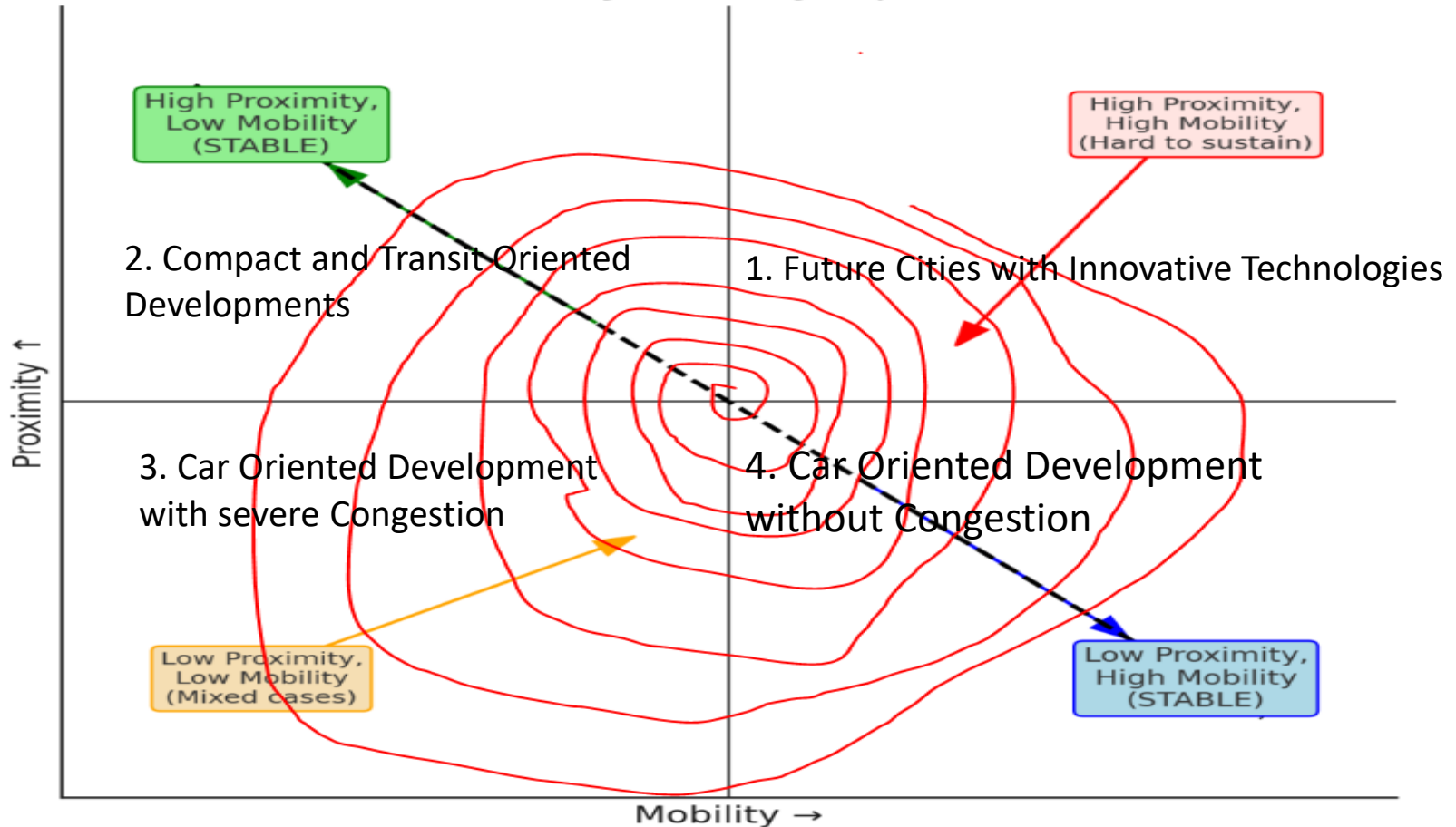
Diagram of 15 Minute City

(S:[Defining the 15-minute city | CNU](#))

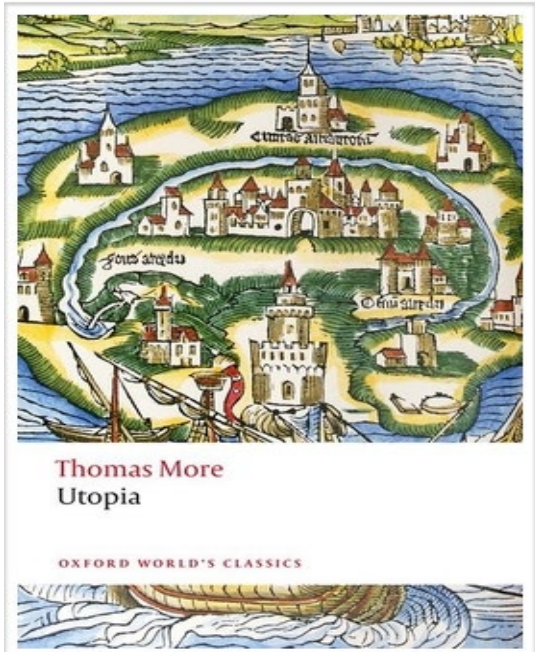


Proximity-Mobility Equilibria

(S: Levinson 2025, [Proximity is the Enemy of Mobility \(and Mobility is the Enemy of Proximity\)](#) – David Levinson)



Utopia (S: Thomas More 1516, Wikipedia)



- The UTOPIA island contains **54 cities**.
- Each city is divided into **four equal parts**.
- The **capital city** is located in **the middle of 54 cities**.
- Each city has not more than **6000 households**.
- Each family consisting of between 10 and 16 adults.
- People are redistributed around the households and towns to **keep numbers even**.
- There is no private property with goods being stored in **warehouses** and people requesting what they need.
- Houses are rotated between the citizens every ten years.
- Citizens must work only six hours a day although many willingly work for longer.

Utopia Concepts City Planning Models (S: [Utopian Concepts and Major Urban Planning Models – Track2Training](#))

1. Utopian Concepts in Urban Planning

- **Definition:** Utopian planning refers to the design of **ideal cities or communities** based on principles of **social justice, harmony, and sustainability** rather than existing realities.
- **Origins:** Philosophers, social reformers, and architects have proposed utopian cities since ancient times. Key objectives include:
 - Organized social structures
 - Healthy living environments
 - Integration of work, leisure, and residence
 - Efficient transportation and public amenities
- **Notable early examples:**
 - **Plato's "Republic":** Concept of a city governed by reason, justice, and communal welfare.
 - **Thomas More's "Utopia" (1516):** Imagined a self-sufficient community with equality, shared resources, and balanced land use.

Utopia Concepts City Planning Models (S: [Utopian Concepts and Major Urban Planning Models – Track2Training](#))

Concept	Originator	Key Feature	Focus
Garden City	Ebenezer Howard	Self-contained, green-belt, mixed-use	Health, community, sustainability
City Beautiful	Daniel Burnham, Charles Mulford Robinson	Monumental buildings, boulevards	Civic pride, aesthetics
Linear City	Arturo Soria y Mata	City along a transportation axis	Accessibility, efficiency
Radiant City	Le Corbusier	High-rise towers, open spaces	Modernism, function
Broadacre City	Frank Lloyd Wright	Decentralized low- density	Integration with nature, individualism

Utopia Concepts City Planning Models (S: [Utopian Concepts and Major Urban Planning Models](#) – Track2Training)



2. Garden City Concept

- **Origin:** Proposed by **Ebenezer Howard (1898)** in England as a response to **industrial urban congestion**.
- **Principle:** Combine the **benefits of town and country** to create a **self-contained, balanced community**.
- **Key Features:**
 1. **Limited population:** Typically 20,000–30,000 people per garden city.
 2. **Zoning:** Separation of **residential, industrial, and agricultural areas**, connected by efficient transport.
 3. **Green belts:** Open spaces surrounding the city to prevent urban sprawl and preserve the natural environment.
 4. **Public amenities:** Parks, schools, markets, and community centers integrated into the urban fabric.
- **Influence:**
 - Letchworth and Welwyn Garden City (England) are early examples.
 - Inspired **suburban planning** worldwide, promoting **sustainable and planned communities**.

Diagram Concept: A circular city with a central core, radial roads, residential and industrial zones, surrounded by a green belt.

Utopia Concepts City Planning Models (S: [Utopian Concepts and Major Urban Planning Models – Track2Training](#))

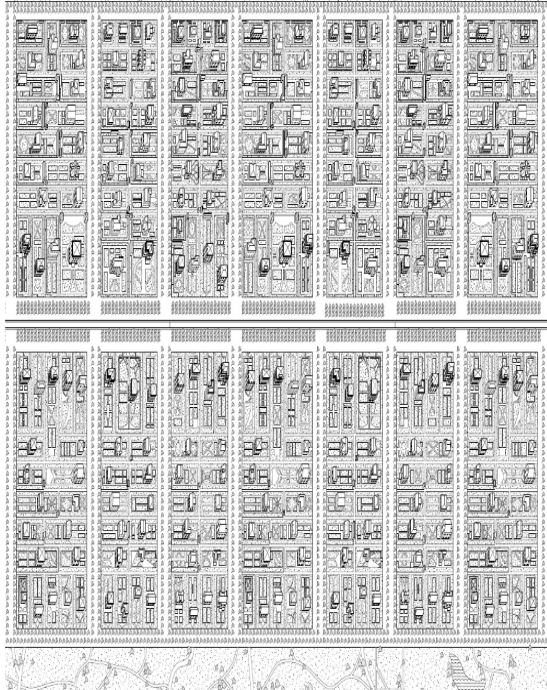


3. City Beautiful Movement

- **Origin:** Late 19th-century United States, influenced by the **World's Columbian Exposition (Chicago, 1893)**.
- **Principle:** Integrate **beauty, order, and monumental architecture** into urban planning to promote **civic pride and social harmony**.
- **Key Features:**
 1. **Wide boulevards and avenues** to improve circulation.
 2. **Monumental civic buildings** like museums, town halls, and libraries.
 3. **Parks and open spaces** for recreation and aesthetic appeal.
 4. **Symmetry and axial planning** in urban design.
- **Influence:**
 - Chicago, Washington D.C., and Denver adopted City Beautiful principles.
 - Inspired urban renewal and the planning of **government districts and civic centers** worldwide.

Impact: Focused less on social reform than aesthetics, but improved **urban infrastructure and public spaces**.

Utopia Concepts City Planning Models (S: [Utopian Concepts and Major Urban Planning Models – Track2Training](#))



4. Linear City Concept

- **Origin:** Proposed by **Spanish urban planner Arturo Soria y Mata (1882–1920)** in Madrid.
- **Principle:** Organize urban development along a **linear axis** to maximize accessibility, reduce congestion, and facilitate expansion.
- **Key Features:**
 1. **Central transportation corridor** (streetcar, tram, or road) along the city's spine.
 2. **Zoning parallel to the axis:** Residential, commercial, and industrial areas arranged in strips.
 3. **Green spaces and parks** integrated along the linear route.
 4. **Modular and expandable design**, allowing cities to grow without losing efficiency.
- **Influence:**
 - Applied in planned suburbs in Europe and Latin America.
 - Inspired modern **transit-oriented development** and corridor-based regional planning.

Diagram Concept: A long, narrow city with a central transport axis, parallel strips for different functions, and green spaces along the route.

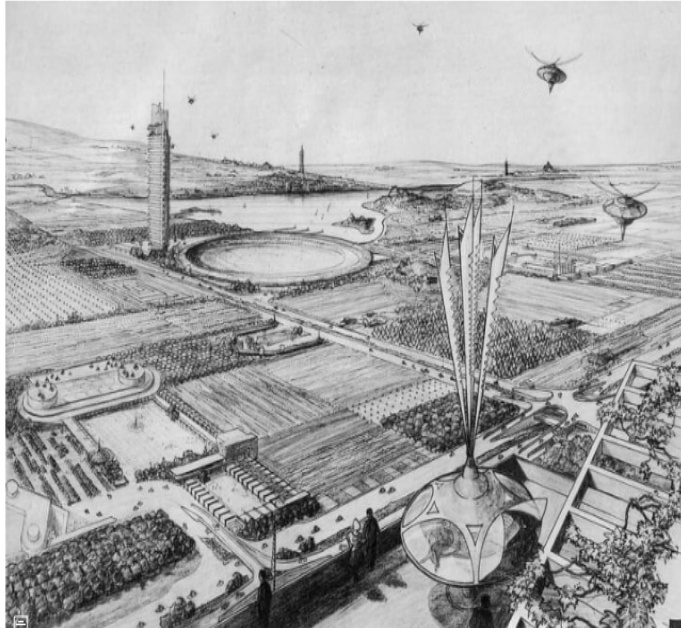
Utopia Concepts City Planning Models (S: [Utopian Concepts and Major Urban Planning Models – Track2Training](#))



5 Radiant City Concept Le Corbusier

- The Radiant City grew out of this new conception of capitalist authority and a pseudo appreciation for workers' individual freedoms.
- The plan had much in common with the Contemporary City - clearance of the historic cityscape and rebuilding utilizing modern methods of production.
- In the Radiant City, however, the pre-fabricated apartment houses, les unités, were at the centre of "urban" life. Les unités were available to everyone based upon the size and needs of each particular family.
- Sunlight and recirculating air were provided as part of the design.
- The scale of the apartment houses was fifty meters high, which would accommodate, according to Corbusier, 2,700 inhabitants with fourteen square meters of space per person.
- The building would be placed upon pilotis, five meters off the ground, so that more land could be given over to nature. Setback from other unités would be achieved by les redents, patterns that Corbusier created to lessen the effect of uniformity.

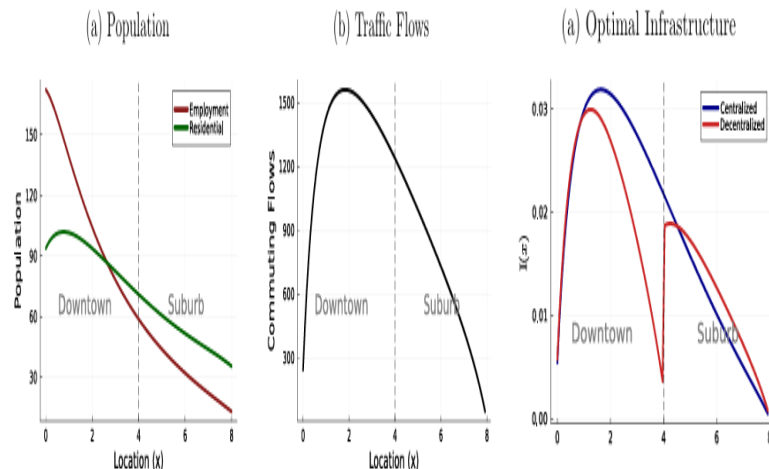
Utopia Concepts City Planning Models (S: [Utopian Concepts and Major Urban Planning Models – Track2Training](#) , Wikipedia)



6. Broadacre City by Frank Lloyd Wright

- Broadacre City was an urban or suburban development concept proposed by Frank Lloyd Wright in 1932.
- Broadacre City was both a planning statement and a socio-political scheme, inspired by [Henry George](#), by which each [U.S.](#) family would be given a one-acre (0.40-hectare) plot of land from the federal lands reserves.
- In a sense it was the exact opposite of [transit-oriented development](#). There is a train station and a few office and apartment buildings in Broadacre City, but the apartment dwellers are expected to be a small minority. All important transport is done by [automobile](#), and the [pedestrian](#) can exist safely only within the confines of the one-acre (0.40-hectare) plots where most of the population dwells

The City Fragmented: A Guide to Understand (S: [The City Fragmented: A Guide to Understanding](#) and Olivia Bordeu 2025)



Note: Commuters travel in both directions because of the idiosyncratic preference shocks for origin-destination pairs. Panel (b) plots the total commuting flows, the sum over both directions. However, most commuters travel towards $x=0$, where wages are higher.

Strategy	Description	Benefits
Mixed-use zoning	Allows for a mix of residential, commercial, and recreational uses	Promotes walkability, reduces sprawl, and increases community engagement
Public transportation investment	Improves access to employment, education, and healthcare	Reduces congestion, pollution, and social isolation
Community-led planning	Empowers local residents to shape their neighborhoods	Increases community engagement, promotes social cohesion, and ensures that development meets local needs
Green infrastructure	Incorporates natural systems into urban design	Mitigates the urban heat island effect, improves air quality, and enhances biodiversity

Networked Cities for Urban Agglomerations (S: Batten 1995)

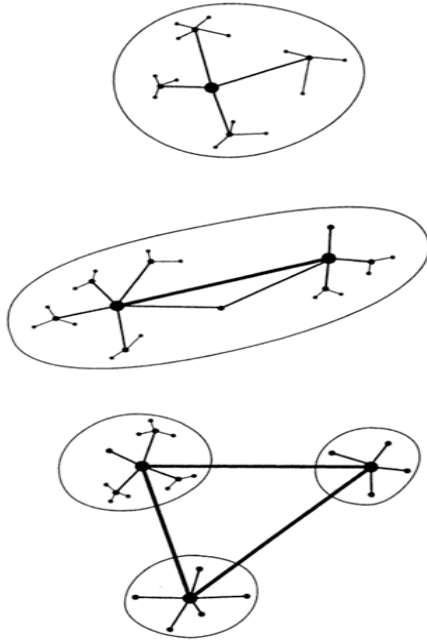


Figure 1. Three urban configurations: the monocentric city, the corridor city (*centre*) and the network city (*below*).

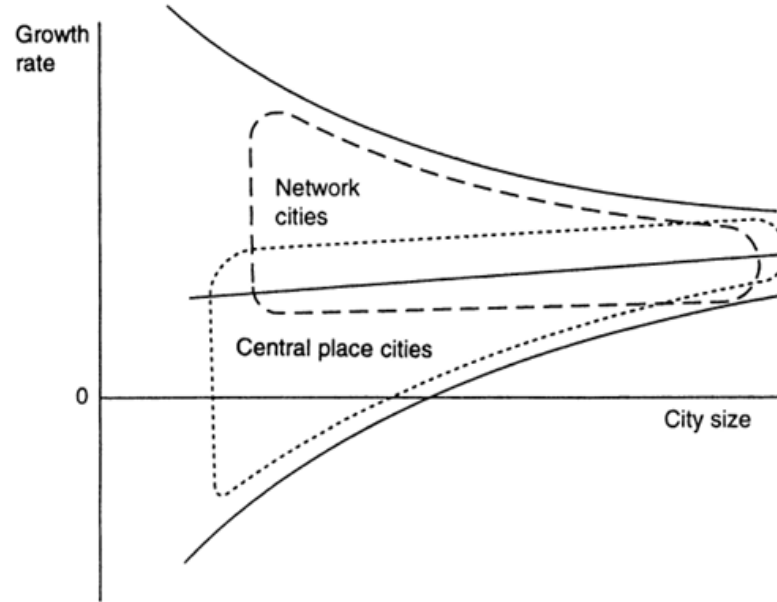


Figure 3. Growth versus size in two systems of cities. *Source:* Hohenberg and Lees (1985).

Mega City Regions (S: Concepts and typology of mega-city regions, Laboratoire Urbanisme Insurrectionnel: PARIS | Enjeu Écologique : la Décroissance urbaine ?)

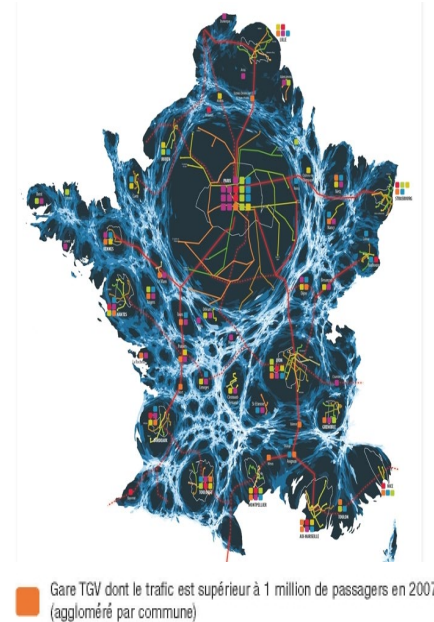
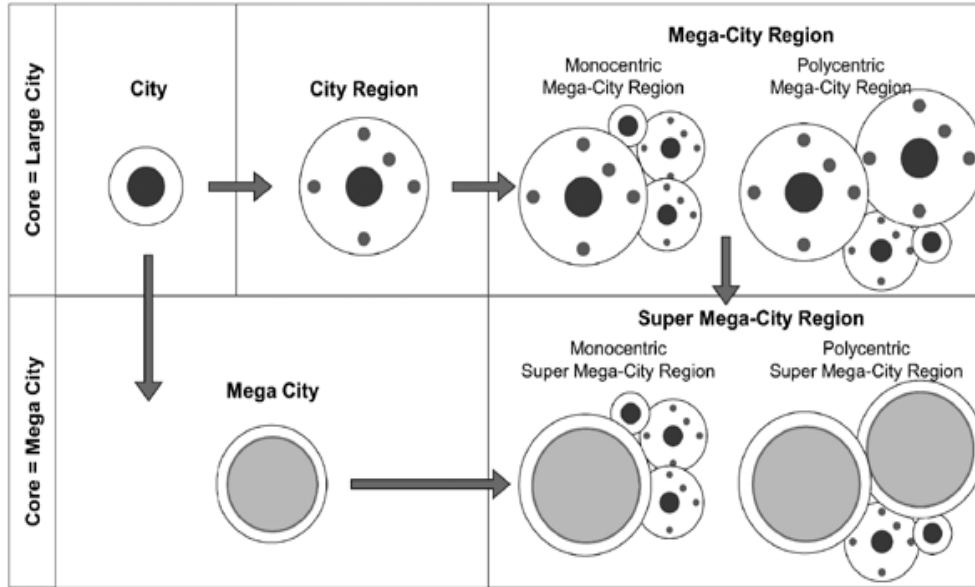
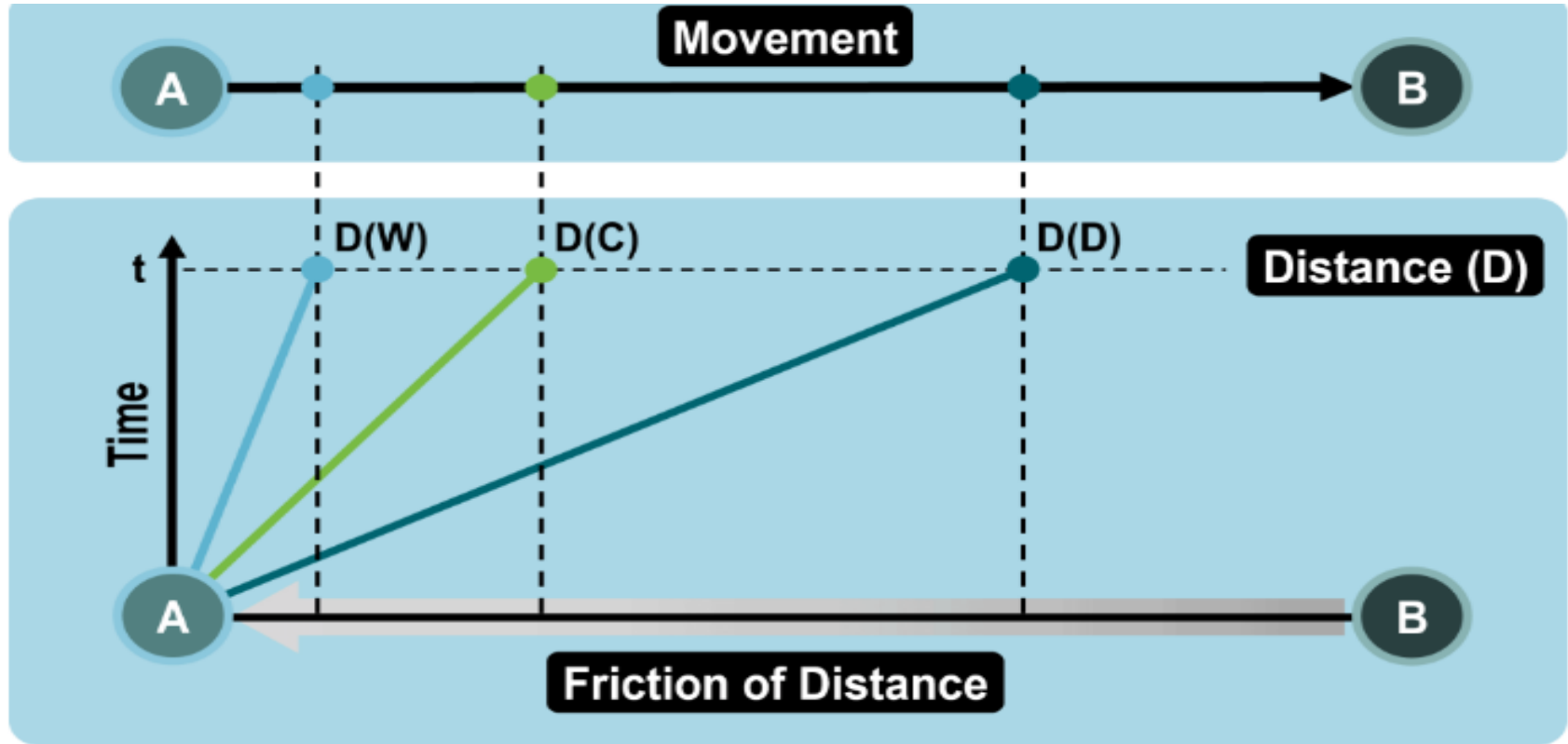


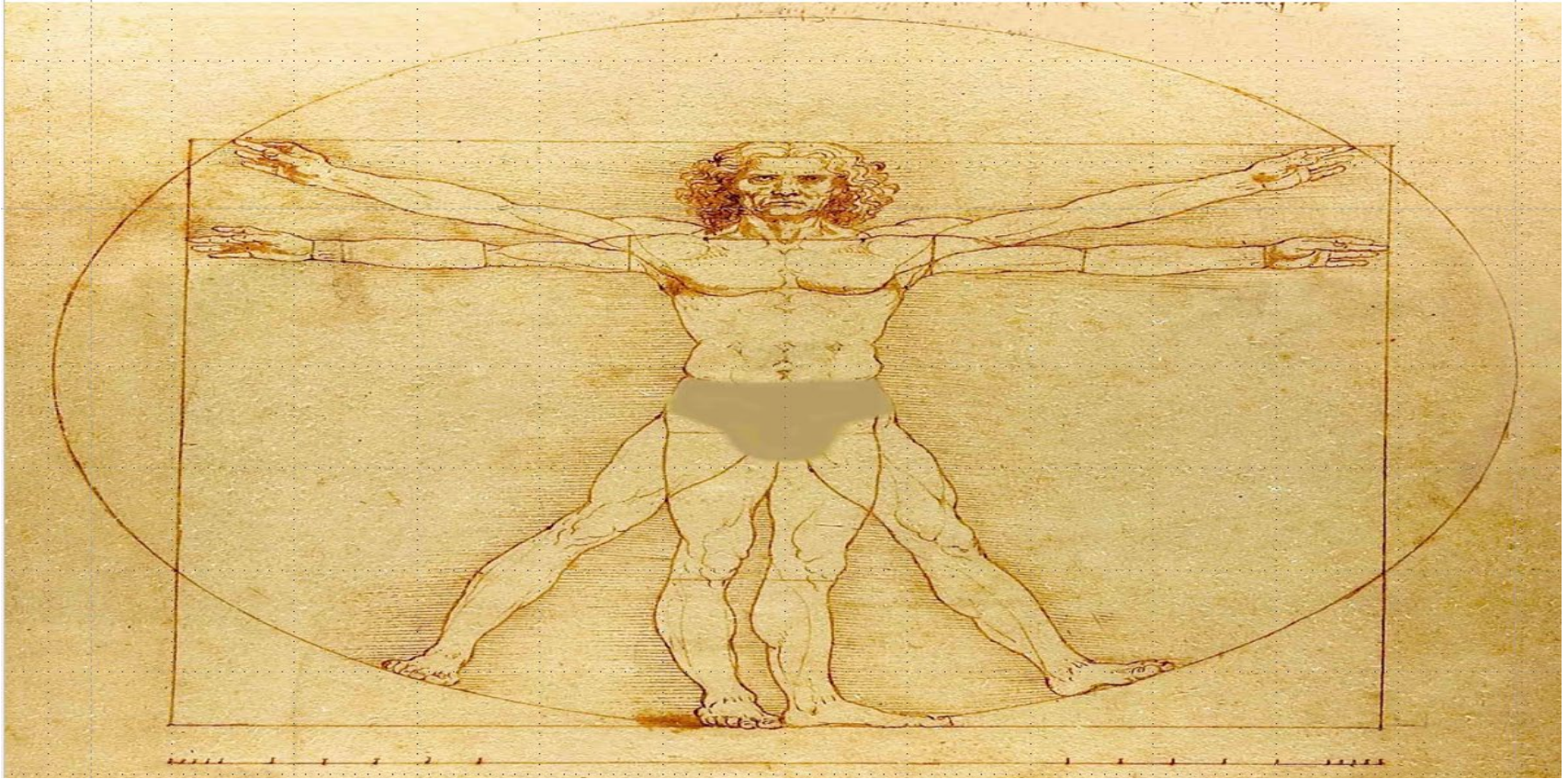
Figure 1.2 Conceptual diagram of cities, city regions, mega-city regions and super megacity regions

Consider Different Spatial Travel Times to Movements



(S: Jean-Paul Rodrigue (2024), The Geography of Transport Systems)

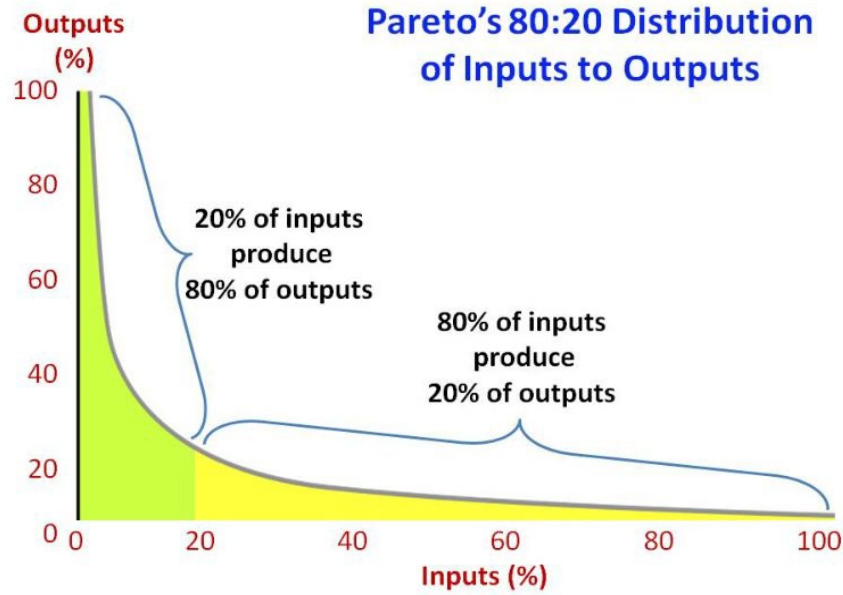
Natural and/or Artificial Systems are Symmetry or Asymmetry ? !



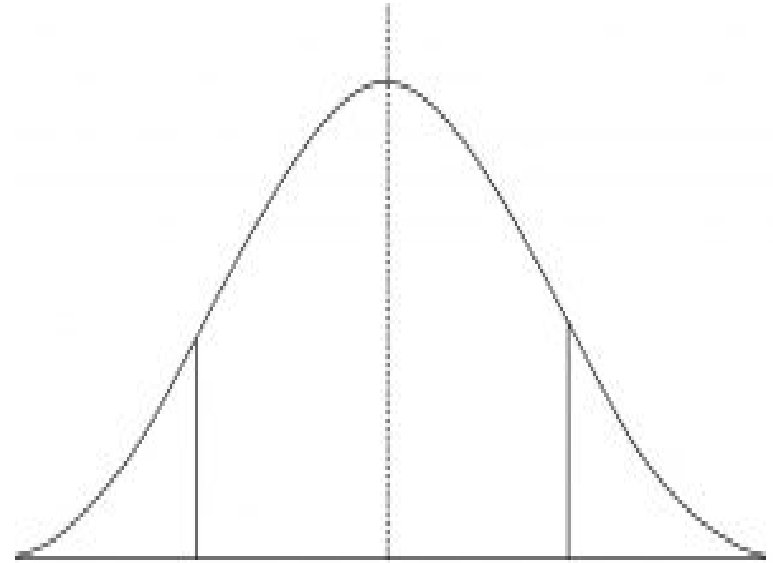
Source : Leonardo da Vinci, *Proportions of Man*, 1490

Asymmetry in pareto curve

- The World is “Pareto”, Not “Bell Curve.”
- “Most things are incredibly skewed in their distribution and follow the Pareto principle, the 80/20 rule”
- Shlomo Maital (1942 -)

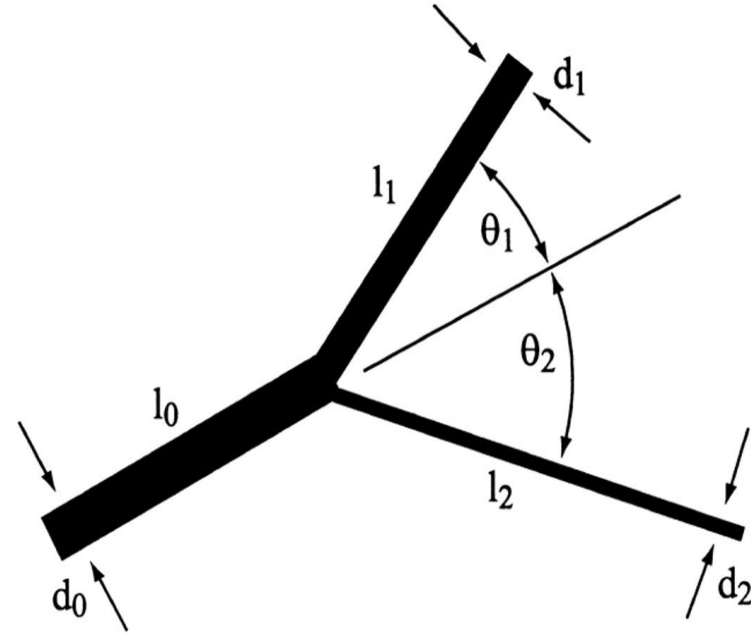
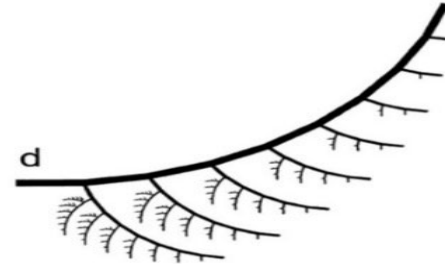
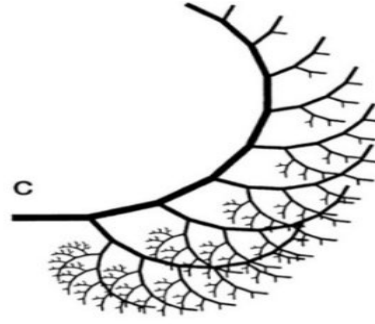
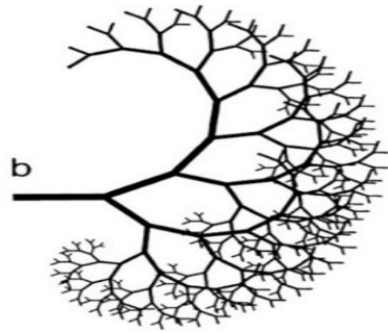
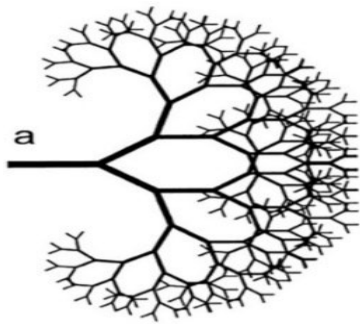


The Pareto curve



The Bell curve

Optimal Blood Transport Systems depending on the value of the bifurcation (Arteries Asymmetry, Capillary Symmetry)

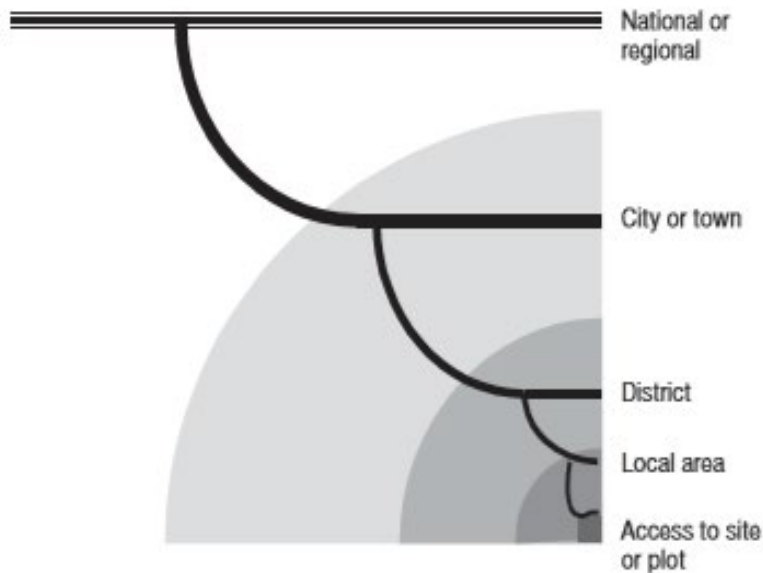


Lengths and diameters, and angles that the two branches make with the direction of the parent vessel

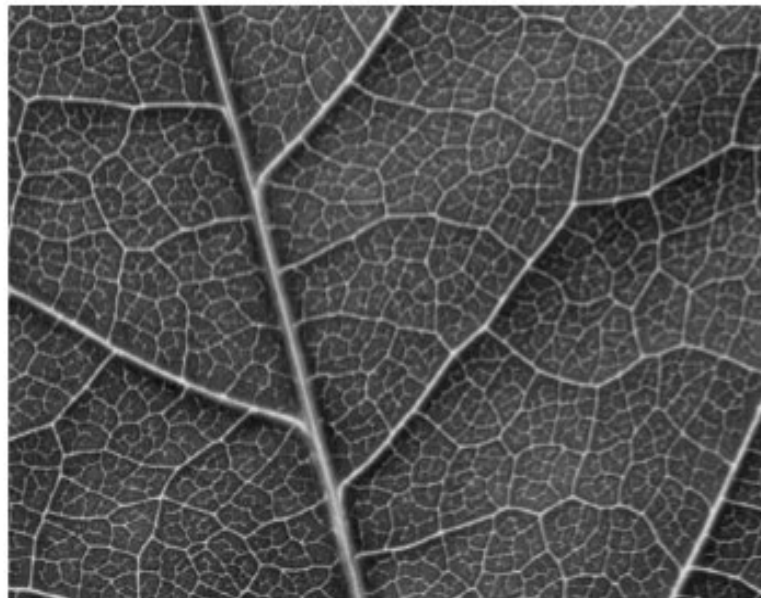
Hierarchy Street Patterns (S. Marshall, Streets and Patterns, 2005)

66

STREETS & PATTERNS



3.20 • A hierarchy of roads and areas. Road types are closely associated with the spatial scale of the area they serve.

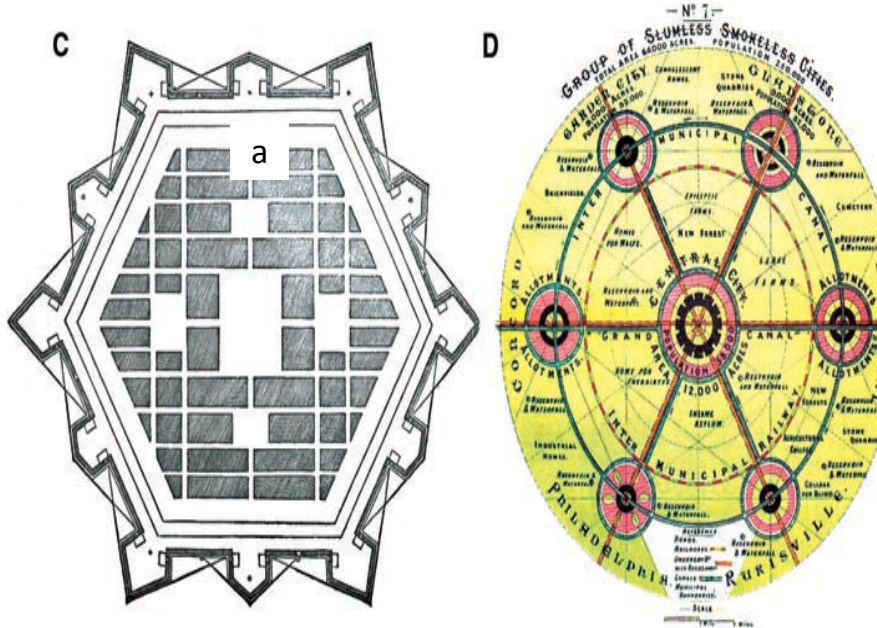


3.21 • The 'arterial' pattern of a leaf. We can recognise that the pattern of veins on a leaf possesses arteriability although we may know nothing of the 'flows' along them. The assignation of links to the strategic road network is similarly abstract.

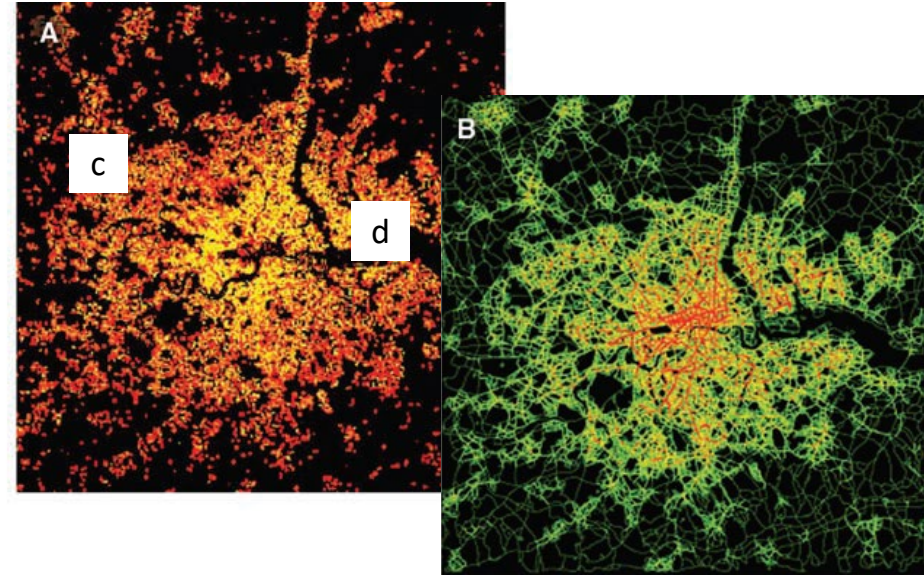
Asymmetry in morphological perspective - Fractal cities

- Intriguingly and importantly, cities have fractal structure
- Such as the maps of the London transportation network and spatial employment patterns

S: Batty, M (2008). The Size, Scale, and Shape of Cities



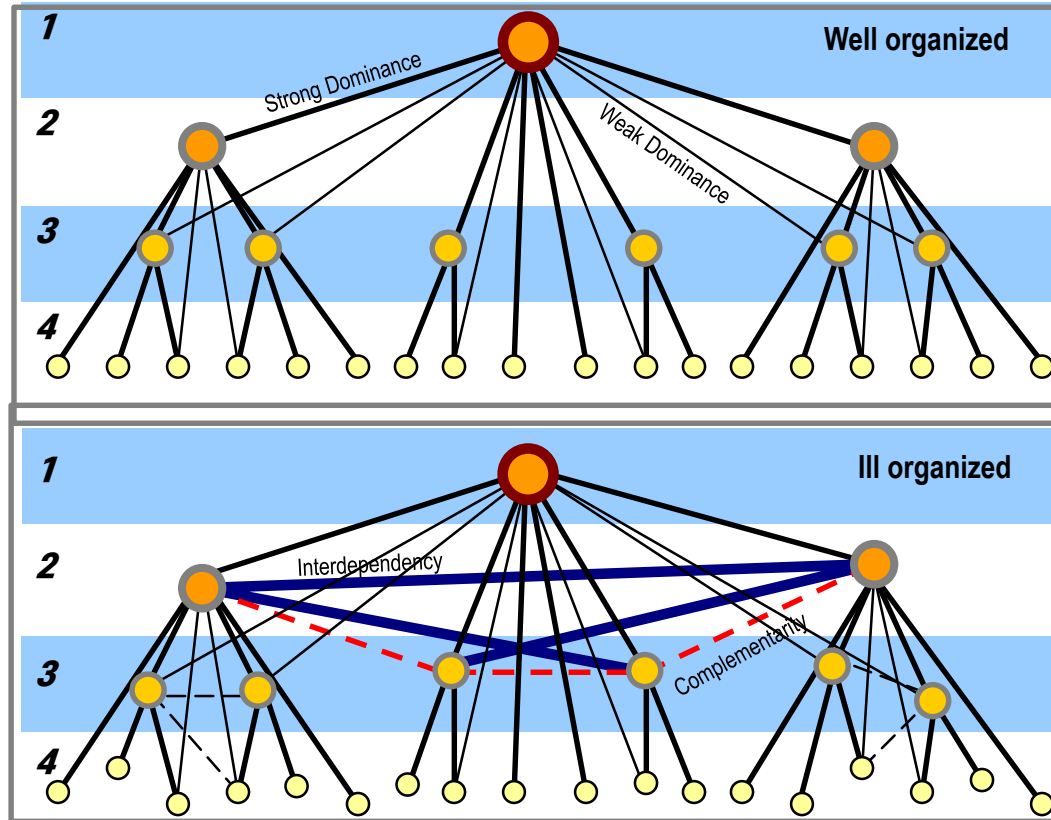
- (a) An idealized geometric city.
- (b) Howard's garden city of tomorrow



- (c) Population morphology of London
- (d) The road network in London colored by level of connectivity

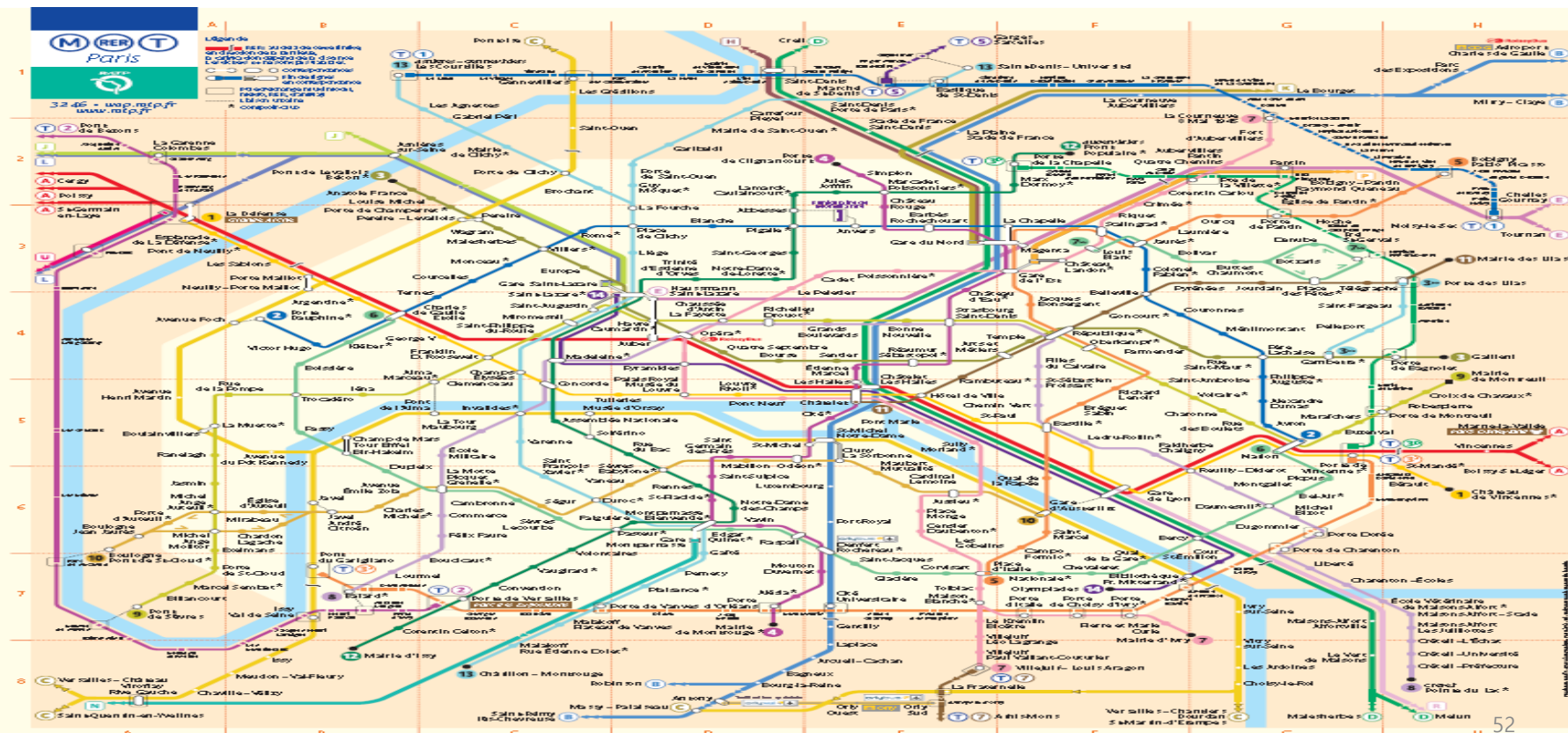
Urban Hierarchy

(Functional relationship between activity locations)



Subway route map in Paris, France

- Unlike the Paris network, the express railway in Seoul is less covered.
- Discuss strategies to address the alienated areas of the express railway.

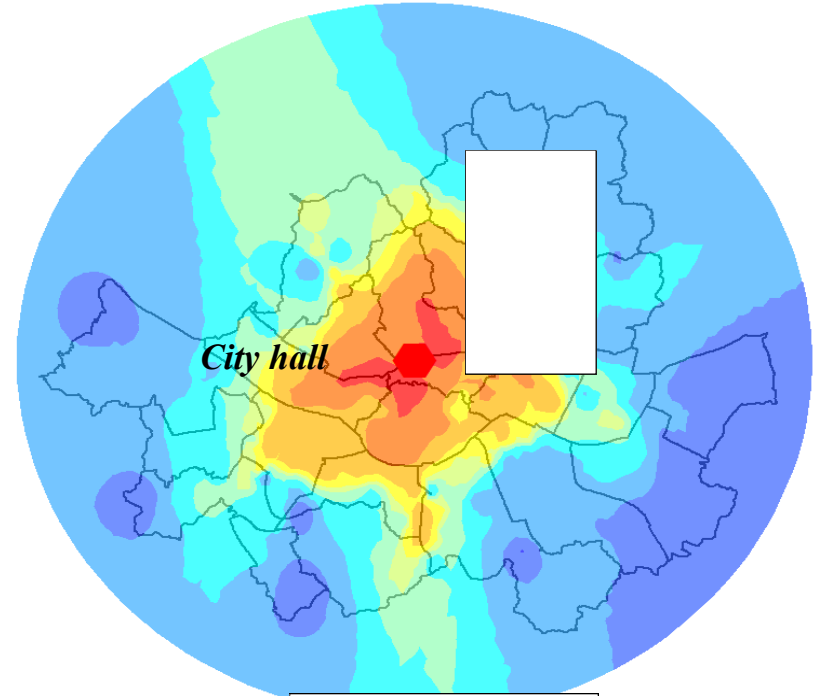
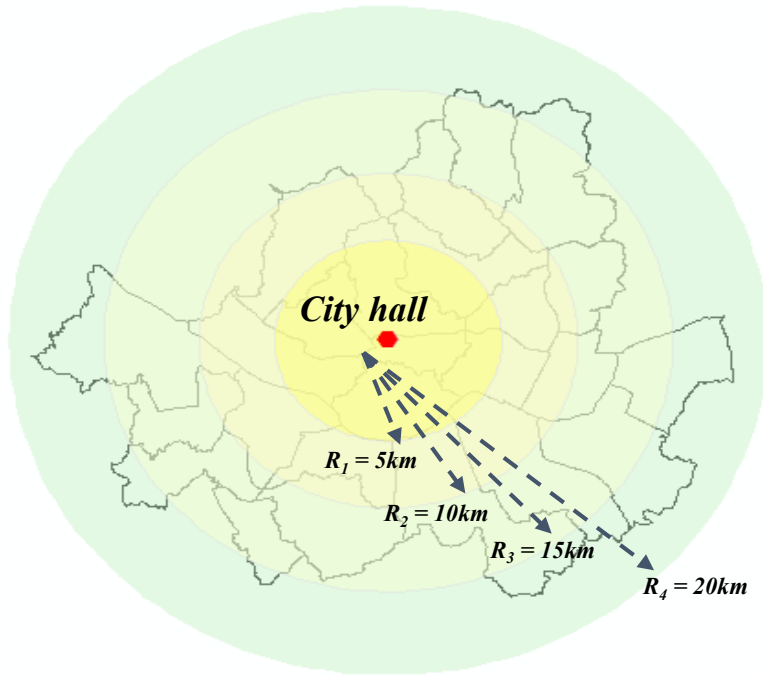


Hausmann's Transformation of Paris (1854-89) (S:Pinterest.com)



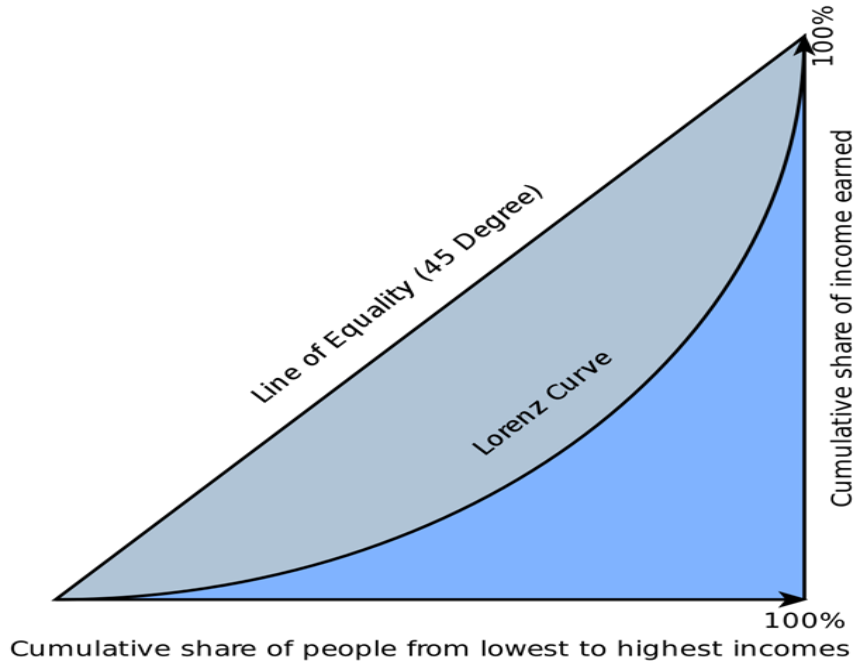
Isometric and Asymmetry Travel Time Pattern

- There is a difference in the supply of public transportation in the city.
- This phenomenon shows the asymmetrical characteristics of travel time.



Inequity index - Lorenz curve and GINI coefficient

- The inequality of regional public transportation service was calculated by GINI coefficient.
- A Gini coefficient of 1 (or 100%) expresses maximal inequality among values



GINI index by using Lorenz curve

$$G_A = 1 - \sum_{k=1}^n (X_k - X_{k-1})(Y_k + Y_{k-1})$$

where

G_A : Gini coefficient of Group A

X_k : Cumulative percentage of population A

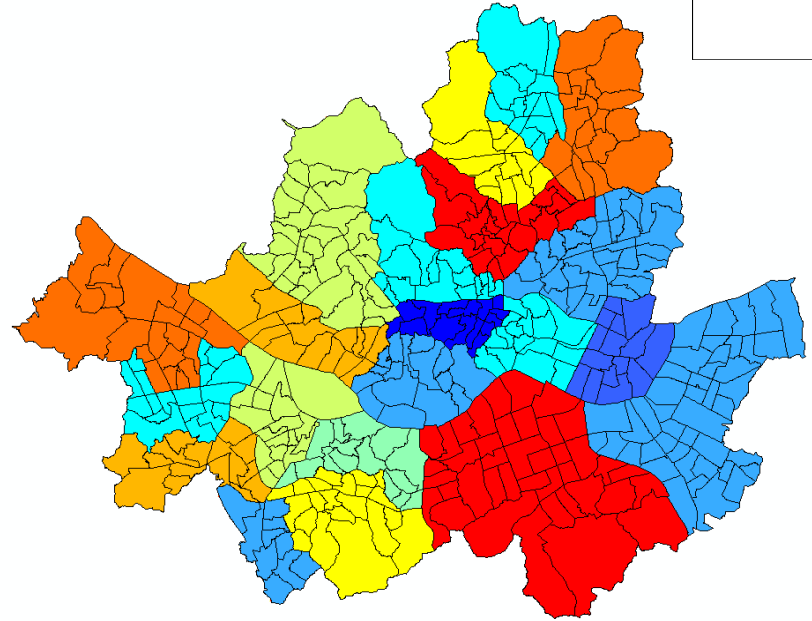
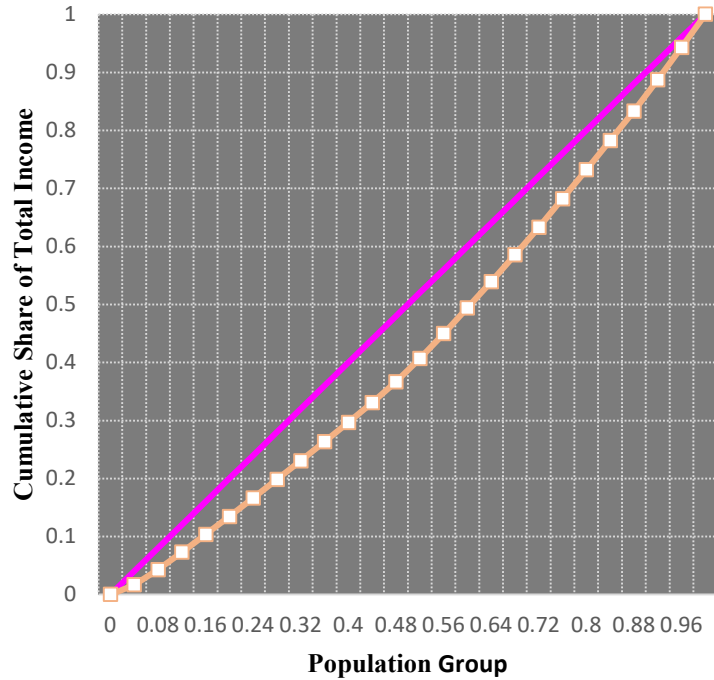
X_k : Cumulative percentage of transit access

$X_0=0, X_n=1, Y_0=0, Y_n=1$

k = Order by size of public transportation access

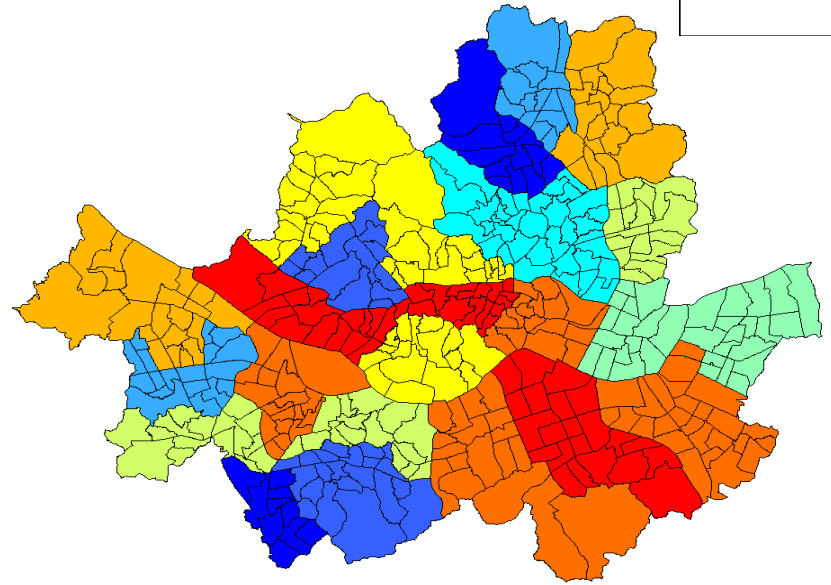
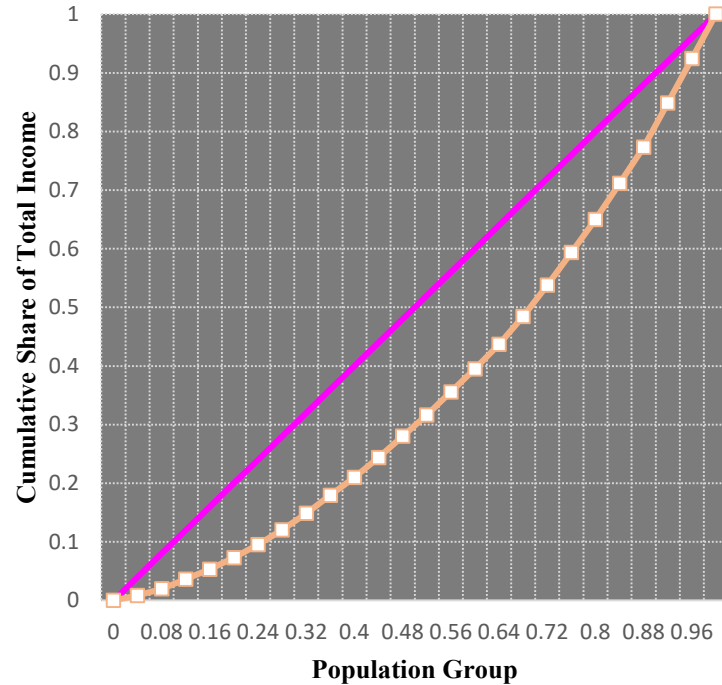
Analysis of bus service supply

- We analyze the number of bus stop to develop a deprivation index
- Gini coefficient of bus stop of each district is 0.14

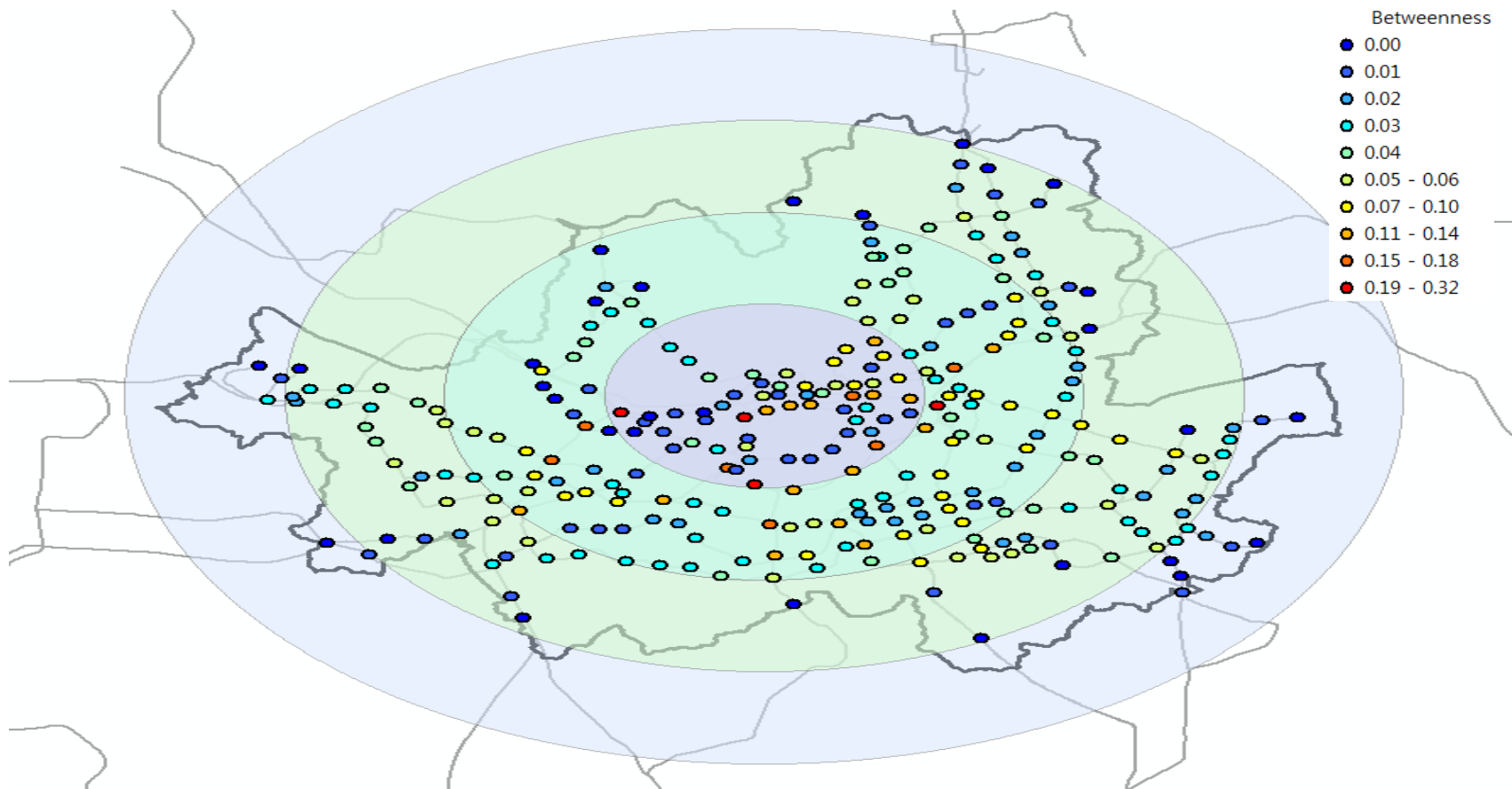


Analysis of subway service supply

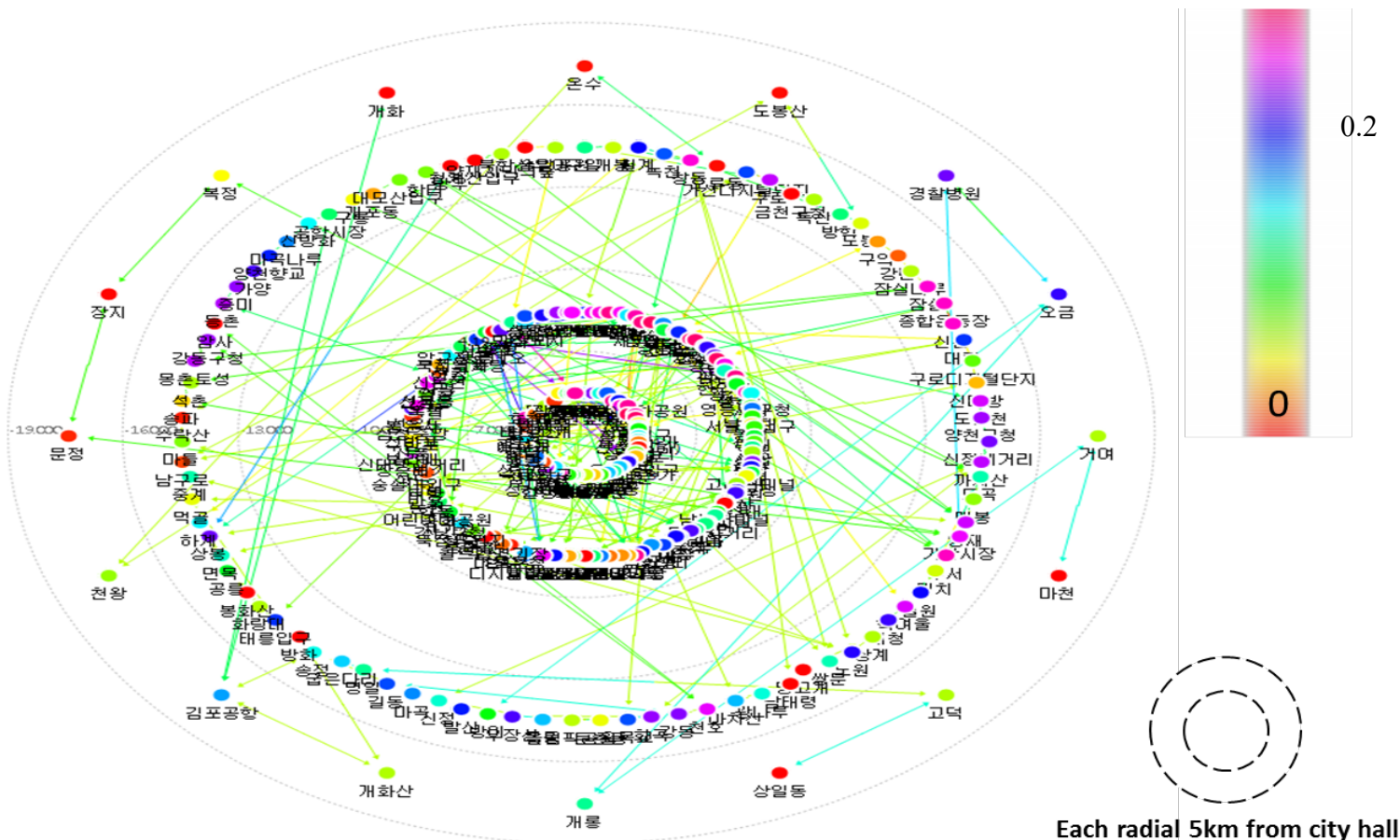
- We analyze the number of subway station to develop a deprivation index
- Gini coefficient of subway station of each district is 0.28



Betweenness centrality(BC) analysis based in 2018



Betweenness centrality(BC) analysis based in 2018



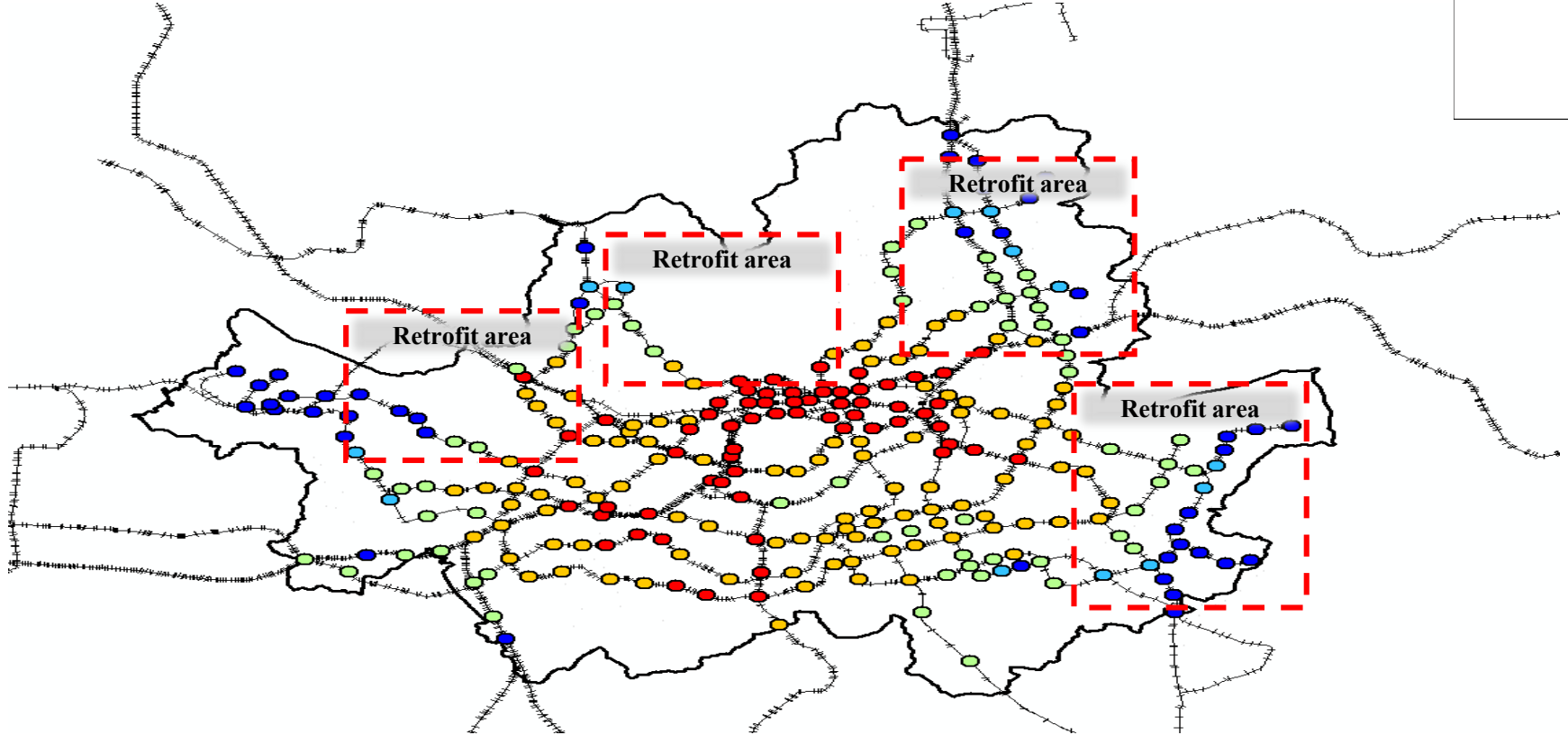
Layer	Stations
5km	60
10km	124
15km	91
20km+	16
Total	291

Time-based Concentrated Network Representation from City hall

Result of the shortest path analysis (2018)

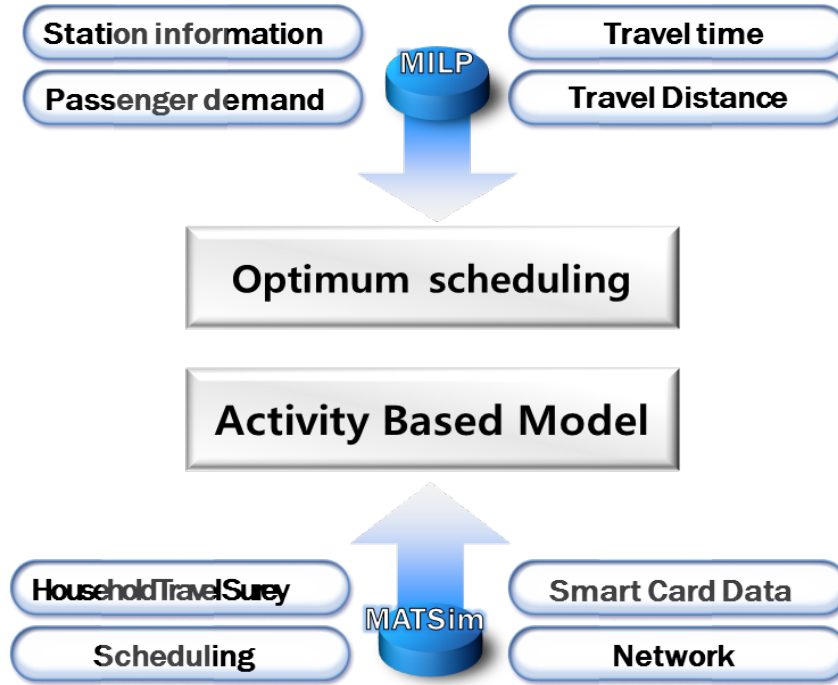
- There are four most retrofit areas in Seoul.
- It is located in the western, the northwestern, the northern, and southeastern part.

Mapping the shortest path result 2018

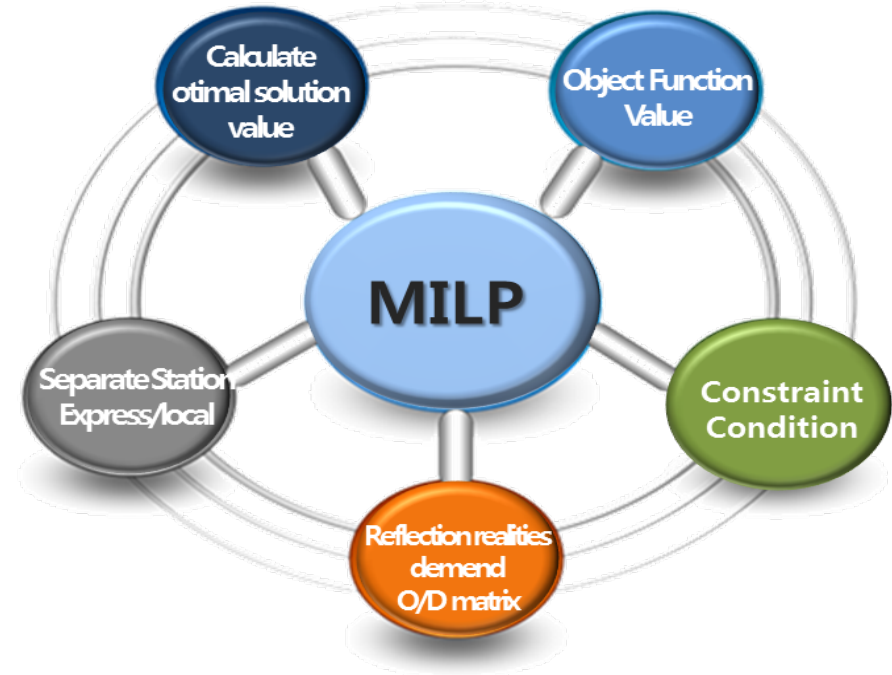


Optimization of Scheduling of Skip and Stop Systems in Subways

- Mapping of cause-effect relationship between parameters
- Five topics about future's urban transportation



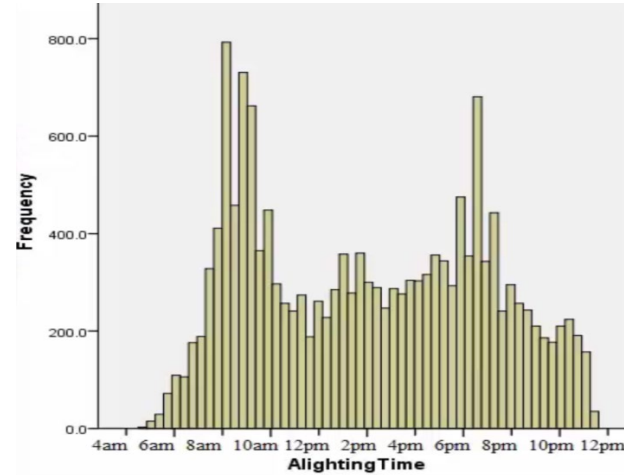
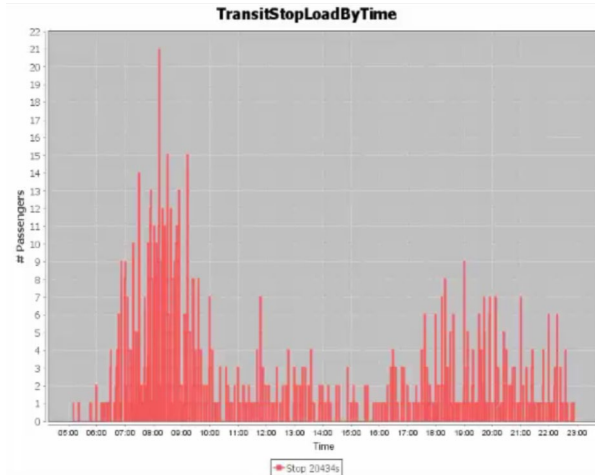
Data use plan of MILP and MATsim



Five topics of MILP

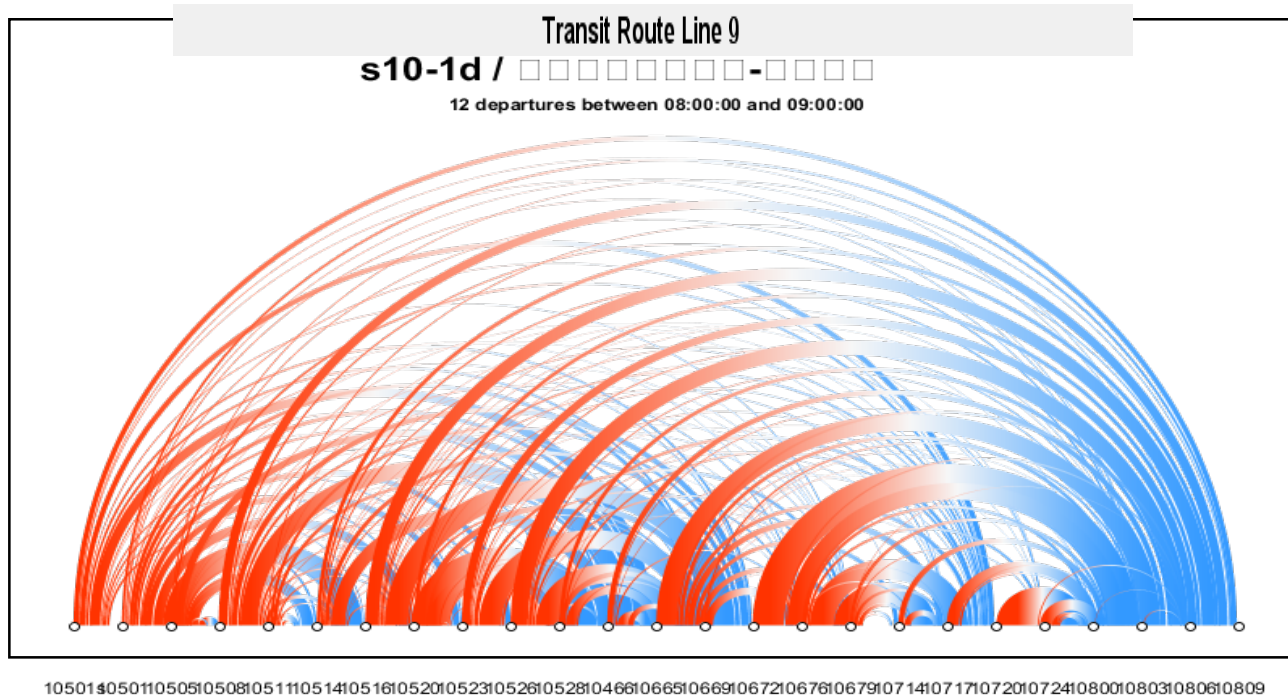
Boarding Analysis

- Example of evaluation index of MATSim analysis.
 - Simulation results: The number of waiters in stops, user travel time and travel distance.
- This results are able to be used to evaluate index of optimum train scheduling.



Schema Analysis for Skip and Stop Systems

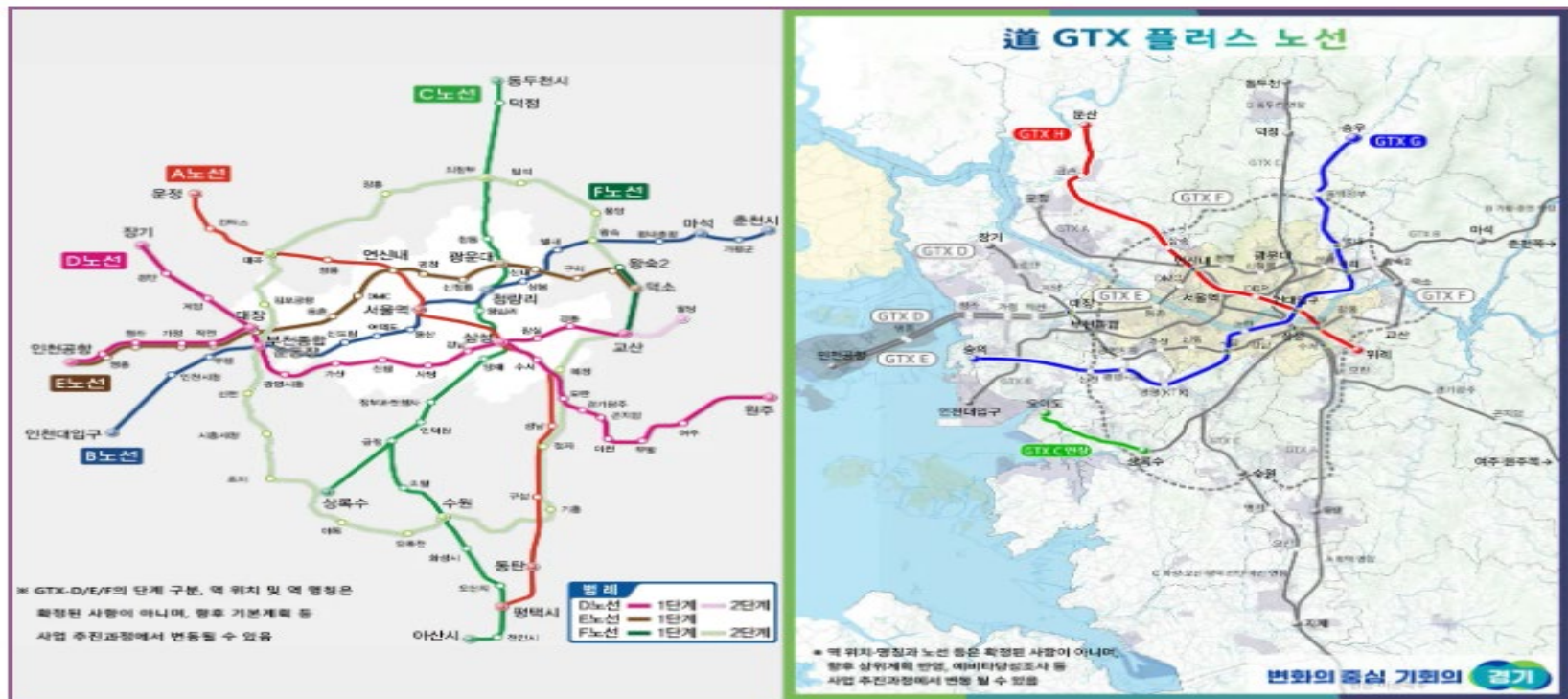
- The result of MATSim analysis
 - Travel patterns between stations
 - The number of people who are get in and out



Planning Great Train eXpress (GTX) (S: 수도권 광역급행철도 - 나무위키)

GTX 수도권 광역급행철도

A선 부분개통	B선 건설 중	C선 건설 중	D선 추진 중
E선 계획 중	F선 계획 중	G선 계획 중	H선 계획 중



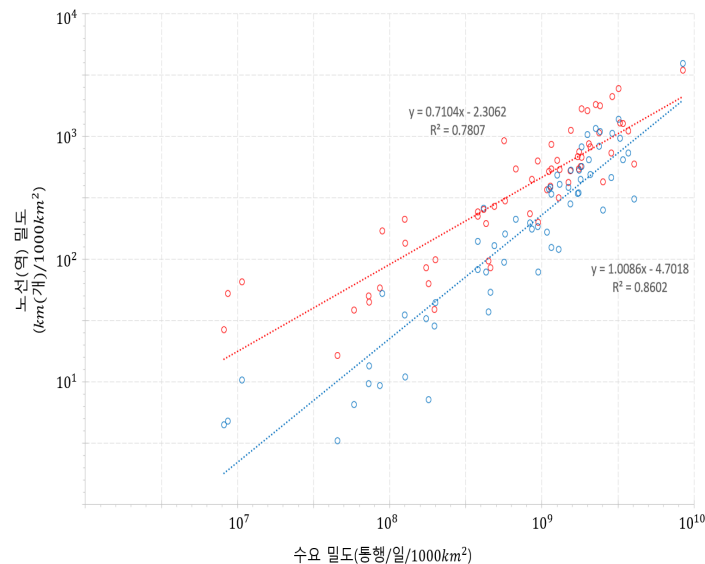
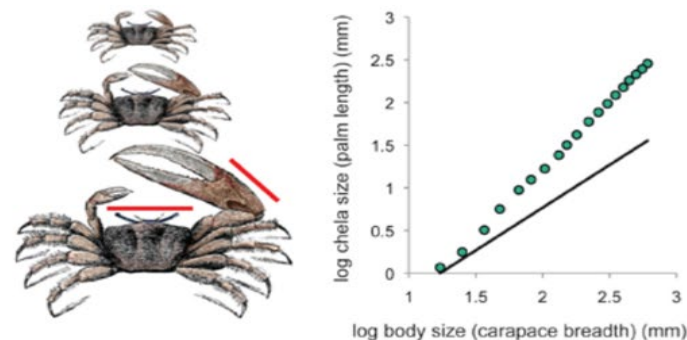
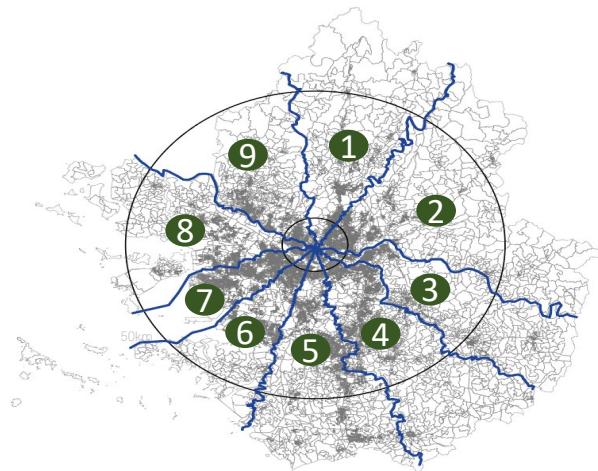
Scale Law and Allometric Growth in Rail Networks based on Land Use

Development (Kim, S. 2024)

$$L = \mu T^\alpha$$

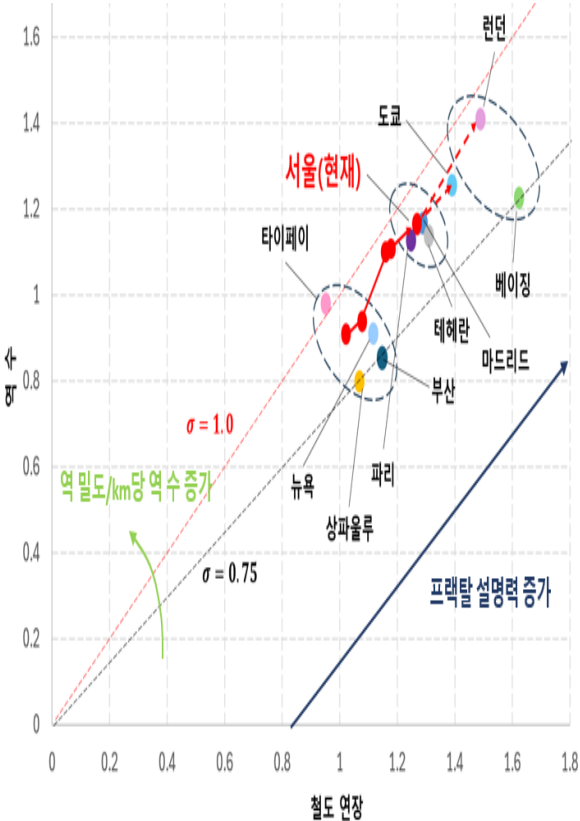
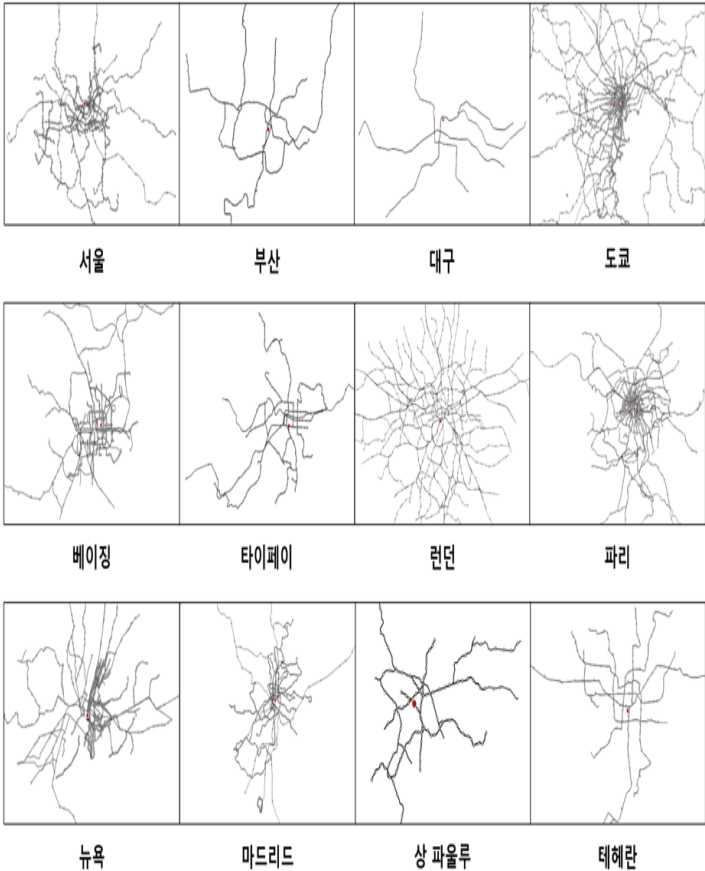
L, T 는 물리량, α 는 스케일링 지수,
 μ 는 비례상수,

Shingleton, A. (2010). Allometry: The Study of Biological Scaling.



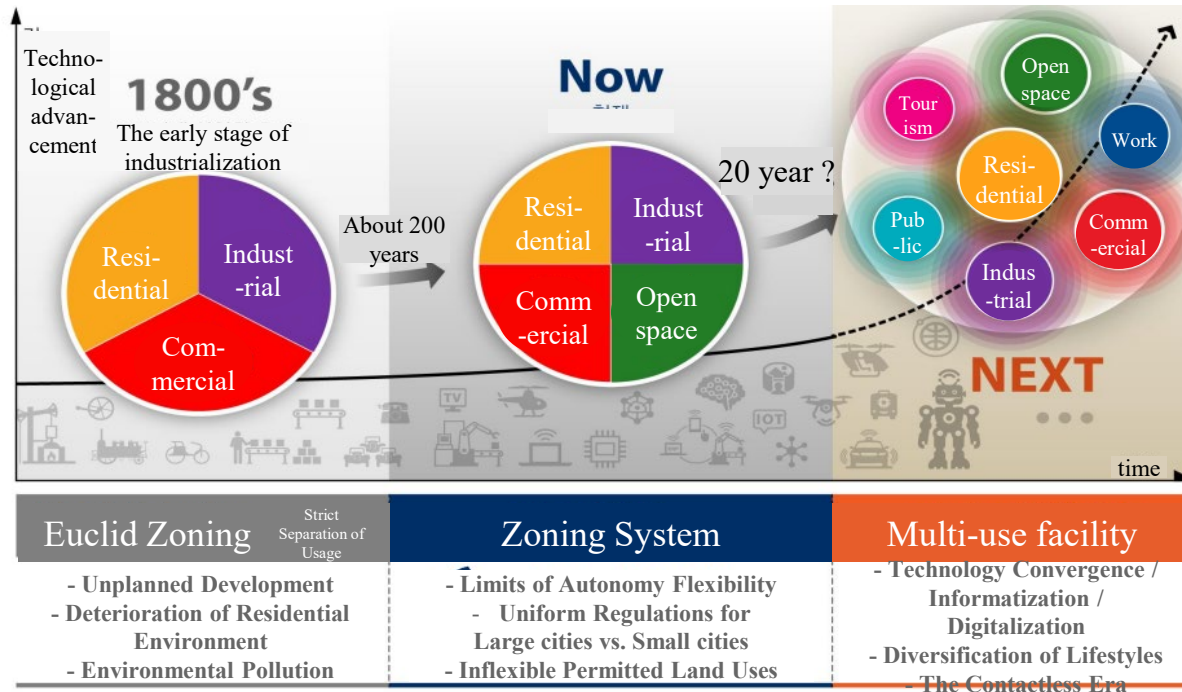
Scale Law and Allometric Growth in Rail Networks based on Land Use Development

(Kim, S. 2024)



Evolution of Zoning Systems: From Single-Use to Multi-Use Spaces

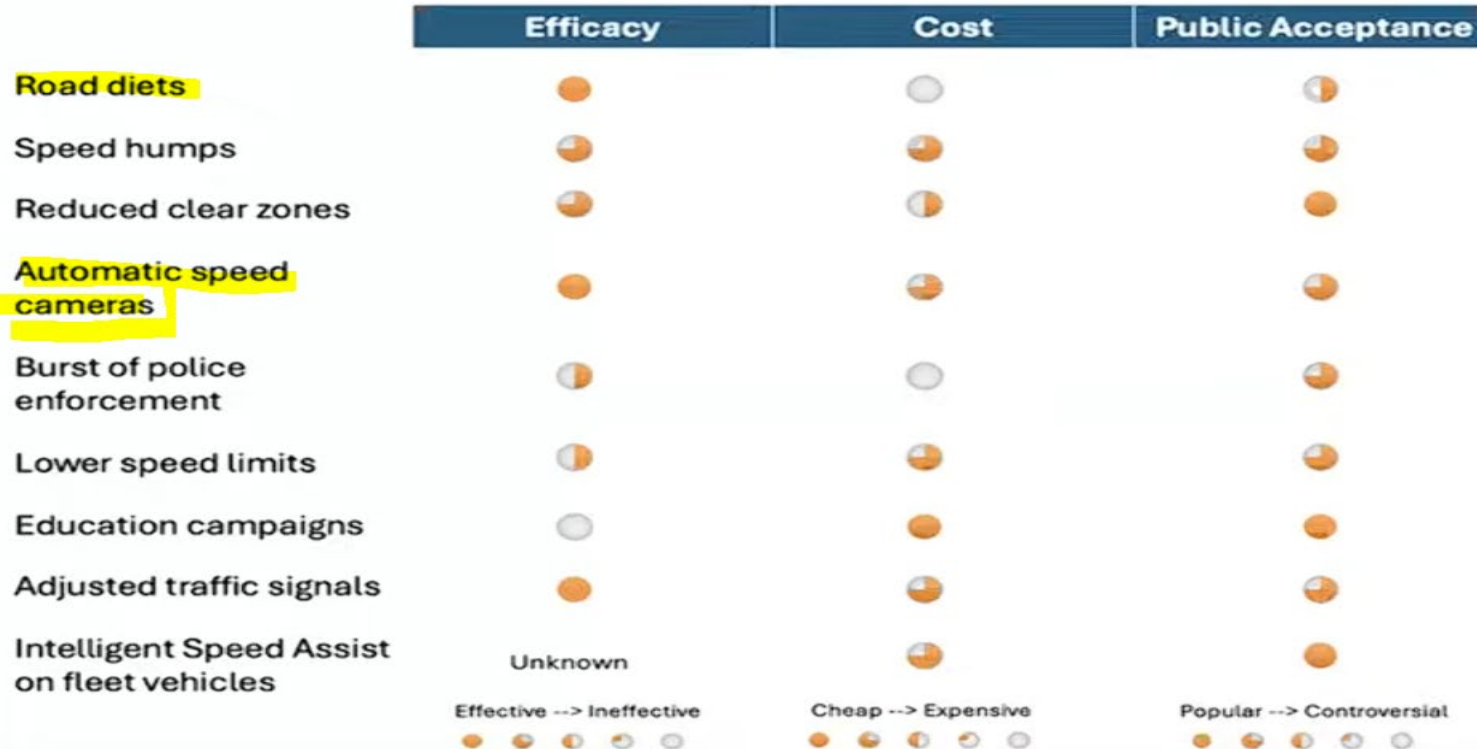
The **Evolution of Zoning Systems** is moving from functionally single-use areas toward flexible, multi-use structures driven by technological advancement and diversification of lifestyles



Evaluations of Road Speed Reduction Schemes

(S: www.davidzipper.com)

Figure 2. Summary of city DOT officials' perceptions of anti-speed tactics



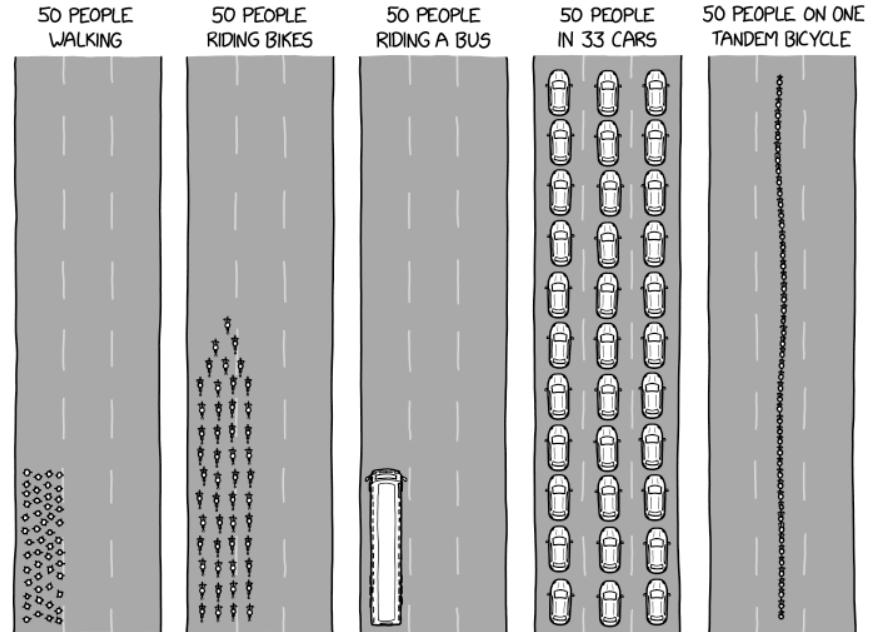
Rethinking road space allocation

Rethinking road space allocation is essential to prioritize the efficient movement of people over vehicles by encouraging shifts towards optimal and sustainable transportation modes.



S: <https://goingcarfree.blogspot.com/2015/03/more-road-space.html>

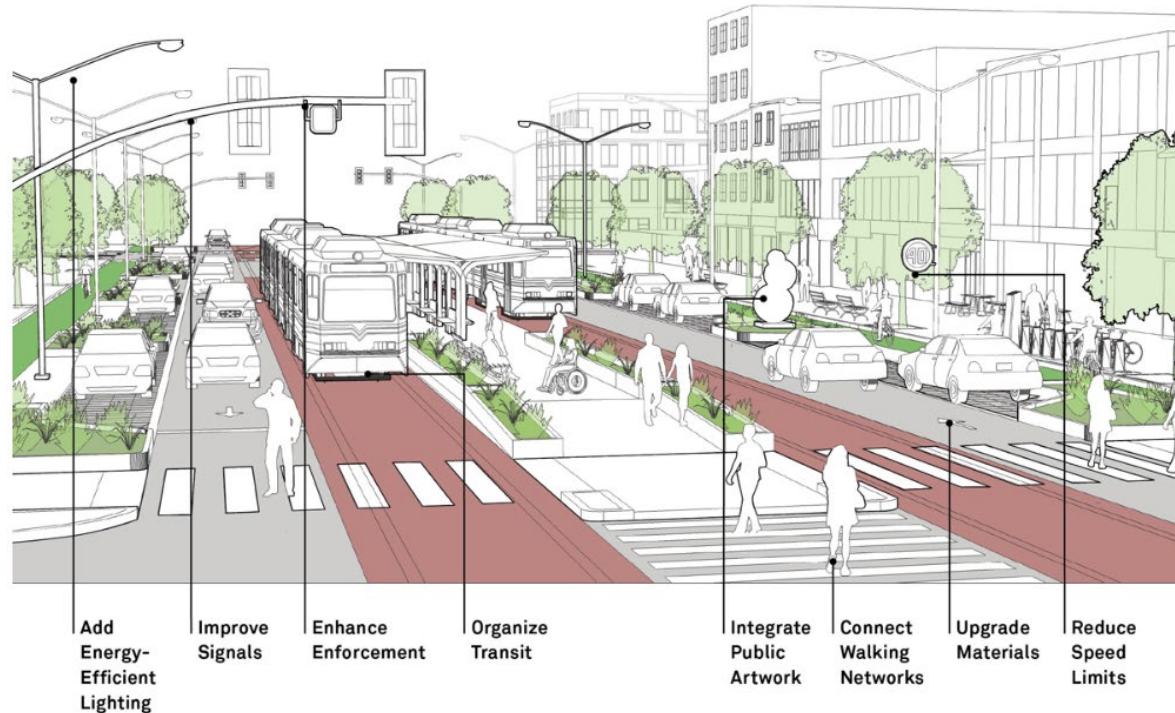
ROAD SPACE COMPARISON



S: <https://xkcd.com/2684/>

Rethinking road space layout

The Rethinking road space layout is based on the principle that streets are public spaces for people, moving beyond classifying streets solely by their function of traffic movement



Rethinking road space layout: Healthy Streets

The **Healthy Streets approach** in London provides key indicators for creating street environments that support walking and cycling



Guide to the Healthy Streets Indicators

Delivering the Healthy Streets Approach

MAYOR OF LONDON



TRANSPORT
FOR LONDON
EVERY JOURNEY MATTERS

Healthy Streets Surveys

Capturing how members of the public experience the street

MAYOR OF LONDON



TRANSPORT
FOR LONDON
EVERY JOURNEY MATTERS



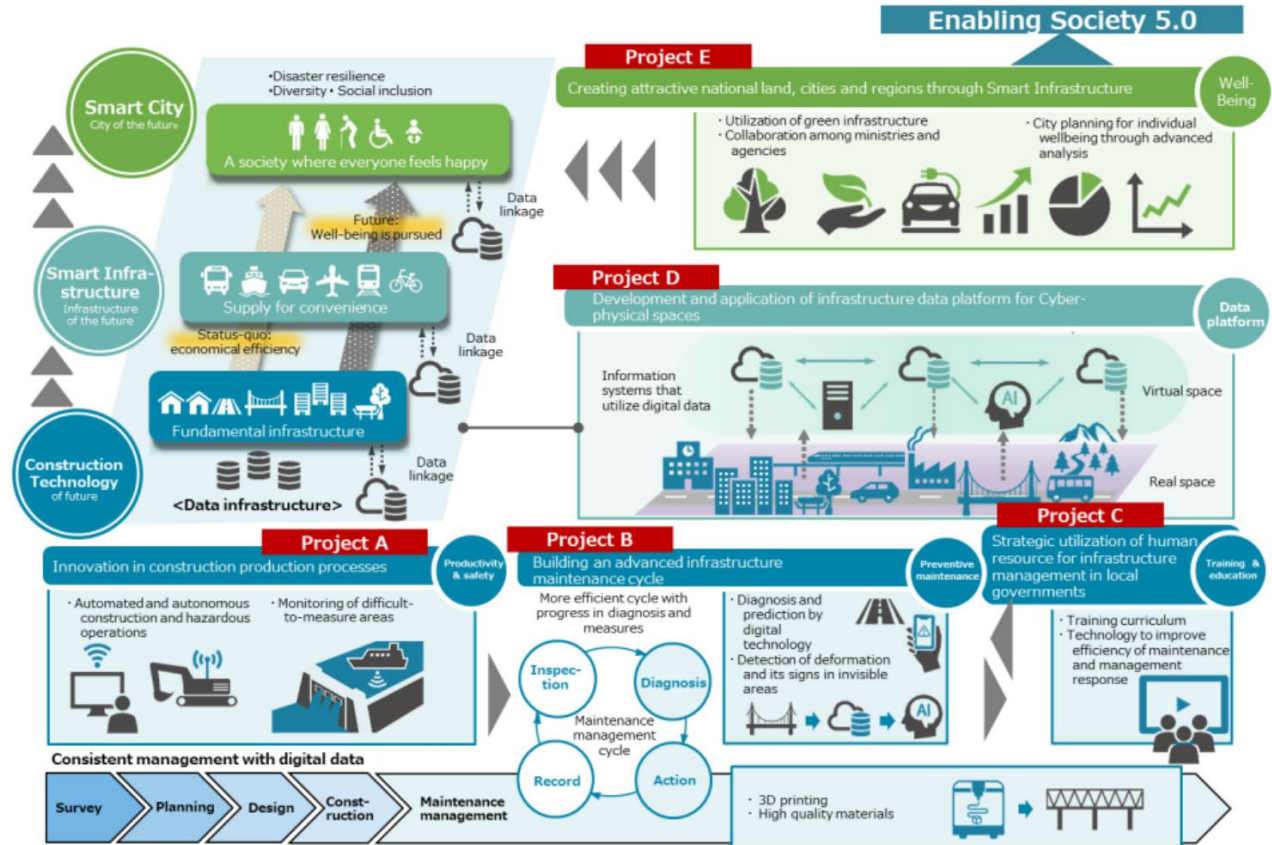
S: <https://www.healthystreets.com/case-studies/london-policy-into-practice>

S: <https://citychangers.org/healthy-streets-everyday-keeps-the-cars-away/>

Tokyo, Strategic Innovation Promotion Program (SIP)



Tokyo's Strategic Innovation Promotion Program (SIP) is an integrated management system utilizing advanced technologies and data to innovate urban infrastructure operation and maintenance for smart cities.



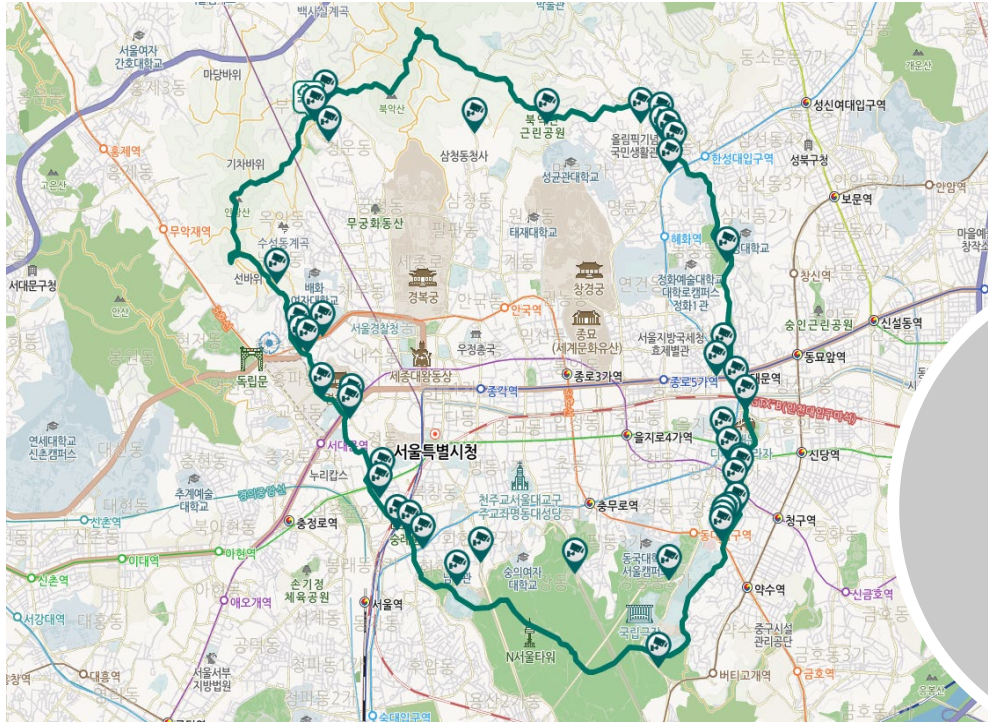
Seoul, Road space reduction at Cheonggyecheon Stream

The restoration of **Seoul, Road space reduction at Cheonggyecheon Stream** transformed a road-focused area into a relaxing, walkable street by significantly reducing vehicular space



Seoul, Green Traffic Zone

Seoul's Green Traffic Zone system manages traffic volume, environment, and congestion, using an integrated platform



Traffic volume
Management

Considering
atmospheric
environment

Traffic
Congestion
Management

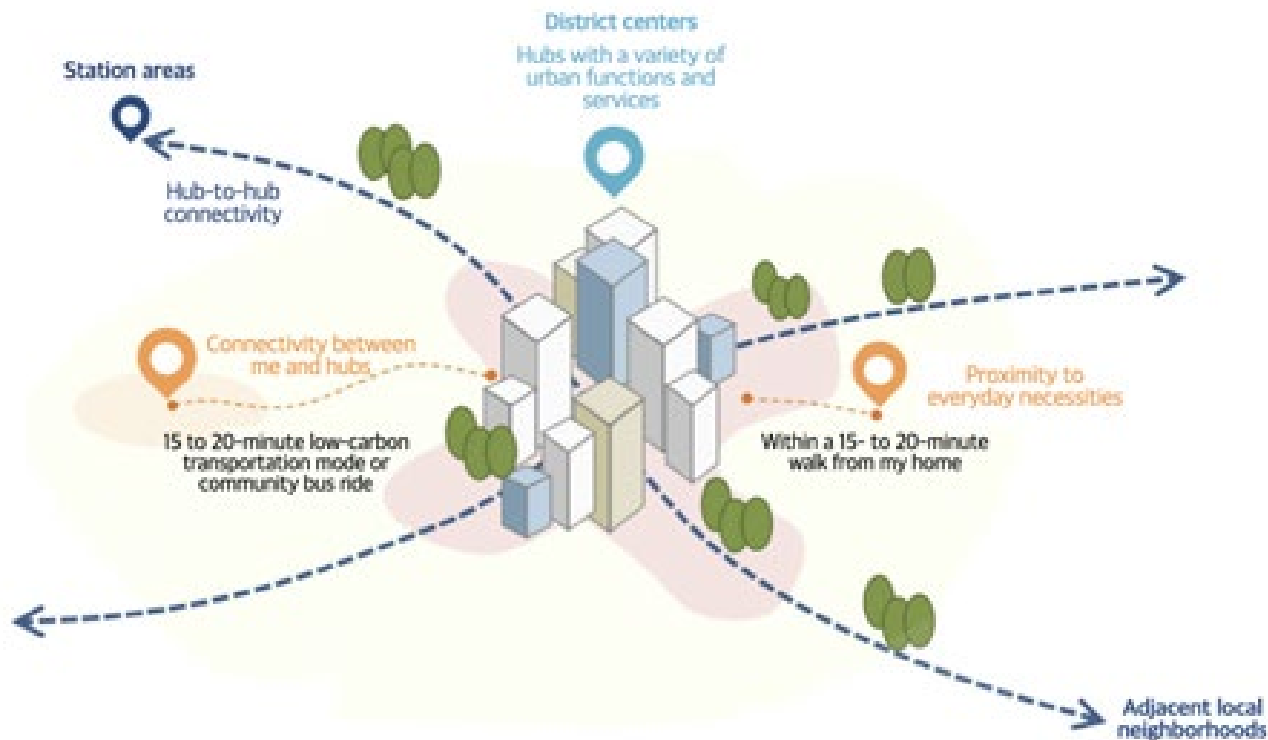
Seoul, Walkable Neighborhood Unit

Walkable Neighborhood Unit (MicroSeoul community) illustrates the vision of Seoul's future city.



Seoul, Walkable Neighborhood Unit

The Spatial Scope of a Walkable Neighborhood Unit as one of the core concepts of the Seoul 2040 plan.

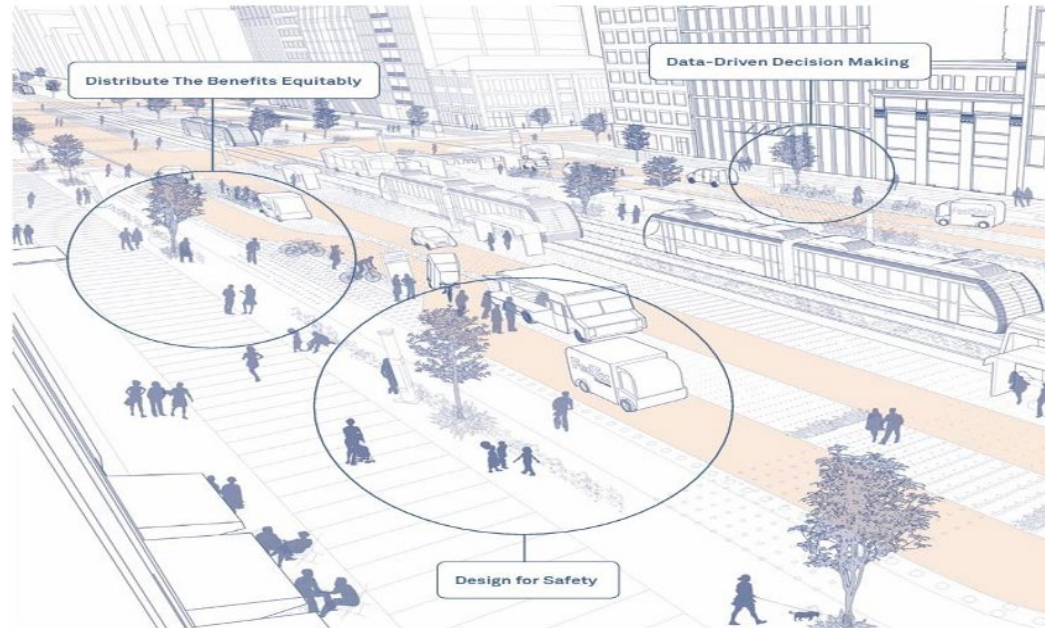


A Blueprint for Autonomous Urbanism

Urban design principles must be strategically developed to ensure a transition towards truly people-centered and resilient future streetscapes

Principles on the Future Street

Move People Not Cars, How AVs could Help, or Hurt, Cities



Automation and the Future of Unmanned Logistics

Increasing automation levels fundamentally transform mobility by moving vehicle operations and logistics terminal management towards unmanned systems.

Automation Level	Modes	Terminals
  None	All driving functions assumed by user.	All functions assumed by manually operated equipment.
  Basic	Driving assistance (e.g. adaptive cruise control), but user responsible for core driving functions.	Operation assistance (location of drop-off, storage and pick up), but manually operated equipment.
  Partial	Some driving tasks (e.g. steering, acceleration, deceleration). User monitors environment and ready to take control.	Planning and managing the use of equipment and storage space (Warehouse and yard management systems).
  Conditional	Perform most driving tasks and monitors driving environment. User must be ready to take control at request.	Semi-automatic equipment (cranes, gantries, storage stacks). Automated access to facilities (automated gates).
  High	Performs all driving tasks and monitors controlled driving environment. User does not need to take control.	Integration between automated handling and storage systems (Fully automated terminal or warehouse). Automated pick-up and deliveries.
  Full	Autonomous vehicle; Performs all driving functions under all environments. User provides destination but does not control vehicle.	Autonomous terminal; responds to demand (modal, intermodal, flows).

Intelligent Moving Cities: Technological Leap and Social Integration of Autonomous Mobile Robots

1970s		Horse and Cart Horse-drawn carriages played a crucial role in transporting goods from rural to urban areas, essential for local economic activity
1980s		Steam trains Steam locomotives rapidly developed in the 1980s, becoming key in the Industrial Revolution by efficiently transporting more cargo and passengers
1990s		90s Car Beginning in the 90s, early automobiles popularized personal transportation and, with road development, drove major societal and economic changes
2000s		Aircraft Aviation advances since the late 20th century have made air travel crucial for global mobility, shortening distances and boosting business and tourism.
2010s		Electric Hybrid Environmental and tech advances have made electric vehicles efficient, eco-friendly alternatives to traditional cars, transforming the auto market
Current		Autonomous vehicle Autonomous vehicles use AI and sensors to reduce accidents, optimize traffic, and enhance comfort, reshaping the future of transportation

Chapter 1: Systematic analysis of the need for Autonomous Robots (S:

<https://www.istockphoto.com/kr>,
<https://www.toyota.co.kr>,
<https://waymo.com>)
 S. Lee, S. (2025)



Chapter 2: Advances in Autonomous Driving Robots: Past, Present and Future (S: <https://www.darpa.mil/about/innovation-timeline>)

Sensor

It receives external stimuli and reacts to them.

Controller

It interprets the commands in turn and sends the necessary signals to direct the actions of each device.

Actuator

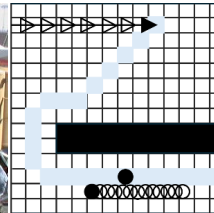
It is a part that performs certain movements corresponding to human joints and arms.

Power supply

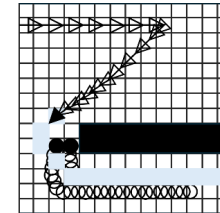
It gives all components the energy they need to operate efficiently



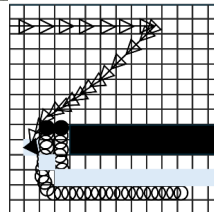
Chapter 4: Core Hardware Components and Functional Requirements (S: <https://hutechrobotics.com>)



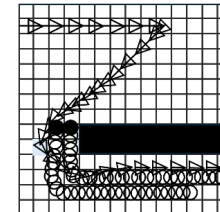
(a) Robot and obstacles approaching



(b) Avoiding two moving obstacles



(c) Just after avoiding



(d) Simulation path

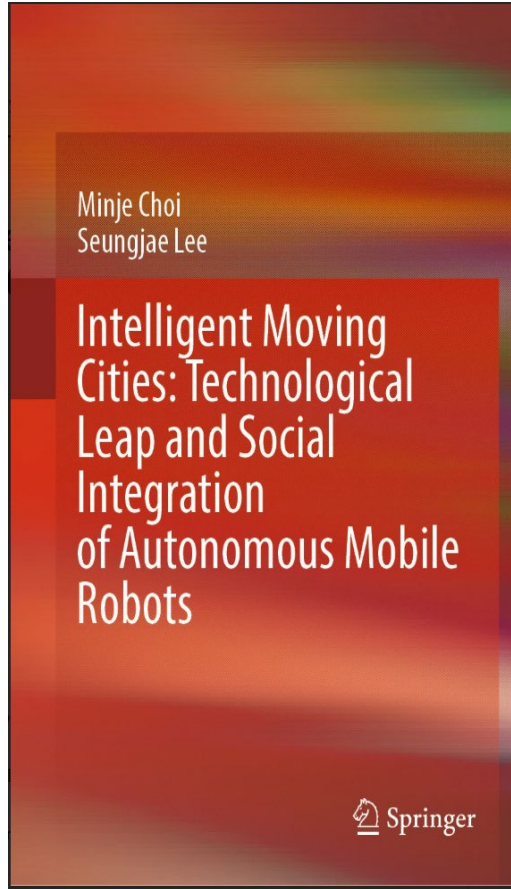
Chapter 3: Technical Considerations for Autonomous Navigation and Route Navigation (S: Jung et al., 2010)

Minje Choi
Seungjae Lee

Intelligent Moving Cities: Technological Leap and Social Integration of Autonomous Mobile Robots

Springer

How to Make it in the framework of Autonomous Mobile Robots



Chapter 1: Need for Autonomous Robots

Explains why cities need autonomous mobile robots, focusing on urban challenges and integration with existing infrastructure.

Chapter 2: Technology Evolution

Covers the development of core technologies like AI, sensors, and computer vision, and their current and future applications.

Chapter 3: Navigation Systems

Explores how robots plan and adapt routes in dynamic urban environments using real-time data and smart algorithms.

Chapter 4: Hardware and Scalability

Discusses robot design for durability, efficiency, and long-term urban use, including material and energy considerations.

Chapter 5: Real-World Applications

Highlights uses in transport, safety, logistics, and accessibility, showing how AMRs benefit urban life.

Chapter 6: Urban Integration

Examines how AMRs fit into smart city planning, sustainability goals, and infrastructure development.

Chapter 7: Challenges and Ethics

Looks at legal, social, and ethical issues, including labor impacts and policy needs for AMR adoption.

Chapter 8: Environmental Impact

Analyzes sustainability, energy use, and strategies for reducing the environmental footprint of AMRs.

Appendix

Describes how AMRs and AI assess pedestrian environments for improved urban planning.

AI future cities with mobility

(S : Seoul metropolitan government)



Discussion

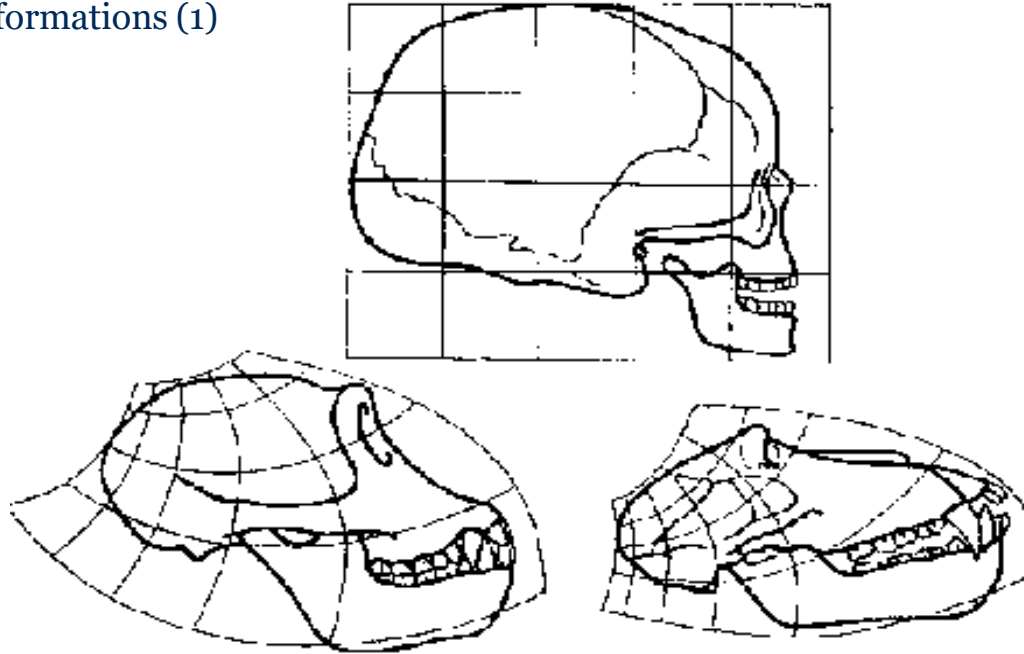


III. Discussion-

How to Transform Asymmetry

D'Arcy Wentworth Thompson's *On Growth and Form* (1917)

Transformations (1)



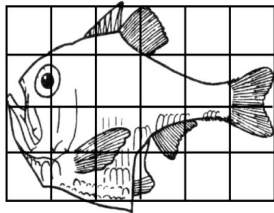
Skulls of a human, a chimpanzee and a baboon
and transformations between them

III. Discussion-

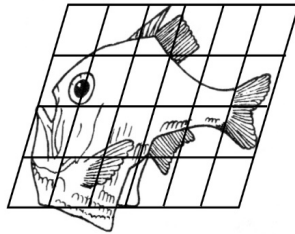
How to Transform Asymmetry

D'Arcy Wentworth Thompson's *On Growth and Form* (1917)

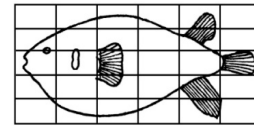
Transformations (2)



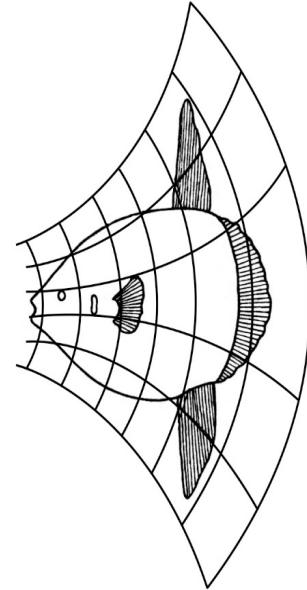
Argyroplecus



Sternoptyx



Diodon



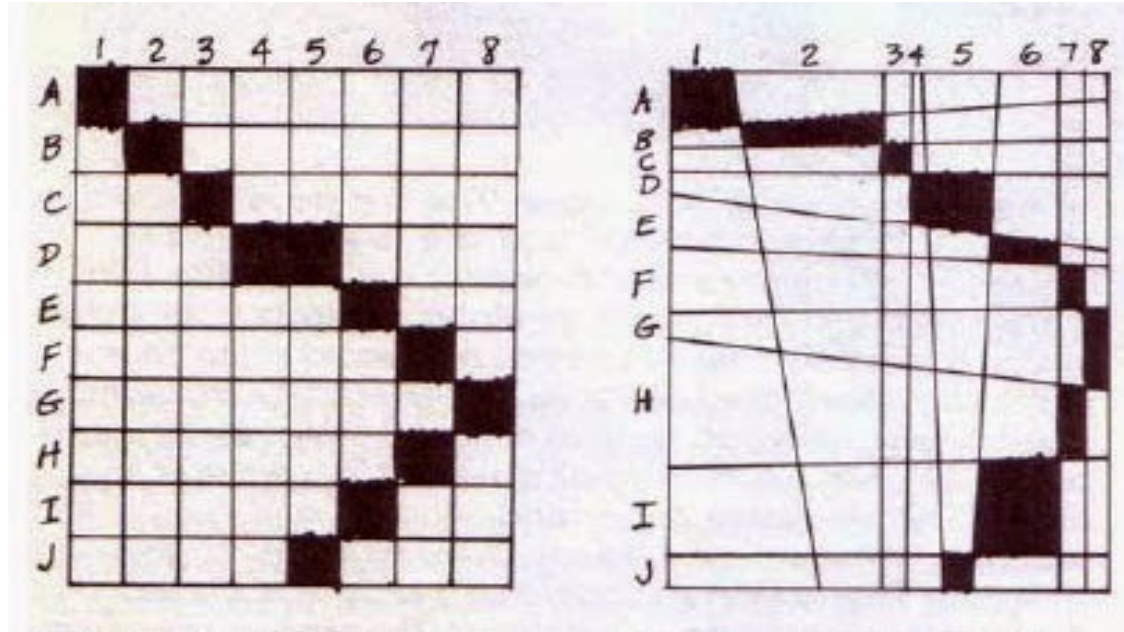
Orthogoriscus

III. Discussion-

Principle of Transformation

D'Arcy Wentworth Thompson's *On Growth and Form* (1917)

Transformations (3)



III. Discussion-

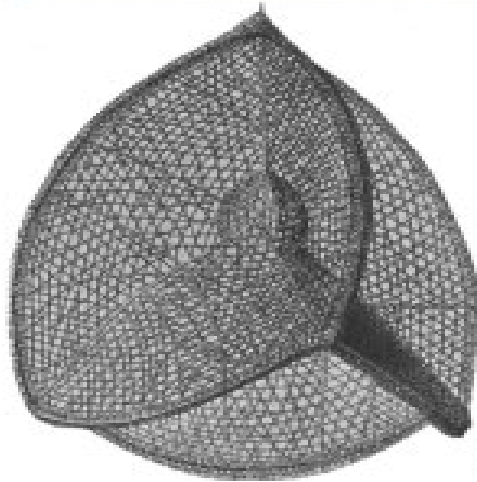
How to Manage in Restricted Splaces

D'Arcy Wentworth Thompson's On Growth and Form (1917)

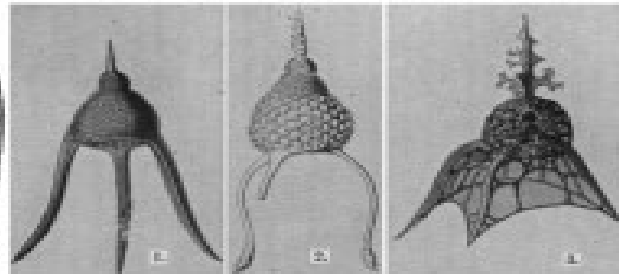
Transformations (4)

It behoves us always to remember that in Physics it has taken great men to discover simple things. They are great names indeed which we couple with the explanation of the path of a stone, the droop of a chain, the tints of a bubble, the shadows in a cup.

On Growth and Form D'Arcy Wentworth Thompson



RADIOLARIAN SKELETONS.

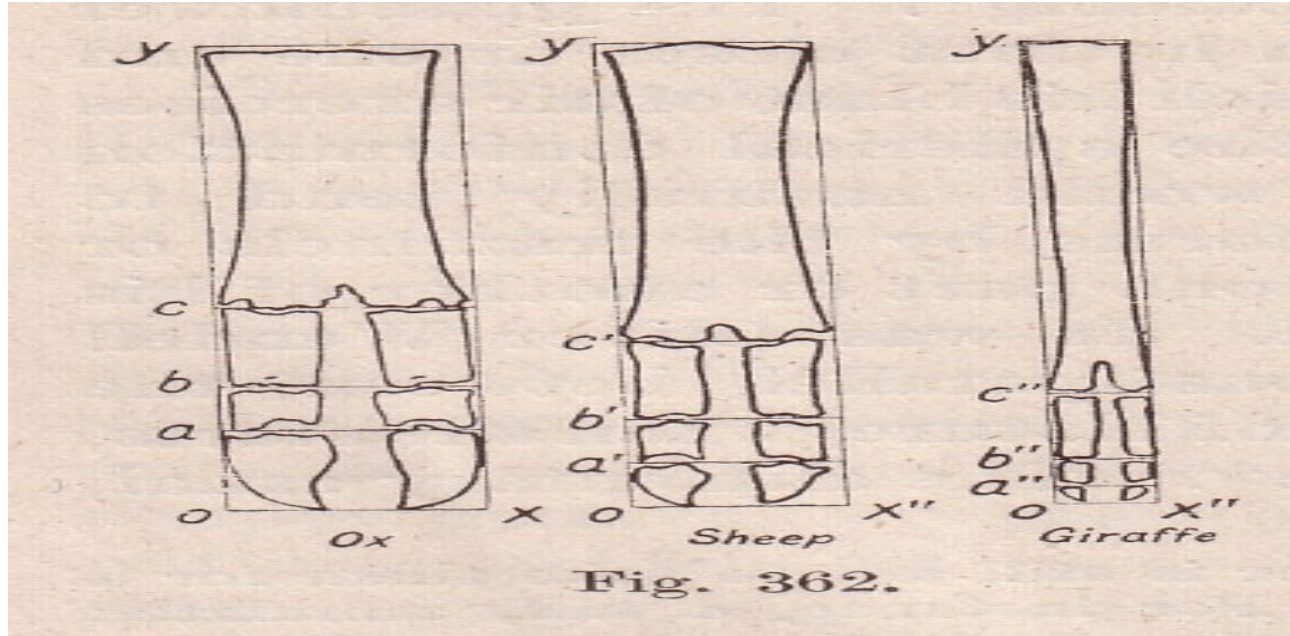


III. Discussion-

How to Build Greater Infrastructure

D'Arcy Wentworth Thompson's *On Growth and Form* (1917)

Transformations (5)



III. Discussion-

How Road Network Forms Affect Activities in Cities ?

(S: Chungho Kim, UoS, 2024)

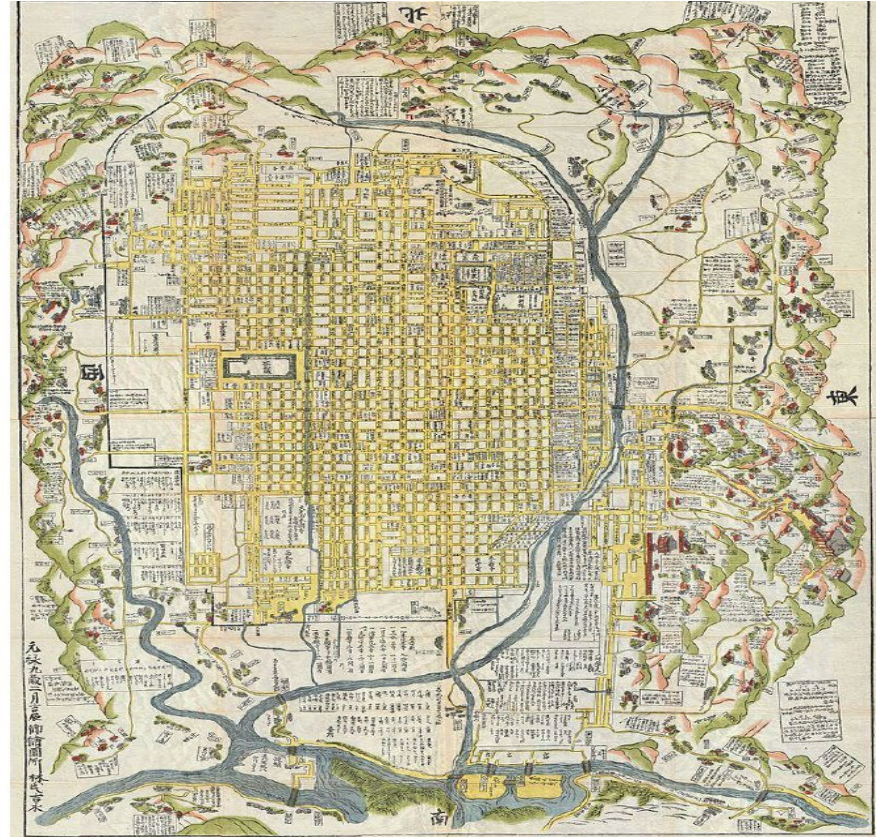
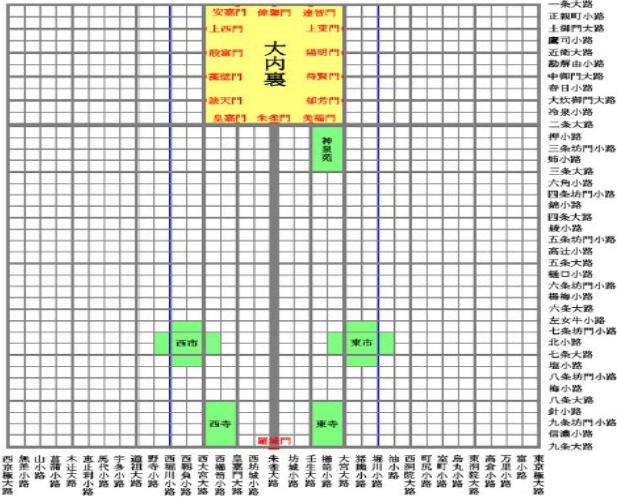
Kyoto (Heian-kyō, 平安京), 794~1868

(Source: <https://en.wikipedia.org/wiki/Heian-ky%C5%8D>)

120m x 120m

4.5km

5.2km



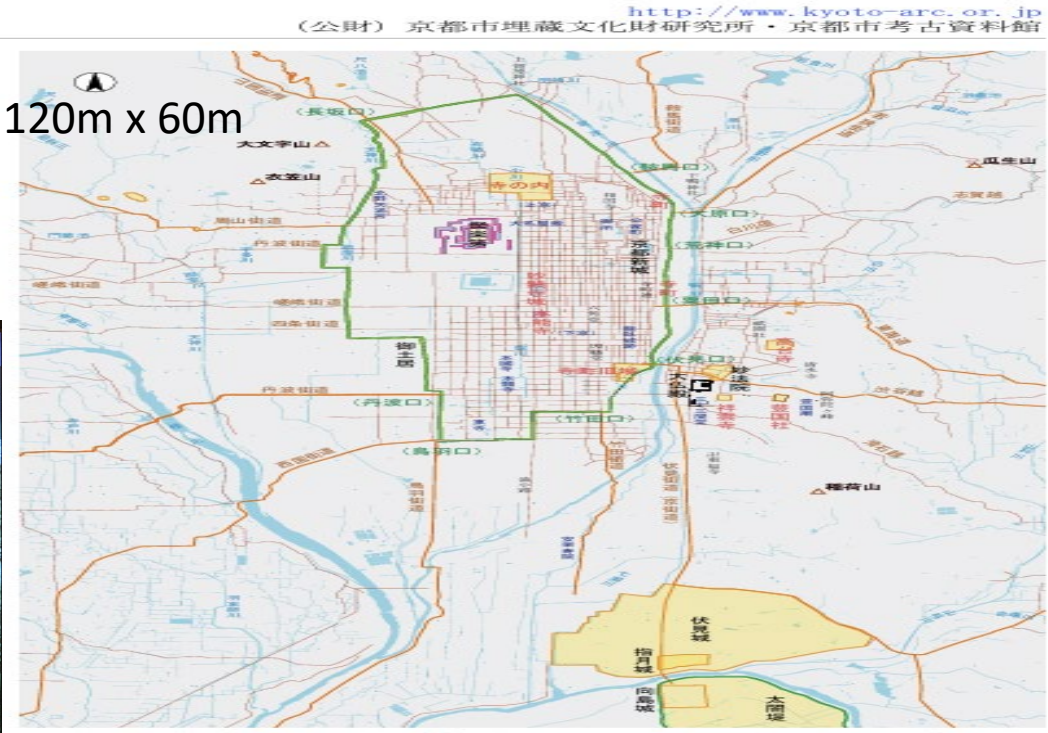
III. Discussion- Reconstruction of Kyoto by Toyotomi Hideyoshi (1537-1598:1590).

(S: <https://www.kyoto-arc.or.jp/news/leaflet/400.pdf>, NHK)

豊臣秀吉と「都づくり」



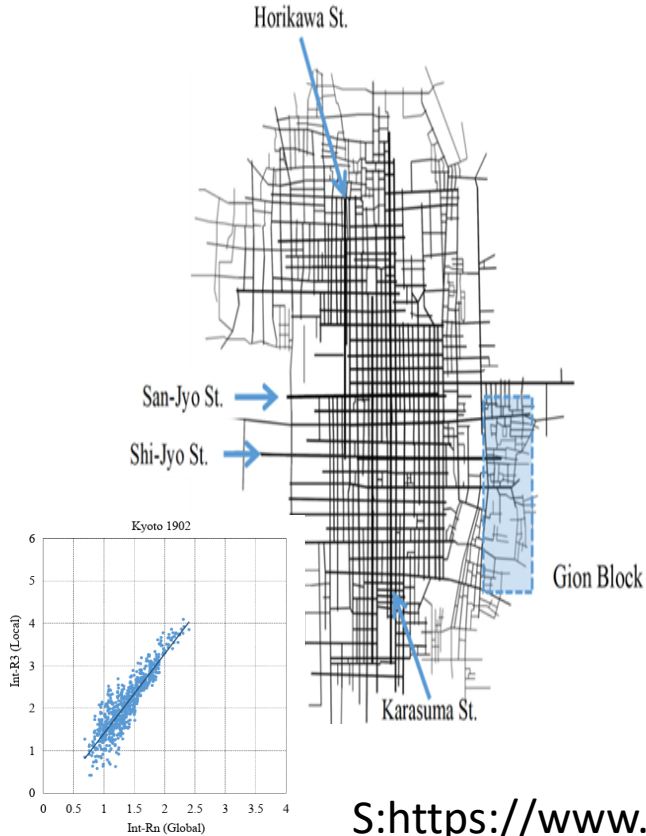
条橋・五条橋の架設、長方形街区



桃山時代の京都（「安土・桃山時代」『平安京百景 京都市平安京創生館展示図録』を調整）

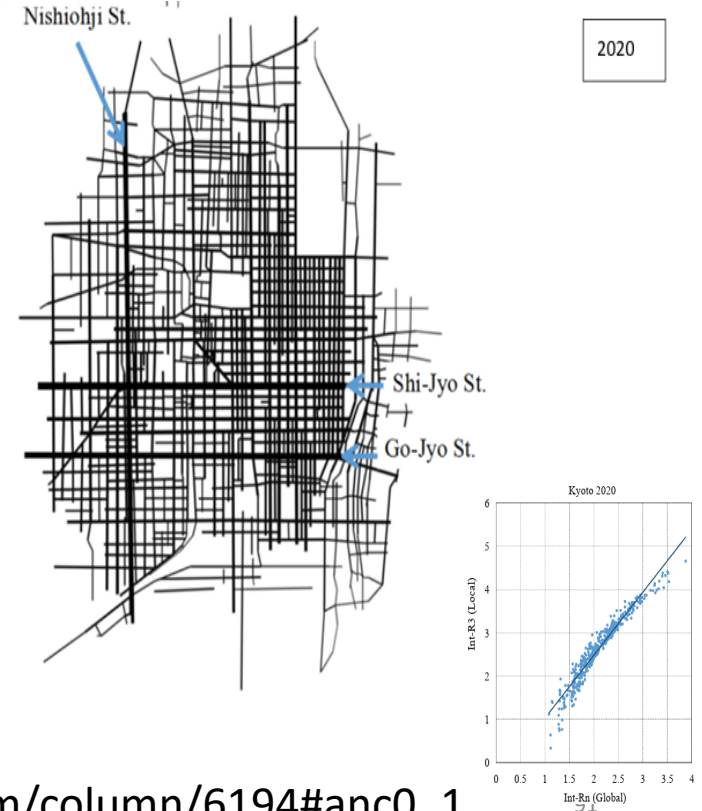
III. Discussion-How Accessibility Measures using Space Syntax have been changed in 1902 and 2020

in Kyoto ? (S: Yuan Kun, Osaka Univ, 2023)



1902

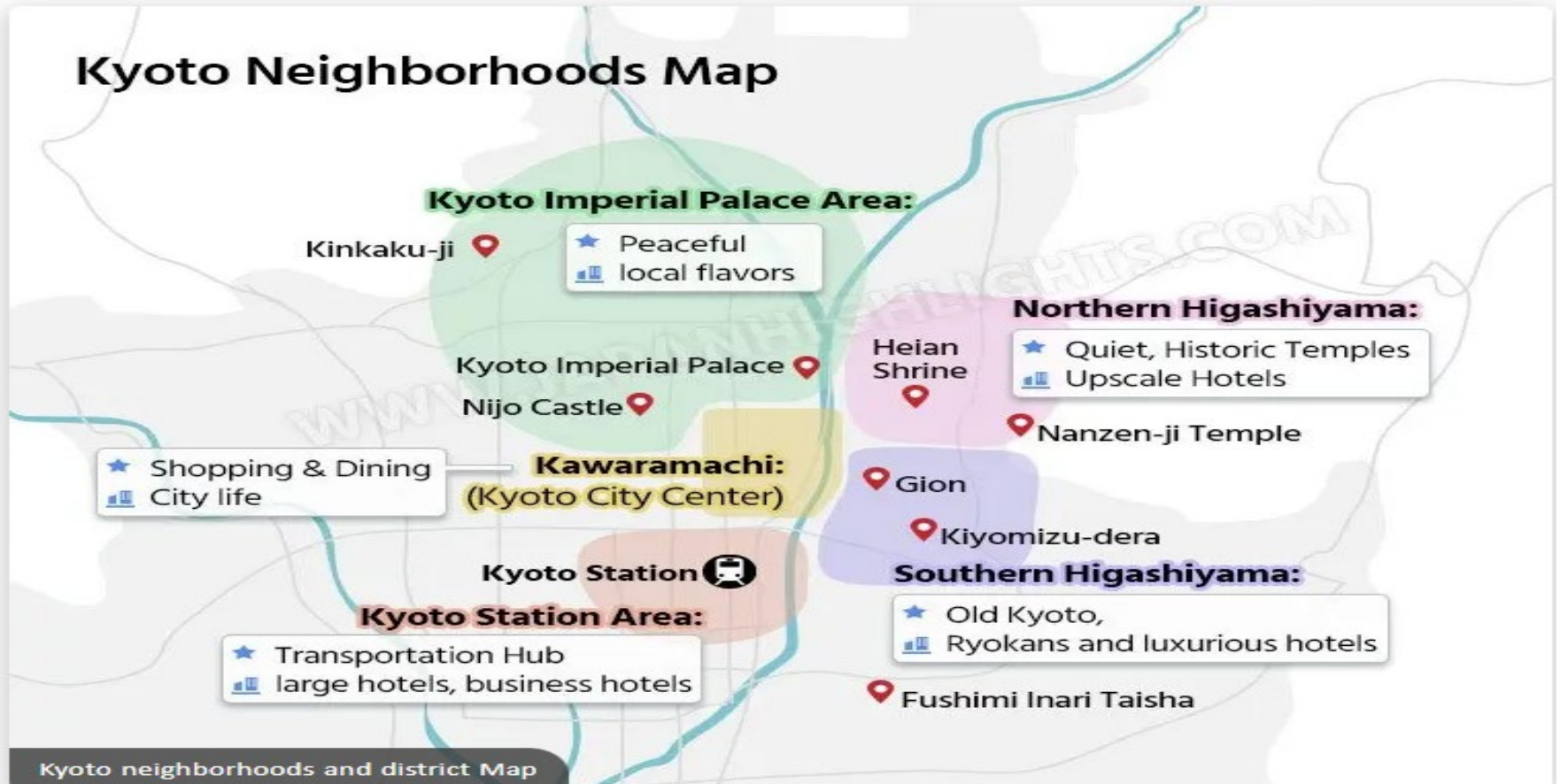
During WWII, buildings were forcibly cleared to **create firebreaks**. Streets turned into fire lanes, leading to post-war road widening.



2020

S:https://www.okamoto-kimono-en.com/column/6194#anc0_1

III. Discussion-How Tourists are Formed in Kyoto ? (S: [Kyoto Map - Tourist Maps with Attractions](#))



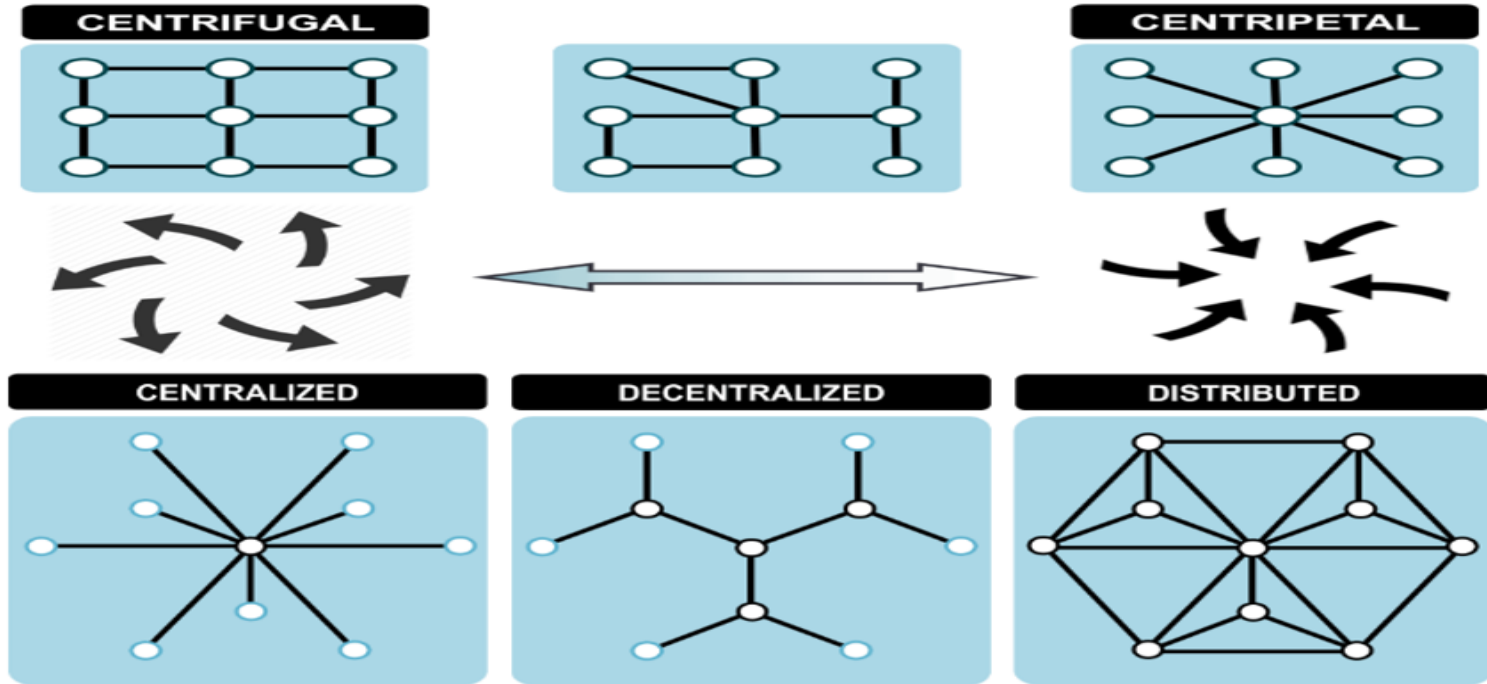
III. Discussion- How People has a Mind Map in Kyoto ?(S: Kyoto and Walkability. Lessor Feasley)

“Agaru / Sagaru” and “Higashi-iru / Nishi-iru”



A quick mind-map sketch of Kyoto's geography, transit, and grid. The black square is Kyoto Station, the nexus of the regional rail, bullet train, bus system, and subway. It serves as a buffer between the superblock sprawl of Osaka and Kyoto. The urban downtown fades into denser traditional blocks as one travels north.

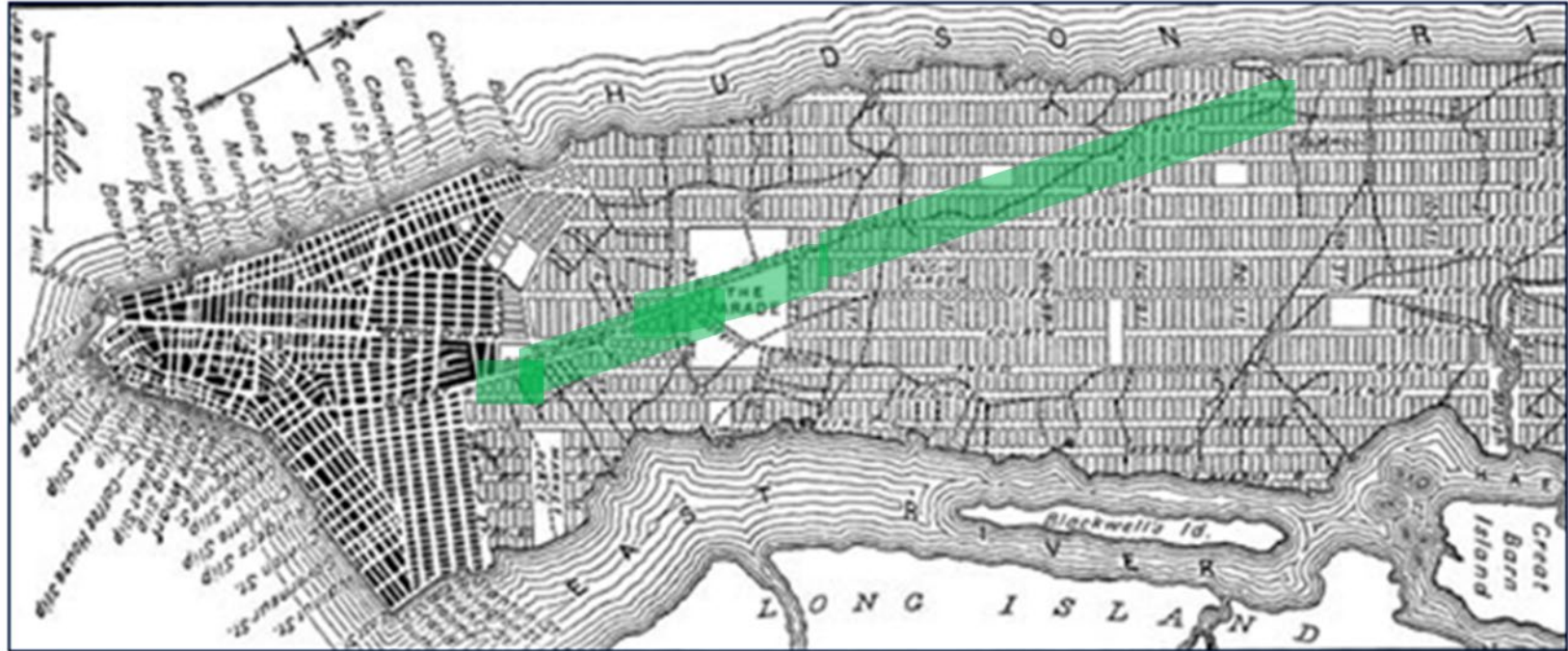
III. Discussion- How Network Structures Affect Walkability ?



(S: Jean-Paul Rodrigue (2024), The Geography of Transport Systems)

III. Discussion- Broadway, New York as a North-South Diagonal Avenue in the Grid Network

(S: [How the 1916 Zoning Law Shaped Manhattan's Central Business Districts](#))



The Commissioners' Plan of 1811 provisional map, released in 1807

Major Intersections with Broadway, New York (Time Square)

III. Discussion-

(S: [Time Square, NY city, USA editorial stock image. Image of skyscraper](#))



III. Discussion- Major Intersections with Broadway, New York (Time Square)

(S: [Times Square in New York editorial photo. Image of urban - 59787261](#))



III. Discussion-

Green Light Projects for Mid Town, New York, 2020's)

(S: [Green Light for Midtown Project - piloting city development](#))



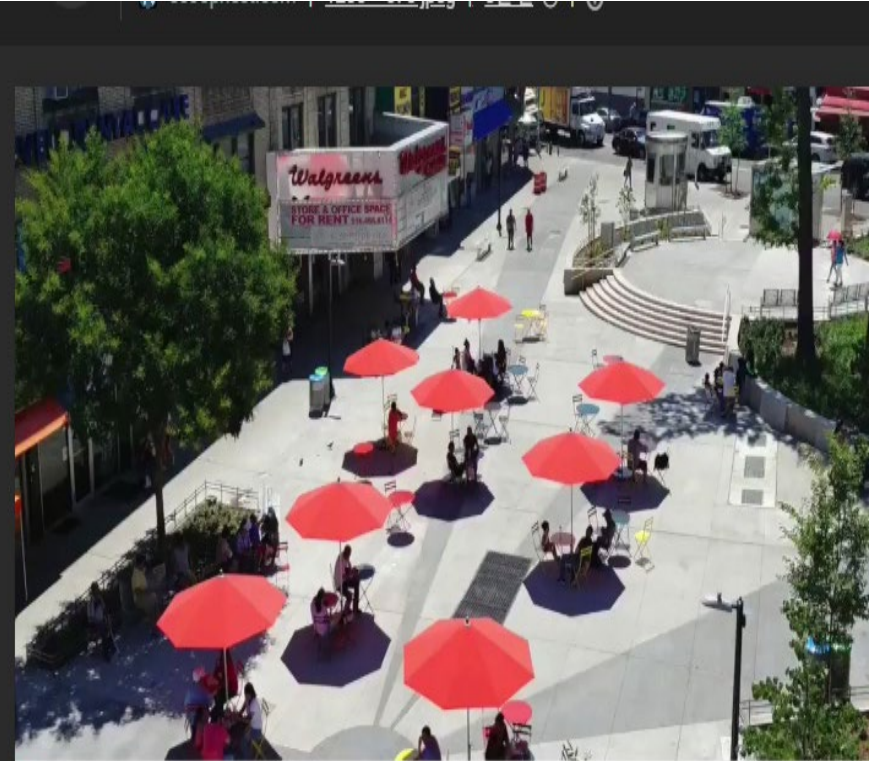
After:



III. Discussion-

Green Light Projects for Mid Town, New York, 2020's)

(S: [Green Light for Midtown Project - piloting city development](#))



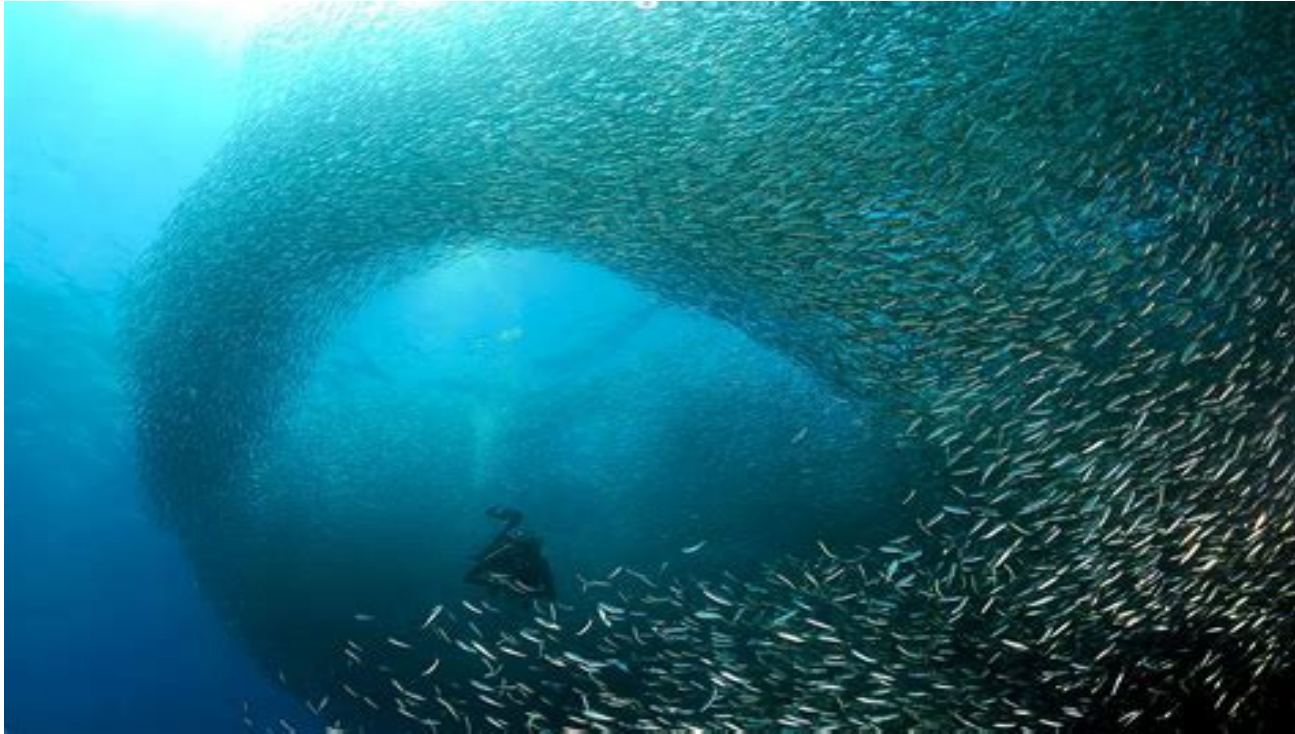
 X NYC Corona Plaza, Qns

#plazasXnyc

III. Discussion-

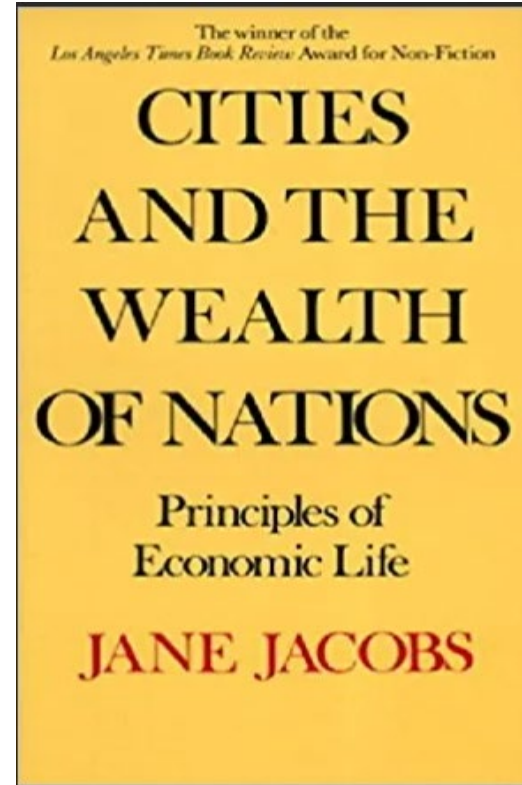
How to Avoid Crowd: Interactions`

Sardine's Shoaling and Schooling by minimizing interactions in crowds



Jane Jacobs (1916-2006)(S: Wikipedia)

- *The Death and Life of Great American Cities*
- *The Economy of Cities*
- *The Question of Separatism: Quebec and the Struggle over Sovereignty*
- *Cities and the Wealth of Nations*
- *Systems of Survival*
- *The Nature of Economies*
- *Dark Age Ahead*



III. Discussion- Social Costs and Benefits

(S: <https://www.tutorchase.com/notes/cie-a-level/economics/7-4-3-deadweight-loss-from-externalities>)

Social Costs and Benefits

- **Social Costs:** These encompass the entire burden that society bears due to an economic activity. It includes the sum of private costs incurred by businesses or individuals and the external costs borne by the community or environment.
- **Social Benefits:** These refer to the total advantages society gains from an economic activity. This includes both the direct benefits to consumers or producers and the indirect benefits enjoyed by the broader community.

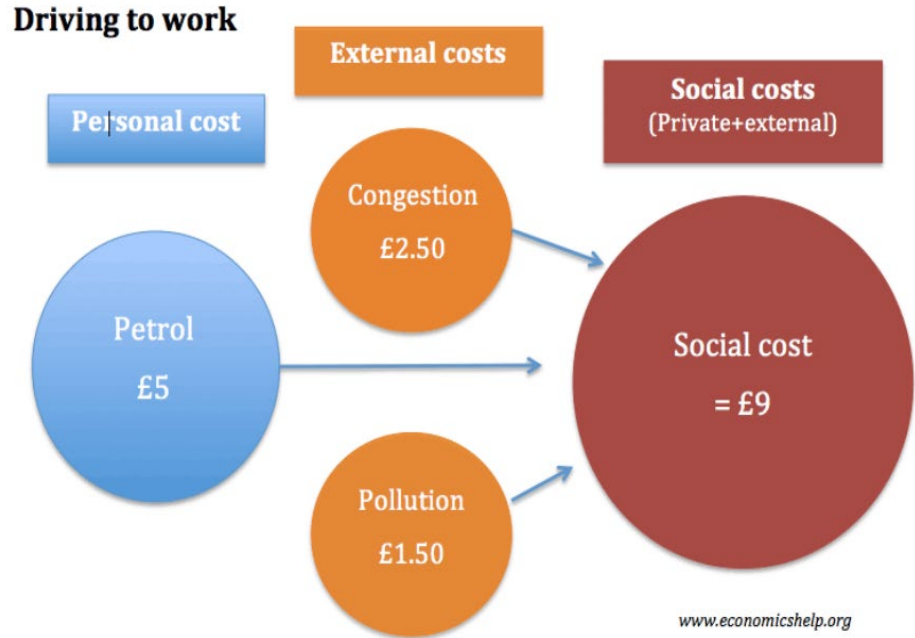


Image courtesy of [economicshelp](https://www.economicshelp.org)

III. Discussion- External Costs and Benefits

(S: <https://www.tutorchase.com/notes/cie-a-level/economics/7-4-3-deadweight-loss-from-externalities>)

External Costs and Benefits

- **External Costs (Negative Externalities):** These are unintended negative consequences of economic activities that affect third-party individuals or the environment, such as air pollution from a factory affecting local residents.
- **External Benefits (Positive Externalities):** These are the unintentional positive effects of an economic transaction on third parties, like the benefit to local biodiversity from a company's green space.

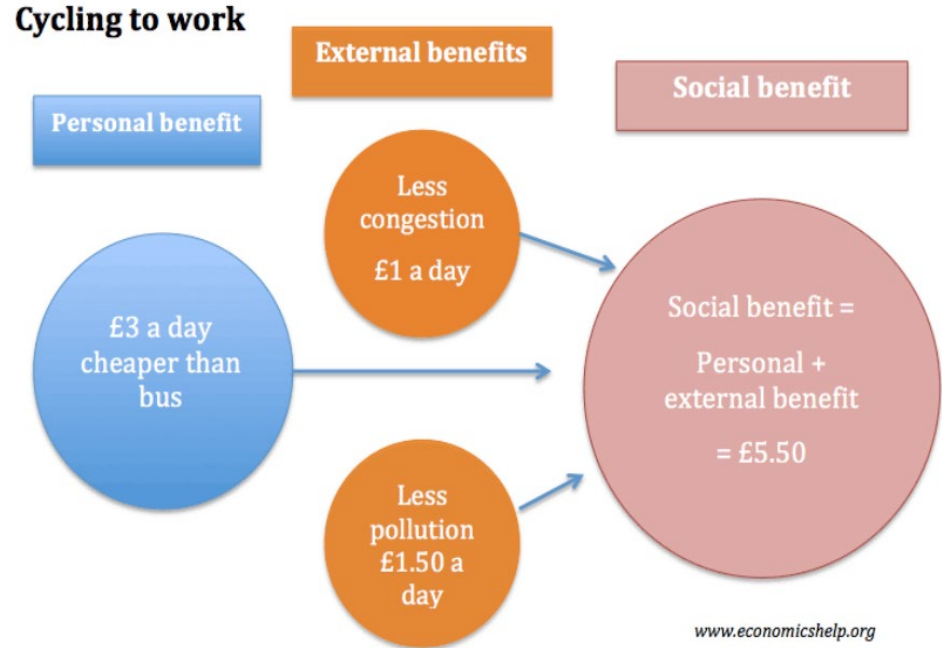


Image courtesy of [economicshelp](http://www.economicshelp.org)

III. Discussion- **Marginal Concepts in Economic Decision-Making**

(S: <https://www.tutorchase.com/notes/cie-a-level/economics/7-4-3-deadweight-loss-from-externalities>)

- **Marginal Social Cost (MSC):** This measures the additional cost to society of producing one more unit of a good or service. It is critical in assessing the point where the cost to society outweighs the benefit.
- **Marginal Private Cost (MPC):** This reflects the additional cost incurred by a producer for one extra unit of output. It's crucial for businesses in decision-making and pricing strategies.
- **Marginal Social Benefit (MSB):** This indicates the additional benefit society receives from one more unit of a good or service, encompassing both direct and indirect benefits.
- **Marginal External Benefit (MEB):** Represents the extra benefit to non-consumers or third parties for each additional unit consumed.